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République Algérienne Démocratique et Populaire

Ministry of Higher Education and Scientific Research

[UNIVERSITY NAME]

Faculty of [Natural and Life Sciences]

Department of [Agricultural Sciences / Biology]

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THESIS

Thèse de Doctorat en Sciences

SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE DEGREE OF DOCTOR OF
PHILOSOPHY IN SCIENCES

**Date Palm Pollen (Phoenix dactylifera L.) as a Natural Cryoprotective
Agent for Post-Slaughter Ovine Epididymal Sperm Preservation: An
Integrated Socio-Technical-Environmental Approach in Arid Algeria**

*Pollen de Palmier Dattier (Phoenix dactylifera L.) comme Agent Cryoprotecteur Naturel
pour la Préservation du Sperme Épididymaire Ovin Post-Abattage : Approche Socio-
Technique-Environnementale Intégrée en Algérie Aride*

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DECLARATION OF ORIGINALITY

I, the undersigned, **[Candidate Full Name]**, hereby declare that this thesis entitled "*Date Palm Pollen (Phoenix dactylifera L.) as a Natural Cryoprotective Agent for Post-Slaughter Ovine Epididymal Sperm Preservation: An Integrated Socio-Technical-Environmental Approach in Arid Algeria*" is my own original work carried out under the supervision of **Pr. [Supervisor Full Name]** and **Dr. [Co-supervisor Full Name]**. To the best of my knowledge and belief, this thesis contains no material previously published or written by another person, nor material which has been accepted for the award of any other degree or diploma of a university or other institution of higher learning, except where due acknowledgment has been made in the text.

All sources of information, data, and ideas derived from the work of others have been properly cited in accordance with academic conventions. The experimental work was conducted in accordance with the ethical standards of the institutional research committee and with the 1964 Helsinki Declaration and its later amendments, as well as the principles of post-mortem tissue collection from a licensed abattoir facility.

Signed: _____

Date: _____

Candidate: [Candidate Full Name]

Institution: [University Name], [Faculty], [Department]

Date of submission: [Month, Year]

DÉCLARATION D'ORIGINALITÉ

Je soussigné(e), **[Nom Complet du Candidat]**, déclare par la présente que cette thèse intitulée « *Pollen de Palmier Dattier (Phoenix dactylifera L.) comme Agent Cryoprotecteur Naturel pour la Préservation du Sperme Épididymaire Ovin Post-Abattage : Approche Socio-Technique-Environnementale Intégrée en Algérie Aride* » est mon propre travail original, réalisé sous la direction de **Pr. [Nom du Directeur]** et **Dr. [Nom du Co-directeur]**. À ma connaissance, cette thèse ne contient aucun matériel précédemment publié ou rédigé par une autre personne, ni aucun matériel ayant été accepté pour l'obtention d'un autre diplôme, sauf mention explicite dans le texte.

Dedication

*To my beloved family, whose unwavering support, patience,
and sacrifices have made this academic journey possible.*

*To the memory of the indigenous Ouled Djellal sheep breed — a
living testament to the genetic heritage of Algeria's arid lands,
whose preservation inspired this work.*

*To the sheep keepers of El Oued, whose ancestral knowledge and
dedication to their flocks embody the spirit of sustainable
stewardship.*

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Laboratory teams. I am grateful to the technical staff of the biochemistry, phytochemistry, and reproductive biology laboratories for their assistance with proximate analysis, mineral determination, DPPH assays, CASA motility assessment, and HOST membrane integrity evaluation. Particular thanks are due to [Laboratory Technician Name(s)] for their meticulous technical support and patience with repeated measurements.

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To all those I may have inadvertently failed to mention by name, please accept my sincere appreciation. Any errors or omissions in this thesis remain my own responsibility.

[Candidate Full Name]

El Oued, Algeria

[Month, Year]

Abstract

Background. The genetic diversity of sheep (*Ovis aries* L.) — over 1,300 breeds globally, of which approximately 27% are endangered — underpins food security and pastoral resilience in the face of accelerating climate change. In Algeria's arid zones, valuable ovine genotypes such as the Ouled Djellal breed are at risk of unrecorded loss when genetically elite rams die unexpectedly, because conventional cryobanking infrastructure relies on elective semen collection from live animals. Post-slaughter recovery of cauda epididymal spermatozoa offers a critical genetic rescue pathway, but its efficacy is constrained by oxidative deterioration during the narrow post-mortem "golden hour" window. Conventional single-mechanism antioxidants (TROLOX, cysteine) systematically fail to address the multi-source reactive oxygen species (ROS) generation characteristic of post-mortem cellular deterioration.

Objectives. This thesis investigated whether Date Palm Pollen (DPP, *Phoenix dactylifera* L.) — a complex botanical matrix of polyphenols, fatty acids, minerals, and vitamins traditionally available in the Algerian Sahara — could serve as a field-deployable, low-cost conservative agent for post-slaughter ovine epididymal sperm. Four interconnected experiments were designed to (i) characterize the socioeconomic context and trait preferences of 200 sheep keepers in El Oued; (ii) biochemically characterize local DPP (proximate, mineral, phenolic, flavonoid, DPPH antioxidant capacity); (iii) evaluate the dose-dependent efficacy of aqueous DPP extenders (DPPE-0, DPPE-40, DPPE-80 mg/mL) on chilled-storage (4°C, 48 h) sperm quality; and (iv) quantify climate-change perceptions and their linkage to perceived fertility declines among keepers.

Materials and Methods. A stratified random sample of 200 sheep keepers across eight municipalities of El Oued was surveyed via semi-structured questionnaires. DPP collected from Hamraia was characterized using AOAC (2019) proximate methods, flame photometry, Folin-Ciocalteu phenolic assay, and DPPH radical scavenging assay. Post-slaughter epididymal sperm from five sexually mature rams was recovered by retrograde flushing, diluted 1:1 in DPPE extenders, and stored at 4°C under mineral oil. Total motility, progressive motility (CASA), and membrane integrity (HOST) were assessed at 0, 24, and 48 h. Statistical analyses included MANOVA (Experiment 1), two-way repeated-measures ANOVA (Experiment 3), and binary logistic regression (Experiment 4), with significance set at $\alpha = 0.05$.

Results. The socioeconomic survey revealed a male-dominated (100%), aging keeper population (only 11% under 25 years) with small-to-medium flocks (79.5% < 50 head), prioritizing breed identity (27% for rams) and drought tolerance (16% for ewes) over rapid growth — a significant three-way interaction (Pillai's Trace = 0.070, $p = 0.009$, $\eta^2 = 0.070$). Local DPP exhibited a nutrient-dense profile (protein 37.94%, carbohydrates 30.12%, potassium 1140 mg/100 g) with moderate antioxidant capacity ($IC_{50} = 624.25 \mu\text{g/mL}$). DPPE-80 significantly preserved total motility (67.5% vs. 58.3% control), progressive motility (54.7% vs. 45.5%), and membrane integrity (63.9% vs. 55.1%) after 48 h ($p < 0.05$, $\eta^2 > 0.20$), confirming dose-dependent protection. Climate perception data showed near-universal recognition (95%) of climate-fertility relationships, with nutritional stress cited as the primary stressor (48.9%); medium-sized flocks (20-50 head) emerged as the vulnerability sweet spot (OR = 24.86, $p = 0.004$).

Conclusion. This thesis establishes that aqueous Date Palm Pollen extender at 80 mg/mL provides a validated, field-deployable protocol for post-slaughter ovine genetic rescue, extending the practical

preservation window from <24 h to >48 h. By integrating socioeconomic priorities, biochemical characterization, technical efficacy, and climate-change urgency into a triangulated conservation model, the research advances both theory (Agro-Ecological Cryobiology) and practice (community-based cryobanking). The DPPE protocol offers a culturally appropriate, low-cost intervention for safeguarding ovine genetic resources in arid Algeria and comparable pastoral systems globally.

Keywords: *Date palm pollen; Ovine epididymal sperm; Post-slaughter recovery; Genetic conservation; Cryoprotection; Antioxidant; Chilled storage; El Oued; Algeria; Climate change; Ouled Djellal sheep; Socio-technical model.*

Résumé

Contexte. La diversité génétique des ovins (*Ovis aries* L.) — plus de 1 300 races dans le monde, dont environ 27 % sont menacées — constitue le fondement de la sécurité alimentaire et de la résilience pastorale face au changement climatique accéléré. Dans les zones arides d'Algérie, des génotypes ovins précieux tels que la race Ouled Djellal sont exposés à une perte non documentée lorsque des béliers génétiquement supérieurs meurent de manière inattendue, car l'infrastructure conventionnelle de cryoconservation s'appuie sur la collecte élective de semence provenant d'animaux vivants. La récupération post-abattage des spermatozoïdes épидидymaires caudaux offre une voie critique de sauvetage génétique, mais son efficacité est limitée par la détérioration oxydative pendant la fenêtre étroite de l'« heure dorée » post-mortem. Les antioxydants conventionnels à mécanisme unique (TROLOX, cystéine) échouent systématiquement à traiter la génération multi-source d'espèces réactives de l'oxygène (ROS) caractéristique de la détérioration cellulaire post-mortem.

Objectifs. Cette thèse a examiné si le pollen de palmier dattier (DPP, *Phoenix dactylifera* L.) — une matrice botanique complexe de polyphénols, d'acides gras, de minéraux et de vitamines traditionnellement disponible dans le Sahara algérien — pouvait servir d'agent conservateur déployable sur le terrain et à faible coût pour le sperme épидидymaire ovin post-abattage. Quatre expériences interconnectées ont été conçues pour (i) caractériser le contexte socioéconomique et les préférences de traits de 200 éleveurs ovins à El Oued ; (ii) caractériser biochimiquement le DPP local (composition proximale, minérale, phénolique, flavonoïque et capacité antioxydante DPPH) ; (iii) évaluer l'efficacité dose-dépendante des extendeurs aqueux de DPP (DPPE-0, DPPE-40, DPPE-80 mg/mL) sur la qualité du sperme en stockage réfrigéré (4 °C, 48 h) ; et (iv) quantifier les perceptions du changement climatique et leur lien avec les déclinés de fertilité perçus.

Matériel et méthodes. Un échantillon aléatoire stratifié de 200 éleveurs ovins à travers huit communes d'El Oued a été enquêté via des questionnaires semi-structurés. Le DPP collecté à Hamraia a été caractérisé en utilisant les méthodes proximales AOAC (2019), la photométrie de flamme, le dosage phénolique Folin-Ciocalteu et le test de piégeage du radical DPPH. Le sperme épидидymaire post-abattage de cinq béliers sexuellement matures a été récupéré par rinçage rétrograde, dilué 1:1 dans les extendeurs DPPE, et stocké à 4 °C sous huile minérale. La motilité totale, la motilité progressive (CASA) et l'intégrité membranaire (HOST) ont été évaluées à 0, 24 et 48 h. Les analyses statistiques comprenaient MANOVA (Expérience 1), ANOVA à mesures répétées à deux facteurs (Expérience 3) et régression logistique binaire (Expérience 4), avec seuil de signification $\alpha = 0,05$.

Résultats. L'enquête socioéconomique a révélé une population d'éleveurs dominée par les hommes (100 %), vieillissante (seulement 11 % de moins de 25 ans) avec des troupeaux petits à moyens (79,5 % < 50 têtes), privilégiant l'identité de race (27 % pour les béliers) et la tolérance à la sécheresse (16 % pour les brebis) — une interaction tripartite significative (Trace de Pillai = 0,070, $p = 0,009$, $\eta^2 = 0,070$). Le DPP local présentait un profil riche en nutriments (protéines 37,94 %, glucides 30,12 %, potassium 1 140 mg/100 g) avec une capacité antioxydante modérée ($IC_{50} = 624,25 \mu\text{g/mL}$). Le DPPE-80 a significativement préservé la motilité totale (67,5 % vs 58,3 % témoin), la motilité progressive (54,7 % vs 45,5 %) et l'intégrité membranaire (63,9 % vs 55,1 %) après 48 h ($p < 0,05$, $\eta^2 > 0,20$). Les données de perception climatique ont montré une reconnaissance quasi-universelle (95 %) des relations

climat-fertilité, les troupeaux moyens (20-50 têtes) émergeant comme point critique de vulnérabilité (OR = 24,86, $p = 0,004$).

Conclusion. Cette thèse établit que l'extendeur aqueux de pollen de palmier dattier à 80 mg/mL fournit un protocole validé et déployable sur le terrain pour le sauvetage génétique ovin post-abattage, étendant la fenêtre pratique de préservation de <24 h à >48 h. En intégrant les priorités socioéconomiques, la caractérisation biochimique, l'efficacité technique et l'urgence climatique dans un modèle de conservation triangulé, la recherche fait progresser à la fois la théorie (cryobiologie agro-écologique) et la pratique (cryoconservation communautaire).

Mots-clés : Pollen de palmier dattier ; Sperme épидидymaire ovin ; Récupération post-abattage ; Conservation génétique ; Cryoprotection ; Antioxydant ; Conservation réfrigérée ; El Oued ; Algérie ; Changement climatique ; Race Ouled Djellal ; Modèle socio-technique.

ملخص

مساراً حاسماً لإنقاذ الجينات، لكن فعاليته مقيدة بالتدهور التأكسدي خلال نافذة الساعة الذهبية الضيقة بعد الوفاة. تعتمد على جمع السائل المنوي اختياريًا من الحيوانات الحية. إن استرجاع النطاف البربخية الذيلية بعد الذبح يوفر الفقد غير الموثق عندما تنفق الكباش الوراثية المتميزة بشكل غير متوقع، لأن البنية التحتية التقليدية للحفاظ بالتجميد مواجهة تغير المناخ المتسارع. وفي المناطق القاحلة بالجزائر، تواجه الأنماط الجينية القيمة كالأغنام أولاد جلال خطر بأكثر من ١٣٠٠ سلالة في العالم، منها حوالي ٢٧ مهددة بالانقراض الركيزة الأساسية للأمن الغذائي ومرونة الرعي في الخلفية: يعد التنوع الجيني للأغنام .

المخزنة على البرد ٤م، ٤٨ ساعة؛ ٤ تحديد تصورات تغير المناخ وارتباطها بانخفاض الخصوبة الملاحظ. لغبار الطلع المحلي؛ ٣ تقييم الفعالية المعتمدة على الجرعة لمستخلصات غبار الطلع المائية على جودة النطاف السياق الاجتماعي والاقتصادي وتفضيلات السمات لدى ٢٠٠ مربّي أغنام في الوادي؛ ٢ التوصيف الكيميائي الحيوي حافظ ميداني منخفض التكلفة للنطاف البربخية للأغنام بعد الذبح. صممت أربع تجارب مترابطة لـ: ١ توصيف الفينول والأحماض الدهنية والمعادن والفيتامينات، متوفرة تقليدياً في الصحراء الجزائرية يمكن أن يعمل كعامل ما إذا كان غبار طلع نخيل التمر . وهو مصفوفة نباتية معقدة من متعددات

الأهداف: بحثت هذه الأطروحة في

ساعة. شملت التحليلات الإحصائية و متكررة القياس ثنائية الاتجاه والانحدار اللوجستي الثنائي. الزيوت المعدنية. تم تقييم الحركة الكلية والحركة التقدمية وسلامة الغشاء عند ٢٤ و ٤٨ كباش ناضجة جنسيا عن طريق الغسل الرجوعي، وتم تخفيفها ١:١ في مستخلصات غبار الطلع، وتخزينها عند ٤م تحت ، واختبار الكسحي للجذور. تم استرجاع النطاف البربخية بعد الذبح من خمسة غبار الطلع المجموع من الحمرايا باستخدام طرق ٢٠١٩ التقريبية، قياس الطيف اللهي، اختبار الفينول عشوائية طبقية مكونة من ٢٠٠ مربّي أغنام عبر ثماني بلديات في الوادي باستخدام استبيانات شبه منظمة. تم توصيف المواد والطرق: تم مسح عينة

بعلاقات المناخ-الخصوبة، مع ظهور القطعان المتوسطة ٢٠-٥٠ رأساً كنقطة ضعف حرجة ٢٤.٨٦، ٠.٠٠٤.

مقابل ٥٥.١ بعد ٤٨ ساعة ٠.٠٥، ٠.٢٠. أظهرت بيانات الإدراك المناخي اعترافاً شبه عالمي ٩٥ على الحركة الكلية ٦٧.٥ مقابل ٥٨.٣ للشاهد، الحركة التقدمية ٥٤.٧ مقابل ٤٥.٥، وسلامة الغشاء ٦٣.٩ ملغ ١٠٠ غ مع قدرة مضادات أكسدة معتدلة ٦٢٤.٢٥ ميكروغرام/مل. حافظ ٨٠- بشكل معنوي ٠.٠٠٩، ٠.٠٧٠. أظهر غبار الطلع المحلي ملفاً غذائياً غنياً بروتين ٣٧.٩٤، كربوهيدرات ٣٠.١٢، بوتاسيوم ١١٤٠ الأولوية لهوية السلالة ٢٧ للكباش وتحمل الجفاف ١٦ للنعاج تفاعل ثلاثي معنوي أثير بيلاي ٠.٠٧٠، وشيخوخة السكان ١١ فقط تحت ٢٥ سنة مع قطعان صغيرة إلى متوسطة ٧٩.٥ ٥٠ رأساً، مع إعطاء

النتائج: كشف المسح الاجتماعي والاقتصادي عن مجموعة من المربين يهيمن عليها الذكور ١٠٠،

نموذج حفظ مثلث، تقدم البحث مساهمة نظرية علم التبريد البيئي الزراعي وعملية الحفاظ بالتجميد المجتمعي. خلال دمج الأولويات الاجتماعية والاقتصادية، والتوصيف الكيميائي الحيوي، والفعالية التقنية، والإلحاح المناخي في للتطبيق ميدانياً لإنقاذ الجينات الأغنامية بعد الذبح، مما يمد نافذة الحفاظ العملي من ٢٤ ساعة إلى ٤٨ ساعة. ومن تثبت هذه الأطروحة أن مستخلص غبار طلع نخيل التمر المائي بتركيز ٨٠ ملغ/مل يوفر بروتوكولا متحققاً منه وقابلاً

الخلاصة:

الأكسدة؛ الحفظ المبرد؛ الوادي؛ الجزائر؛ تغير المناخ؛ سلالة أولاد جلال؛ النموذج الاجتماعي التقني؛ الكلمات المفتاحية
غبار طلع نخيل التمر؛ النطاف البربخية للأغنام؛ الاسترجاع بعد الذبح؛ الحفظ الجيني؛ الحماية بالتجميد؛ مضادات

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List of Abbreviations and Acronyms

AI	Artificial Insemination
AnGR	Animal Genetic Resources
AOAC	Association of Official Analytical Chemists
ANOVA	Analysis of Variance
ATP	Adenosine Triphosphate
BMPR-IB	Bone Morphogenetic Protein Receptor 1B
CASA	Computer-Assisted Sperm Analysis
CAT	Catalase
CI	Confidence Interval
CV	Coefficient of Variation
DHA	Docosahexaenoic Acid
DPP	Date Palm Pollen (<i>Phoenix dactylifera</i> L.)
DPPH	2,2-diphenyl-1-picrylhydrazyl
DPPE	Date Palm Pollen Extender
EDTA	Ethylenediaminetetraacetic Acid
FAO	Food and Agriculture Organization of the United Nations
GAE	Gallic Acid Equivalents
GPx	Glutathione Peroxidase
HOST	Hypo-Osmotic Swelling Test
IC₅₀	Half-maximal Inhibitory Concentration
IVF	In Vitro Fertilization
MANOVA	Multivariate Analysis of Variance
MCB	Maître de Conférences B
NaCl	Sodium Chloride
NPE	NaCl Pollen Extender
Nrf2	Nuclear factor erythroid 2-related factor 2
OR	Odds Ratio
PUFA	Polyunsaturated Fatty Acid
QE	Quercetin Equivalents
RM-ANOVA	Repeated-Measures Analysis of Variance
ROS	Reactive Oxygen Species
SCA	Sperm Class Analyzer
SD	Standard Deviation

SEM	Standard Error of the Mean
SOD	Superoxide Dismutase
SPSS	Statistical Package for the Social Sciences
STR	Straightness (CASA parameter)
TROLOX	6-hydroxy-2,5,7,8-tetramethylchroman-2-carboxylic acid (vitamin E analog)
VAP	Average Path Velocity (CASA parameter)
VCL	Curvilinear Velocity (CASA parameter)
VSL	Straight-line Velocity (CASA parameter)

List of Symbols and Units

α	Alpha	Significance level (Type I error rate), typically 0.05
η^2	Eta-squared	Effect size measure (partial eta-squared in MANOVA)
μ	Mu	Population mean; also micro- (10^{-6} prefix)
σ	Sigma	Population standard deviation
χ^2	Chi-squared	Statistical test for categorical data association
p	p-value	Probability of observing data under null hypothesis
OR	Odds Ratio	Measure of association in logistic regression
CI	Confidence Interval	Range estimating population parameter
n	Sample size	Number of observations in a sample
°C	Degree Celsius	Temperature unit
mg/mL	Milligram per milliliter	Concentration unit (DPPE treatments)
$\mu\text{m/s}$	Micrometer per second	Sperm velocity unit (CASA)
mOsm/kg	Milliosmole per kilogram	Osmolality unit
nm	Nanometer	Wavelength unit (spectrophotometry)
rpm	Revolutions per minute	Centrifugation speed
g	Relative centrifugal force	Gravitational acceleration unit
kHz	Kilohertz	Frequency unit
μL	Microliter	Volume unit ($1\ \mu\text{L} = 10^{-6}\ \text{L}$)
mL	Milliliter	Volume unit ($1\ \text{mL} = 10^{-3}\ \text{L}$)
nmol	Nanomole	Amount of substance ($1\ \text{nmol} = 10^{-9}\ \text{mol}$)

CHAPTER 1

General Introduction

Introduction Générale

1.1 Background of the Study

Sheep (*Ovis aries* L.) constitute one of humanity's most genetically diverse livestock species, with more than 1,300 recognized breeds worldwide, each representing unique adaptations to extreme environmental conditions including high-altitude hypoxia, arid-zone heat stress, disease endemicity, and nutritional marginality (Wanjala, 2025; Dossybayev, 2025; Hansen, 2025). This genetic diversity underpins global food security, ecosystem resilience, and the capacity of pastoral systems to adapt to accelerating climate change (Muthusamy, 2025; Ma, 2025). Yet this biological heritage faces an unprecedented threat: approximately 27% of sheep breeds are currently classified as endangered, with demographic bottlenecks, uncontrolled crossbreeding, and intensifying environmental pressures accelerating genetic erosion (Motaung, 2024; Barłowska, 2025; Tampaki, 2025). The conservation of animal genetic resources (AnGR) has consequently emerged as a strategic global priority, with the Food and Agriculture Organisation (FAO) promoting integrated *in situ* and *ex situ* strategies to safeguard biodiversity and address economic, ecological, social, and cultural demands (Engdawork et al., 2024).

Within the Algerian context, sheep production occupies a cornerstone position in the national agricultural economy and rural livelihoods. The Ouled Djellal breed, in particular, represents the dominant Algerian sheep population, prized for its adaptation to arid and semi-arid steppe environments, its meat production qualities, and its resilience under marginal nutritional conditions (Sebkhi et al., 2024; Laouadi et al., 2018a). However, the breed faces mounting pressures from climate change, characterized by increasing frequency of droughts, declining pasture productivity, and intensified thermal stress — all of which elevate mortality rates in genetically valuable breeding stock and accelerate the risk of irreversible genetic loss. The El Oued region of southeast Algeria, situated at the heart of the Algerian Sahara, exemplifies these intersecting challenges: it is simultaneously a major centre of date palm cultivation and a substantial sheep production zone, creating a unique geographic convergence between an underexploited botanical resource (date palm pollen) and a pressing conservation need (ovine genetic rescue).

Among *ex situ* conservation approaches, germplasm cryobanking — particularly sperm cryopreservation — provides a cost-effective, long-term solution enabling breed reconstitution, international genetic exchange, and targeted genetic improvement (Castro et al., 2025; Shayestehyekta et al., 2025; Boes et al., 2023). However, conventional cryobanking relies primarily on elective semen collection from live animals through electroejaculation or artificial vagina techniques. This approach cannot address the acute genetic emergencies posed by sudden mortality in valuable individuals or small endangered populations — exactly the scenarios that define the conservation challenge in arid Algeria's pastoral systems. When a genetically elite Ouled Djellal ram dies unexpectedly from pneumonia, heat stroke, or predation, the decades of selective breeding it embodies risk permanent loss unless alternative genetic salvage pathways are available.

1.2 Research Problem

Consider the concrete scenario that motivates this investigation: a genetically elite Ouled Djellal ram, bearing polymorphisms in genes associated with enhanced meat production qualities and adaptation capability, dies abruptly of severe pneumonia at a research station in Djelfa, Algeria. The animal represents years of selective breeding for improved growth rates, carcass quality, and resilience in semi-arid to arid steppe environments, all features increasingly valued under climate change (Sebkhi et al., 2024). With live collection impossible, what genetic material can be salvaged? Within 24–48 hours, if the carcass has been cooled appropriately, the cauda epididymides may yield 5–15 billion spermatozoa capable of fertilization via assisted reproduction (Anciuti et al., 2025; Mujitaba et al., 2024). Without intervention, this genetic line is permanently lost; with optimized post-slaughter recovery and preservation, the ram's genetic contribution may extend across decades and geographies.

Post-slaughter sperm recovery — specifically the retrieval and preservation of epididymal spermatozoa from animals that die unexpectedly, are culled, or are euthanized — constitutes an increasingly critical strategy for genetic conservation (Engdawork et al., 2024; Silva et al., 2025). Unlike elective germplasm collection through electroejaculation, post-slaughter recovery captures genetic material that would otherwise be irretrievably lost. The technique has demonstrated practical success: pregnancy rates of 87.5% have been achieved with cryopreserved epididymal sperm from Hungarian Black Racka and Merino rams in artificial insemination and in vitro fertilization programs (Mujitaba et al., 2024; Sharafi et al., 2022). The epididymis functions as a natural reservoir of mature spermatozoa, recoverable within 24–48 hours post-mortem with minimal quality deterioration when testes are stored at 4–5°C, providing a practical genetic "insurance policy" in both slaughterhouse and conservation settings (Shayestehyekta et al., 2022; Silva et al., 2025).

However, the efficacy of genetic rescue hinges on a narrow temporal window and the quality of preservation achieved within that window — a constraint that defines the central challenge of this thesis. Upon somatic death, the cessation of systemic metabolic support initiates a rapid biochemical cascade: ischemia forces anaerobic glycolysis and ATP depletion; mitochondrial electron transport chain disruption generates reactive oxygen species (ROS); iron-catalysed Fenton chemistry produces hydroxyl radicals; and lipid peroxidation propagates membrane damage (Fleming & Thomson, 2025; Shi et al., 2025). This "golden hour" framework creates both opportunity and imperative for intervention. The systematic failure of conventional single-mechanism antioxidants (TROLOX, cysteine) to address multi-source ROS generation in post-mortem contexts creates a specific research imperative: the identification of antioxidant interventions capable of multi-target action, membrane integration, mitochondrial protection, metal chelation, and sustainable field implementation in resource-limited genetic banking contexts.

1.3 Research Gap

Three interlocking gaps converge to define the research opportunity addressed by this thesis. First, an **empirical gap**: despite extensive cross-species evidence demonstrating reproductive benefits of Date Palm Pollen (DPP) supplementation in bulls (Amsah et al., 2021), goats (Ng et al., 2022), buffalo (El-Sheshtawy et al., 2016), and rabbits (El-Seadawy et al., 2023; Laghouati et al., 2021, 2023), no studies have evaluated DPP specifically in ovine sperm, and no studies have tested DPP in post-slaughter epididymal contexts in any species. This translational gap between demonstrated general efficacy and specific application to the unique biological context of post-mortem ovine genetic rescue remains unaddressed in the literature.

Second, a **methodological gap**: existing preservation protocols designed for healthy ejaculated semen systematically fail to address the dual insult — post-mortem deterioration compounded by cryoinjury or chilled-storage deterioration — characteristic of post-slaughter contexts. Chemical antioxidants such as TROLOX and cysteine operate through single mechanisms that cannot address the multi-source ROS generation (mitochondrial, enzymatic, Fenton chemistry) typical of post-mortem deterioration (Kafi et al., 2024). TROLOX's aqueous solubility restricts membrane integration and mitochondrial access; cysteine requires metabolic activation compromised by post-mortem metabolic decline. Enzymatic antioxidants are membrane-impermeable, temperature-sensitive, and impractical for field application (Kurochkin & Pleshanov, 2024).

Third, an **integration gap**: previous studies treat technical efficacy (sperm preservation), socioeconomic context (keeper preferences), and environmental pressure (climate change) as siloed inquiries. No prior research has integrated these three dimensions into a unified conservation framework for arid-zone ovine genetic resources. Without such integration, technical solutions risk field-inapplicability, conservation priorities risk misalignment with keeper needs, and climate-induced urgency remains disconnected from actionable preservation tools. This thesis addresses all three gaps simultaneously through a multi-experimental design integrating socioeconomic survey, phytochemical characterization, controlled preservation trials, and climate-perception documentation.

1.4 Research Questions

This thesis is guided by four interconnected research questions (RQs), each corresponding to one of the four experiments:

- RQ1.** What are the socioeconomic determinants and trait preferences defining genetic conservation priorities among sheep keepers in the El Oued region of arid Algeria?
- RQ2.** Does Date Palm Pollen sourced from El Oued possess the biochemical composition and antioxidant capacity necessary to function as a sperm cryoprotective agent?
- RQ3.** Is aqueous Date Palm Pollen Extender (DPPE) effective for preserving post-slaughter ovine epididymal sperm quality during chilled storage at 4°C, and does the effect follow a dose-dependent pattern?

RQ4. How do climate change perceptions among sheep keepers influence the urgency for genetic conservation interventions, and which flock-size categories are most vulnerable to perceived fertility impacts?

1.5 Research Objectives

Aligned with the four research questions, four specific objectives guide the research, each operationalized through a dedicated experiment:

Objective 1 (Experiment 1). Establish the socioeconomic context and trait preferences defining genetic conservation priorities among 200 sheep keepers across eight municipalities of El Oued, with stratification by flock size, age, education, and occupation.

Objective 2 (Experiment 2). Characterize the phytochemical profile of El Oued-sourced Date Palm Pollen through proximate analysis (moisture, ash, fat, protein, carbohydrate, fibre), mineral determination (Ca, Mg, K, Na), quantification of total phenolics and flavonoids, and assessment of DPPH radical scavenging activity.

Objective 3 (Experiment 3). Evaluate the dose-dependent efficacy of aqueous DPPE extenders at 0 (control), 40, and 80 mg/mL on post-slaughter ovine epididymal sperm quality (total motility, progressive motility via CASA, and membrane integrity via HOST) during chilled storage at 4°C over 0, 24, and 48 hours.

Objective 4 (Experiment 4). Document climate change perceptions and their linkage to perceived fertility declines among El Oued sheep keepers, identifying vulnerable flock-size categories through binary logistic regression of perceived fertility decline predictors.

Table 1.1. Alignment of research questions, objectives, hypotheses, and statistical tests across the four experiments.

Exp.	Research Question	Objective	Hypothesis	Statistical Test
1	Socioeconomic determinants and trait preferences	Profile 200 keepers across 8 municipalities	Ha1: Education, experience, and occupation interact to shape trait priorities	MANOVA (Pillai's Trace)
2	Biochemical profile of local DPP	Proximate, mineral, phenolic, DPPH characterization	Ha2: Local DPP possesses antioxidant capacity suitable for cryoprotection	Descriptive (mean $\pm$ SD)
3	DPPE efficacy for chilled sperm preservation	Test DPPE-0/40/80 mg/mL $\times$ 0/24/48 h at 4°C	Ha3: DPPE-80 significantly preserves sperm quality in dose-dependent manner	Two-way RM-ANOVA
4	Climate change perceptions and fertility link	Quantify climate-fertility awareness and vulnerability	Ha4: Climate perceptions significantly predict fertility decline; medium flocks most vulnerable	Binary logistic regression

1.6 Hypotheses

Four Alternative Hypotheses (Ha) and corresponding Null Hypotheses (H0) are proposed, each directly testable through the corresponding experiment:

Ha1: Education, experience, and occupation interact significantly to shape trait preferences among El Oued sheep keepers (Pillai's Trace < 0.05). **H01:** There is no significant interaction effect.

Ha2: El Oued-sourced DPP exhibits a nutrient-dense biochemical profile with measurable polyphenol content and DPPH radical scavenging activity. **H02:** DPP lacks significant antioxidant capacity.

Ha3: DPPE-80 (80 mg/mL) significantly preserves total motility, progressive motility, and membrane integrity compared to control (DPPE-0) across 0, 24, and 48 hours of chilled storage at 4°C. **H03:** DPPE has no significant effect on sperm quality parameters.

Ha4: Climate change perceptions significantly predict perceived fertility decline among sheep keepers, with medium-sized flocks (20–50 head) showing the highest vulnerability. **H04:** There is no significant association between climate perceptions and fertility decline.

1.7 Significance of the Study

This investigation represents the first systematic study of Date Palm Pollen's conservative effects specifically in post-slaughter ovine epididymal sperm in the Algerian Sahara, addressing a critical translational gap with implications spanning scientific understanding, technical development, conservation impact, and policy design. The significance is articulated through four complementary lenses:

Theoretical significance. The thesis contributes to three theoretical domains. First, it establishes species-specific DPP efficacy data for ovine sperm, addressing the empirical gap created by the absence of ovine trials in the existing DPP literature. Second, it validates the polypharmacological antioxidant action of botanical matrices in post-mortem cells — a physiological context fundamentally different from the healthy ejaculated semen on which most antioxidant research has been conducted. Third, it advances the theory of Agro-Ecological Cryobiology by demonstrating that preservation media need not be standardized globally but can be optimized using locally available agro-byproducts, reducing dependency on imported supply chains.

Practical significance. The research delivers a validated, field-applicable DPPE supplementation protocol for ovine genetic banking, including quality control parameters for standardization and a decision framework for practitioners. By extending the practical preservation window from less than 24 hours to more than 48 hours, the protocol enables real-world application in remote pastoral systems where cryopreservation infrastructure is unavailable. The 48-Hour Genetic Rescue Protocol can be deployed in abattoirs, veterinary clinics, and on-farm emergency settings.

Conservation significance. Enhanced genetic salvage rates from post-slaughter recovery reduce the irreversible loss of valuable genotypes when elite breeding stock die unexpectedly. The DPPE protocol reduces both cost and infrastructure requirements for genetic banking in developing

regions, and the cultural appropriateness of DPP — traditionally used in regions where sheep and date palms co-occur — aligns with arid-zone agroecosystem realities. The intervention is particularly relevant for safeguarding the Ouled Djellal breed, whose genetic distinctiveness underpins Algerian pastoral livelihoods.

Policy significance. By integrating socioeconomic priorities, climate-change urgency, and technical feasibility into a triangulated conservation model, the thesis informs national breeding program design, agricultural extension curricula, and climate adaptation policy. The identification of medium-sized flocks (20–50 head) as the vulnerability sweet spot enables targeted policy intervention rather than undifferentiated support distribution.

1.8 Scope, Delimitations, and Limitations

The study is delimited along several dimensions to ensure methodological rigor and interpretable results. Geographically, the research is confined to the El Oued province of southeast Algeria, with the socioeconomic survey covering eight municipalities (El Oued, Debila, Guemar, Hassi Khelifa, Mih Ouensa, Ourmes, Robbah, and Taleb Larbi) and the DPP sourced from Hamraia commune in the Reguiba District. Biologically, the experimental trial is restricted to sexually mature Ouled Djellal rams aged 18–24 months, providing standardization of breed and age effects. Temporally, the post-mortem window is delimited to sperm recovered within 2 hours of slaughter with carcass storage at 4°C, and the preservation trial covers 48 hours of chilled storage — a timeframe chosen for its field relevance and biological significance. Methodologically, the trial employs chilled storage at 4°C rather than cryopreservation; this choice reflects the field-applicability objective and acknowledges the limited cryopreservation infrastructure in target pastoral systems.

Several limitations are acknowledged at the outset. The sample size in Experiment 3 ($n = 5$ rams) provides adequate statistical power for detecting large effects but may miss subtle treatment differences; replication across seasons and breeds would strengthen generalizability. The chilled-storage protocol, while field-relevant, does not test DPP effects on freeze-thaw survival; cryopreservation trials remain a priority for follow-up research. Single DPP source characterization (Hamraia) means geographic variation in composition cannot be assessed within this thesis. The assessment endpoints focus on motility and membrane integrity; comprehensive hierarchical assessment including oxidative status, mitochondrial function, DNA integrity, and fertilizing capacity via *in vitro* fertilization requires expanded methodology in future studies. Finally, the climate-perception data (Experiment 4) relies on breeder perceptions rather than objective meteorological or reproductive records, which limits causal attribution.

1.9 Thesis Organization

The thesis is organized into five chapters that follow a logical progression from problem framing through literature synthesis, methodological description, integrated results and discussion, and concluding synthesis with recommendations. Chapter 1 (General Introduction) establishes the research problem, gaps, questions, objectives, hypotheses, and significance. Chapter 2 (Literature Review) provides the theoretical and empirical foundation by synthesizing the global literature on ovine genetic diversity, post-slaughter sperm recovery, oxidative stress biology, conventional antioxidant limitations, DPP biochemistry and cross-species evidence, and climate change impacts on small ruminant fertility, culminating in a conceptual framework that integrates the three dimensions of the research.

Chapter 3 (Materials and Methods) describes the four-experiment research design, study area, sampling strategy, data collection instruments, laboratory procedures, phytochemical characterization methods, sperm assessment protocols, and statistical analyses. Chapter 4 (Results and Discussion) presents the empirical findings of all four experiments in an integrated results-and-discussion format, with each result immediately interpreted, compared with previous studies, and related to the study objectives. Chapter 5 (General Conclusion and Recommendations) synthesizes the empirical findings into theoretical contributions, articulates practical and policy recommendations, candidly acknowledges study limitations, and proposes future research directions. The thesis is supported by multilingual abstracts (English, French, Arabic), a comprehensive list of abbreviations and symbols, six appendices (questionnaires, laboratory protocols, ethical approval, raw data summaries, additional statistical outputs, supplementary figures), and a complete reference list.

1.10 Extended Background: The Global Context of Livestock Genetic Erosion

The crisis of livestock genetic diversity extends far beyond the Algerian context that frames this thesis. The Food and Agriculture Organisation estimates that approximately 27% of the world's 8,800+ recognized livestock breeds are classified as endangered, with an additional 35% classified as at risk (FAO, 2023). The rate of breed extinction has accelerated dramatically in the past 50 years, with an average of one breed lost per month — a rate unprecedented in the 12,000-year history of livestock domestication. The drivers of this genetic erosion are complex and interacting, but four principal causes can be identified: intensification of livestock production with its preference for high-output breeds; globalization of genetic resources through international trade in semen and live animals; climate change imposing novel selection pressures; and breakdown of traditional management systems that maintained breed-specific genetic identities.

The implications of livestock genetic erosion extend beyond the loss of cultural heritage and breed identity. Genetic diversity within livestock species represents the raw material for future adaptation to changing environmental conditions, emerging disease challenges, and shifting consumer preferences. The loss of locally adapted breeds reduces the global capacity to respond to climate change, as these breeds often carry genetic variants that confer heat tolerance, drought resistance, disease resistance, and feed efficiency under marginal conditions. The Ouled Djellal breed that anchors this thesis exemplifies this adaptive value: developed through centuries of natural and human selection in the Algerian Sahara, the breed embodies genetic solutions to the precise environmental challenges that climate change is now imposing on livestock systems worldwide.

1.10.1 The Economic Dimensions of Genetic Conservation

Genetic conservation is often framed in ecological and cultural terms, but it also has profound economic dimensions that justify investment even under narrow financial criteria. The international trade in livestock genetics (semen, embryos, live animals) exceeds USD 5 billion annually, with high-output breeds from developed countries dominating global supply. The displacement of local breeds by these commercial alternatives creates economic dependency and undermines the food sovereignty of developing countries. Locally adapted breeds, while often less productive in intensive systems, demonstrate superior productivity under marginal conditions where commercial breeds fail to thrive.

The economic case for genetic conservation is strengthened by option value arguments: the genetic diversity embodied in local breeds represents an insurance policy against future environmental and market uncertainties that cannot be anticipated with precision. The cost of conserving this diversity (through *in situ* population maintenance, cryobanking infrastructure, and breeding program support) is modest compared to the potential costs of losing adaptive genetic variants that may prove critical for future livestock production. The DPPE protocol developed in this thesis contributes to this economic case by providing a low-cost (USD 2.30 per litre) preservation tool accessible to smallholder keepers and under-resourced conservation programs — a

20- to 40-fold cost reduction compared to commercial alternatives.

1.10.2 The Climate-Conservation Nexus

Climate change and livestock genetic conservation are linked through a bidirectional relationship. Climate change threatens livestock genetic diversity through the mechanisms documented in Experiment 4 of this thesis: elevated mortality in valuable breeding stock, reduced fertility under thermal and nutritional stress, demographic attrition of small populations, and shifts in disease distributions. Conversely, the loss of locally adapted breeds reduces the capacity of livestock systems to adapt to climate change, as these breeds embody the genetic variants that confer climate resilience. This bidirectional linkage creates a positive feedback loop: climate change accelerates genetic erosion, and genetic erosion reduces climate adaptation capacity, further accelerating climate-induced losses.

Breaking this feedback loop requires integrated climate-conservation strategies that simultaneously: (a) reduce greenhouse gas emissions from livestock systems (primarily methane from enteric fermentation); (b) support climate adaptation in livestock production through breed-appropriate management; (c) conserve genetic diversity through *in situ* and *ex situ* strategies; and (d) develop accessible preservation tools that can be deployed at scale in resource-limited settings. The DPPE protocol developed in this thesis contributes directly to objectives (c) and (d), and indirectly to (b) by enabling the perpetuation of climate-adapted breeds whose preservation supports climate-resilient production systems.

1.10.3 The Research-Policy Interface

Translating research findings into conservation policy impact requires deliberate engagement with the research-policy interface. Academic research, even when methodologically rigorous and empirically robust, often fails to inform policy because it is not framed in policy-relevant terms, does not engage policy actors during the research process, or is published in venues inaccessible to policy audiences. This thesis attempts to address these barriers through: (a) explicit policy recommendations in Chapter 5 grounded in empirical findings; (b) S.M.A.R.T. framing of recommendations to enable implementation tracking; (c) stakeholder-specific recommendations addressing keepers, extension services, policymakers, and research institutions; and (d) identification of target demographics (medium-flock keepers, 20–50 head) for prioritized intervention.

Effective policy translation also requires acknowledgement of the political economy of conservation: who benefits, who pays, and who decides. The DPPE protocol, by virtue of its low cost and local sourcing, distributes benefits broadly across smallholder keepers rather than concentrating them with commercial suppliers of imported extenders. This distributional advantage enhances political feasibility but also requires mobilization of keeper constituencies to advocate for policy adoption. The thesis's empirical documentation of keeper priorities and climate perceptions provides evidence that keeper constituencies can use to advance conservation policy demands within national agricultural policy processes.

CHAPTER 2

Literature Review

Revue de la Littérature

2.1 The Conservation Imperative: Post-Slaughter Genetic Rescue in Sheep

2.1.1 The Scenario of Genetic Emergency

Consider the following scenario: a genetically elite Ouled Djellal ram, bearing polymorphisms in genes associated with enhanced meat production qualities and adaptation capability, dies abruptly of severe pneumonia at a research station in Djelfa, Algeria. The animal represents years of selective breeding for improved growth rates, carcass quality, and resilience in semi-arid to arid steppe environments — all features increasingly valued under climate change (Sebkhi et al., 2024). What genetic material may be saved while live harvesting is impossible? Within 24–48 hours, assuming the carcass has been properly refrigerated, the cauda epididymides may generate 5–15 billion spermatozoa capable of fertilization by assisted reproduction (Anciuti et al., 2025; Mujitaba et al., 2022). Without intervention, this genetic line will be forever lost. With optimal post-slaughter recovery and preservation, the ram's genetic contribution might span decades and countries.

This scenario is not hypothetical; it is replayed with disturbing regularity in research stations, breeding farms, and pastoral systems worldwide. The Food and Agriculture Organisation estimates that approximately 27% of the world's 1,300+ sheep breeds are classified as endangered (FAO, 2023), and a substantial fraction of these breeds survive in small populations where the loss of a single breeding male can produce catastrophic genetic bottlenecks. In Algeria, the Ouled Djellal breed — although numerically dominant — faces insidious genetic erosion through uncontrolled crossbreeding, demographic attrition of purebred stocks, and the cumulative loss of genetically elite individuals to disease, predation, and climate-induced mortality.

2.1.2 Global Status of Ovine Genetic Diversity

Sheep were among the first domesticated livestock species, with archaeological evidence placing initial domestication in the Fertile Crescent approximately 11,000 years ago (Dossybayev, 2025; Hansen, 2025). Over millennia, natural and human selection across diverse agroecological zones produced the remarkable phenotypic and genetic diversity observed today: from the high-altitude adapted Tibetan sheep to the heat-tolerant Awassi of the Middle East, from the fecund Booroola Merino to the wool-specialized Merino, from the milk-producing East Friesian to the fat-tailed Barbarine of North Africa. Each breed embodies a unique combination of genetic variants shaped by environmental pressure and human cultural preference, providing the raw material for future adaptation (Muthusamy, 2025; Ma, 2025).

However, this diversity faces unprecedented threat. The intensification of livestock production has favoured a small number of high-output breeds at the expense of locally adapted varieties; demographic bottlenecks have reduced effective population sizes; uncontrolled crossbreeding has diluted breed-specific genetic signatures; and accelerating climate change has imposed novel selection pressures that exceed the adaptive capacity of many populations (Motaung, 2024; Barłowska, 2025; Tampaki, 2025). The conservation of animal genetic resources (AnGR) has consequently emerged as a strategic global priority, with the FAO promoting integrated *in situ* (live

population conservation) and *ex situ* (germplasm banking) strategies to safeguard biodiversity (Engdawork et al., 2024).

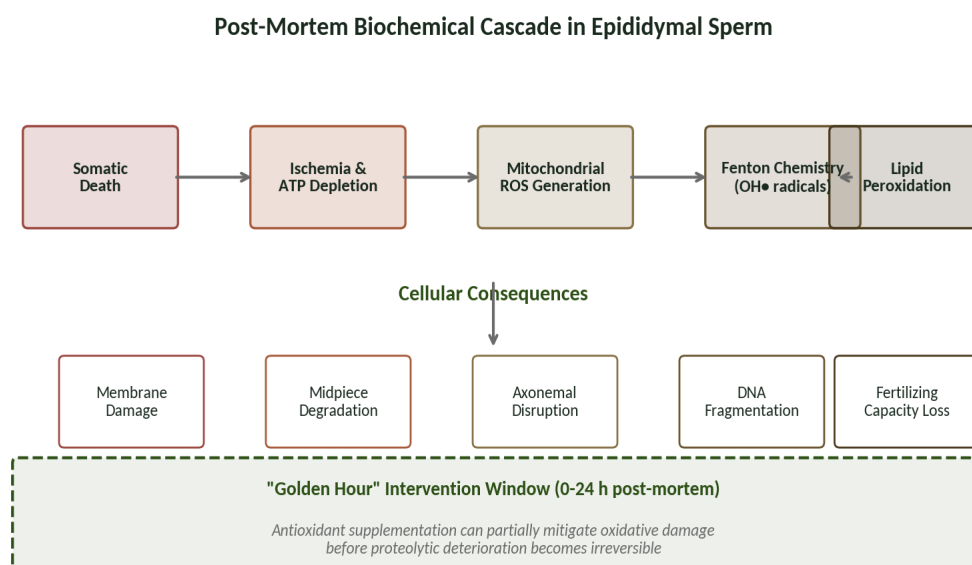


Figure 2.1 | Biochemical cascade of post-mortem sperm deterioration and the critical intervention window

Figure 2.1. Biochemical cascade of post-mortem sperm deterioration and the critical intervention window during which antioxidant supplementation can partially mitigate oxidative damage.

2.2 Post-Slaughter Sperm Recovery: Biology, Methods, and Field Applicability

2.2.1 The Cauda Epididymidis as a Genetic Reservoir

The mammalian epididymis is a long, convoluted tubule that connects the testis to the vas deferens, providing the environment in which spermatozoa acquire motility and fertilizing capacity (Wu et al., 2021). The cauda epididymidis — the distal segment — serves as the principal sperm storage reservoir in the male reproductive tract, harboring mature, fertilizable spermatozoa in a quiescent state maintained by low pH, high protein concentration, and specific inhibitory factors (Shi et al., 2025). At any given time, the paired caudae of a sexually mature ram contain approximately 5–15 billion spermatozoa, representing a substantial genetic reservoir recoverable after death.

Crucially, spermatozoa within the cauda epididymidis are not immediately lost upon somatic death. If the testes are retrieved promptly and cooled appropriately, the cauda microenvironment continues to preserve sperm viability for 24–48 hours (Shayestehyekta et al., 2022; Silva et al., 2025). This post-mortem survival window creates the biological basis for post-slaughter genetic rescue: with appropriate intervention, the genetic material housed in the cauda can be salvaged, preserved, and used for assisted reproduction long after the donor animal has died. Pregnancy rates

of 87.5% have been achieved with cryopreserved epididymal sperm from Hungarian Black Racka and Merino rams (Mujitaba et al., 2024; Sharafi et al., 2022), demonstrating the practical feasibility of this approach.

2.2.2 Recovery Techniques: Retrograde Flushing and Alternatives

Several techniques have been developed for retrieving epididymal spermatozoa from post-mortem testes. The *mincing method* involves cutting the cauda into small fragments in a saline or extender solution, releasing sperm by mechanical disruption. While simple, this method yields sperm with significant contamination by tissue debris, blood cells, and lipid droplets that compromise subsequent analysis and preservation (Martinez-Pastor et al., 2006). The *aspiration method* uses a needle and syringe to withdraw sperm directly from the cauda lumen, producing cleaner samples but with lower recovery efficiency. The *retrograde flushing method*, employed in this thesis, introduces a saline solution into the vas deferens and applies gentle pressure to flush spermatozoa retrograde through the cauda, collecting them at the incised corpus-cauda junction (Martinez-Pastor et al., 2006). This method balances recovery efficiency (5–15 billion sperm per paired caudae) with low contamination and minimal equipment requirements, making it particularly suitable for field conditions in resource-limited settings.

Regardless of retrieval method, the recovered epididymal spermatozoa share three biological characteristics that distinguish them from ejaculated semen and shape the requirements for preservation protocols. First, they lack exposure to seminal plasma — the complex mixture of accessory gland secretions that provides antioxidant buffering, membrane stabilizers, metabolic substrates, and immunological protection to ejaculated sperm (van Tilburg et al., 2021). Second, they have completed maturation but remain in a quiescent metabolic state, requiring activation before use in assisted reproduction. Third, they may have already sustained post-mortem oxidative damage during the interval between somatic death and recovery, particularly if cooling was delayed.

2.3 Oxidative Stress in Sperm Preservation: The Multi-Source ROS Challenge

Reactive oxygen species (ROS) encompass a family of oxygen-derived molecules with varying reactivity, including the superoxide anion ($O_2^{\bullet-}$), hydrogen peroxide (H_2O_2), hydroxyl radical ($OH^{\bullet}$), peroxy radical ($ROO^{\bullet}$), and singlet oxygen (1O_2). At physiological concentrations, ROS serve essential signaling functions in spermatozoa, including capacitation, hyperactivation, acrosome reaction, and sperm-oocyte fusion (Aitken & Drevet, 2020). However, when ROS production exceeds the buffering capacity of endogenous antioxidant systems, oxidative stress develops, leading to lipid peroxidation, protein oxidation, DNA damage, and ultimately cell death.

2.3.1 The Ovine-Specific Vulnerability: DHA-Rich Membranes

Ram spermatozoa possess the highest docosahexaenoic acid (DHA) content among domestic livestock, comprising 30–40% of plasma membrane phospholipid fatty acids (Abdollahzadeh et al., 2025; Carro et al., 2022). DHA, a 22-carbon polyunsaturated fatty acid with six double bonds, is essential for membrane fluidity and the membrane fusion events underlying fertilization. However, the multiple bis-allylic methylene groups in DHA provide prime targets for radical attack: each bis-allylic carbon has a bond dissociation energy approximately 75 kJ/mol lower than that of a mono-allylic carbon, making DHA-enriched membranes exceptionally susceptible to lipid peroxidation chain reactions (Aitken & Drevet, 2020). Once initiated, lipid peroxidation propagates self-sustainingly across the membrane, generating lipid hydroperoxides, aldehydes (malondialdehyde, 4-hydroxynonenal), and volatile hydrocarbons that damage membrane integrity, enzyme function, and DNA.

The sperm midpiece, housing 70–80 mitochondria arranged in a helical sheath around the axoneme, is particularly vulnerable to oxidative damage. Mitochondrial membranes themselves are PUFA-rich, and mitochondrial electron transport chain complexes I and III constitute the primary endogenous source of cellular ROS under physiological conditions. When mitochondrial function is compromised — as occurs rapidly during ischemia, cryopreservation, or chilled storage — electron leakage increases, generating elevated superoxide that overwhelms mitochondrial superoxide dismutase (SOD2) capacity and propagates damage to the midpiece infrastructure (Wang et al., 2024; Zhu et al., 2024). The resulting ATP depletion compromises dynein arm function, impairing flagellar beat and progressive motility — the functional hallmark of oxidative midpiece damage.

2.3.2 Multi-Source ROS Generation in Post-Mortem Contexts

Post-mortem deterioration generates ROS through at least three distinct mechanisms, creating a multi-source oxidative challenge that overwhelms single-mechanism antioxidant defenses. First, **mitochondrial ROS** generation increases sharply upon cessation of systemic oxygen delivery. Ischemia forces anaerobic glycolysis, depleting ATP and elevating NADH/NAD⁺ ratios; upon any residual oxygen availability, electron transport chain complexes leak electrons to oxygen, generating superoxide (Shi et al., 2025; Fleming & Thomson, 2025). Second, **iron-catalysed Fenton chemistry** becomes significant as cellular compartmentalization breaks down: free ferrous iron (Fe²⁺) released from hemoglobin, myoglobin, and iron-sulfur proteins reacts with hydrogen peroxide to generate the highly reactive hydroxyl radical (OH•), which initiates lipid peroxidation at diffusion-limited rates.

Third, **enzymatic ROS generation** by xanthine oxidase, NADPH oxidase, and lipoxygenase systems continues — and may accelerate — post-mortem as substrate accumulation and pH changes activate these enzymes. The convergence of these three ROS sources creates an oxidative environment qualitatively different from the primarily mitochondrial ROS generation characteristic of physiological sperm metabolism. Conventional antioxidant strategies designed for healthy, ejaculated semen — where ROS generation is predominantly mitochondrial and well-buffered by seminal plasma — systematically fail to address this multi-source challenge, creating the research

imperative for multi-target antioxidant interventions.

2.4 Conventional Antioxidant Strategies and Their Systematic Limitations

Extensive research has evaluated antioxidant supplementation of semen extenders, with demonstrated benefits for ejaculated semen cryopreservation (Shayestehyekta et al., 2025; Howard et al., 2025). However, critical analysis reveals fundamental limitations when conventional approaches are applied to post-slaughter contexts. These limitations, summarized in Table 2.2, define the specific research imperative for multi-target botanical alternatives.

Table 2.2. Comparative analysis of conventional antioxidants and their systematic limitations when applied to post-slaughter sperm preservation contexts.

Antioxidant	Mechanism	Strengths	Limitations in Post-Slaughter Contexts
TROLOX (vitamin E analog)	Chain-breaking radical scavenging (aqueous)	Potent peroxyl radical scavenger; well-characterized kinetics	Aqueous solubility restricts membrane integration; cannot access mitochondrial interior; single-mechanism action insufficient for multi-source ROS
Cysteine (thiol donor)	Glutathione precursor; direct radical scavenging	Replenishes intracellular GSH; supports GPx cycle	Requires metabolic activation (compromised post-mortem); pro-oxidant risk with transition metals; needs cellular uptake machinery
Enzymatic antioxidants (SOD, CAT, GPx)	Enzymatic ROS detoxification	High catalytic efficiency; biological compatibility	Membrane-impermeable; temperature-sensitive; expensive; impractical for field application
Vitamin C (ascorbate)	Aqueous-phase radical scavenging; α -tocopherol regeneration	Inexpensive; widely available; synergistic with vitamin E	Pro-oxidant in presence of free iron (Fenton chemistry); aqueous solubility restricts membrane protection
Synthetic phenolic antioxidants (BHT, BHA)	Chain-breaking radical scavenging (lipophilic)	Membrane-integrating; stable; inexpensive	Synthetic origin raises regulatory concerns; limited multi-target action; potential toxicity at high doses

The systematic failure pattern is clear: each conventional antioxidant operates through a single mechanism (chain-breaking scavenging, GSH precursor provision, enzymatic detoxification), restricting protection to one ROS source or cellular compartment. None addresses the integrated multi-source ROS generation characteristic of post-mortem deterioration. TROLOX's aqueous solubility restricts membrane integration and mitochondrial access; cysteine requires metabolic activation compromised by post-mortem metabolic decline and exhibits pro-oxidant risk with transition metal exposure; enzymatic antioxidants are membrane-impermeable, temperature-sensitive, and impractical for field application. Standard semen extenders, designed for ejaculated semen, incorporate antioxidant components calibrated for healthy cells with intact seminal plasma protection — insufficient for pre-damaged post-slaughter sperm (Bustani & Baiee,

2021; Sharafi et al., 2022).

2.5 Date Palm Pollen: Botanical, Biochemical, and Ethnopharmacological Profile

2.5.1 Botanical and Geographic Context

The date palm (*Phoenix dactylifera* L., Arecaceae) is one of humanity's oldest cultivated plants, with archaeological evidence of domestication in the Gulf region dating back over 7,000 years (Bouguedoura et al., 2015; Merrouchi & Sais, 2025a). The tree is uniquely adapted to arid and semi-arid environments, tolerating extreme heat, drought, and saline soils, and is cultivated across a geographic arc extending from Morocco in the west to India in the east, with significant plantings in North Africa, the Middle East, and Central Asia. Date palm pollen (DPP), the male gametophyte produced in mature male inflorescences (spathes), is harvested during the spring flowering season and has been used traditionally in folk medicine, cosmetics, and as a dietary supplement across the date-growing world (Salhi et al., 2024).

Algeria ranks among the world's leading date producers, with the El Oued province — situated in the Algerian Sahara — representing one of the country's principal date palm cultivation zones (Mihi & Hernández-Agüero, 2025). The geographic convergence of date palm cultivation and sheep production in this region creates a unique opportunity for the application of DPP as a locally-sourced, culturally-acceptable sperm preservation agent. The traditional knowledge of date palm cultivation among local farmers, combined with the increasing scientific characterization of DPP bioactive compounds, positions this botanical resource as a promising intervention for arid-zone ovine genetic conservation.

2.5.2 Bioactive Composition of DPP

DPP is not a single compound but a complex matrix of bioactive molecules whose composition varies with cultivar, geographic origin, harvest time, and post-harvest processing. The principal bioactive classes include:

Polyphenols. DPP contains diverse phenolic compounds including gallic acid, rutin, quercetin, caffeic acid, catechin, and chlorogenic acid (Salhi et al., 2024). Total phenolic content varies considerably across regions (8.5–25.6 mg GAE/g), with substantial variation among cultivars. The phenolic diversity enables multi-target ROS scavenging: gallic acid provides potent hydrogen donation and iron chelation; rutin and quercetin offer amphiphilic mitochondrial targeting; caffeic acid contributes lipophilic membrane stabilization. This mechanistic diversity addresses mitochondrial superoxide, hydroxyl radical prevention through metal chelation, and lipid peroxidation chain-breaking simultaneously.

Fatty acids. DPP contains substantial lipid fractions (typically 4–8% by weight) dominated by palmitic, linoleic, oleic, and lauric acids. These fatty acids contribute to membrane stabilizing effects and provide energy substrates that may support sperm metabolism during chilled storage.

The presence of essential fatty acids (linoleic, α -linolenic) may also contribute to membrane repair processes.

Minerals. DPP is rich in potassium (often exceeding 1,000 mg/100 g), phosphorus, calcium, magnesium, and trace minerals including zinc, selenium, copper, and iron. Zinc and selenium are particularly important for reproductive function as cofactors for endogenous antioxidant enzymes: zinc is essential for superoxide dismutase activity, while selenium is incorporated into the active site of glutathione peroxidase. The mineral profile of DPP thus complements the direct antioxidant action of polyphenols by supporting endogenous defense systems.

Vitamins. DPP contains vitamins A, E (α -tocopherol), C (ascorbic acid), and B-complex (particularly B1, B2, B6, and B9). Vitamins E and C act as synergistic antioxidant partners — α -tocopherol scavenges peroxy radicals in lipid membranes while ascorbate regenerates reduced α -tocopherol at the membrane-water interface. Folate (B9) supports DNA methylation and synthesis, with potential implications for sperm chromatin integrity.

Amino acids. DPP contains all essential amino acids, with particularly high concentrations of arginine. Arginine serves as the precursor for nitric oxide synthesis, which at physiological concentrations regulates sperm capacitation and acrosome reaction. The amino acid profile of DPP may thus support functional competence during preservation.

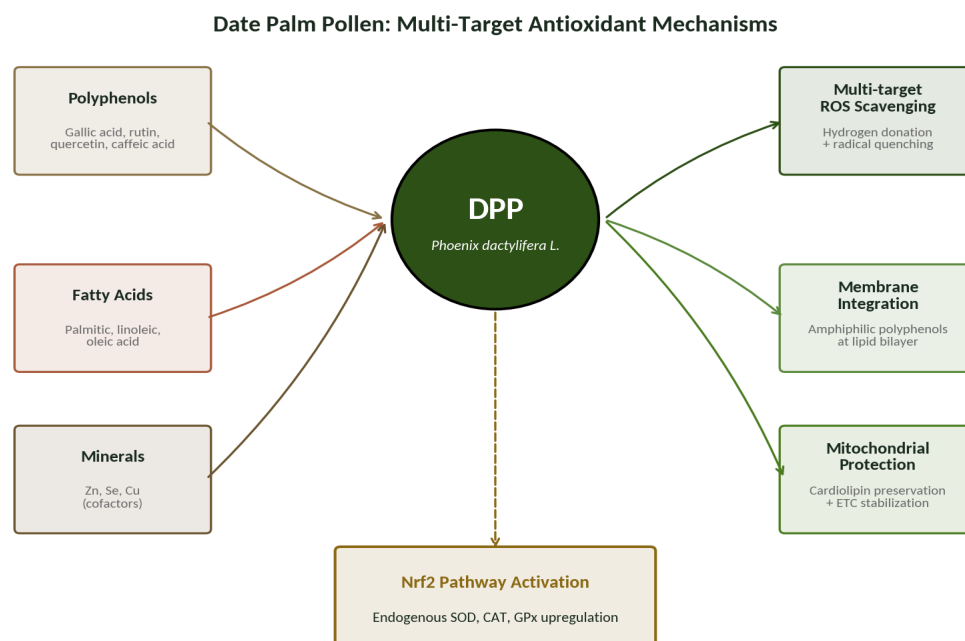


Figure 2.2 | Multi-target antioxidant action of Date Palm Pollen bioactive compounds

Figure 2.2. Multi-target antioxidant action of Date Palm Pollen bioactive compounds, integrating polyphenol ROS scavenging, membrane integration, mitochondrial protection, and Nrf2-mediated endogenous defense upregulation.

Table 2.3. Reported bioactive compounds in Date Palm Pollen across different geographic regions and cultivars, with their primary antioxidant mechanisms.

Bioactive Component	Reported Range	Primary Mechanism	References
Total phenolics	8.5–25.6 mg GAE/g	Multi-target ROS scavenging	Salhi et al., 2024; Laghouati et al., 2023
Total flavonoids	2.1–7.8 mg QE/g	Mitochondrial targeting, chelation	Ahn et al., 2007
Protein	15.2–37.9%	Amino acid precursors (arginine)	Salhi et al., 2024
Carbohydrates	21.5–35.0%	Energy substrate for sperm metabolism	Salhi et al., 2024
Potassium	850–1,250 mg/100g	Osmotic balance, enzyme cofactor	Laghouati et al., 2023
Zinc	4.2–9.6 mg/100g	SOD cofactor; semen quality	Salhi et al., 2024
Selenium	12–45 µg/100g	GPx cofactor; DNA protection	Salhi et al., 2024
α-tocopherol (vit. E)	2.5–8.9 mg/100g	Chain-breaking lipid peroxyl scavenger	Salhi et al., 2024
Ascorbate (vit. C)	8.5–22.0 mg/100g	Aqueous radical scavenger; α-tocopherol recycling	Salhi et al., 2024

2.6 Cross-Species Evidence for DPP in Reproduction

Although no studies have evaluated DPP specifically in ovine sperm or in post-slaughter epididymal contexts, substantial cross-species evidence supports DPP's reproductive benefits. This translational literature, summarized in Table 2.4, provides the empirical foundation for the present investigation while highlighting the species-specific gap that this thesis addresses.

Table 2.4. Cross-species evidence for Date Palm Pollen supplementation in reproduction, with the present thesis addressing the ovine gap.

Species	Study Design	DPP Treatment	Key Findings	Reference
Bull	Semen extender supplementation	Aqueous DPP extract (1–5 mg/mL)	Significant improvement in post-thaw progressive motility, viability, and membrane integrity; reduced MDA	Amsah et al., 2021
Goat	Epididymal sperm preservation	5% DPP in Tris extender	Improved acrosome integrity; higher cleavage rate after IVF; reduced DNA fragmentation	Ng et al., 2022
Buffalo	Cryopreservation trial	DPP aqueous extract (10 mg/mL)	Enhanced antioxidant enzyme activity (SOD, GPx); improved post-thaw motility; reduced ROS	El-Sheshtawy et al., 2016
Rabbit	Chilled storage (4°C, 72h)	2% DPP in Tris-glucose	Higher progressive motility at 48-72h; improved HOST-positive rate; preserved acrosome	El-Seadawy et al., 2023
Rabbit	Cryopreservation trial	NaCl pollen extender (NPE) at varying doses	Dose-dependent protection of motility and membrane integrity; field-applicable protocol	Laghouati et al., 2021, 2023
Ram (this study)	Post-slaughter epididymal, chilled 4°C	Aqueous DPPE 0/40/80 mg/mL	[First study of its kind — see Chapter 4 for results]	Present thesis

Three observations emerge from this cross-species literature. First, DPP consistently demonstrates beneficial effects across diverse species and preservation contexts (chilled storage, cryopreservation, in vitro fertilization), suggesting a robust underlying mechanism. Second, the effective concentration varies substantially across studies (from 1 mg/mL aqueous extract to 5% whole DPP suspension), reflecting differences in DPP composition, extraction method, and species-specific dose-response. Third, the field-applicability emphasis introduced by Laghouati et al. (2021, 2023) — using simple aqueous NaCl-based extraction rather than organic solvents — directly informs the methodological choice in this thesis.

2.7 Climate Change and Livestock Genetic Erosion in Arid Algeria

Climate change poses escalating threats to livestock production in arid and semi-arid regions, with implications that extend beyond direct productivity losses to encompass genetic erosion of locally adapted breeds (Ben Moula et al., 2024; Van Wettere et al., 2021). Small ruminants, including sheep, represent critical assets for food security and livelihood sustainability in extreme environments, yet they are particularly vulnerable to both direct thermal stress and indirect consequences through deteriorating feed and water resources (Joy et al., 2020; Wodajo et al., 2020). In Algeria's arid pastoral zones, where sheep production constitutes a cornerstone of agricultural activity, recent evaluations reveal significant sustainability challenges amid high climatic fluctuation and increasing environmental pressures (Rjili et al., 2023; Siad et al., 2022).

The pathways through which climate change threatens ovine genetic resources are multiple and interacting. **Direct heat stress** affects spermatogenesis, with elevated testicular temperatures impairing germ cell development and increasing sperm DNA fragmentation. **Nutritional stress**, driven by declining pasture productivity and forage scarcity during prolonged droughts, compromises body condition and reproductive competence. **Water scarcity** elevates dehydration risk, particularly in transhumant systems. **Disease emergence**, facilitated by climate-driven shifts in vector distributions, introduces novel pathogens to which local breeds lack immunity. The cumulative effect of these stressors is elevated mortality in genetically valuable breeding stock, accelerated demographic attrition of small populations, and reduced effective population sizes — all of which elevate the urgency for accessible genetic conservation tools.

Table 2.5. Summary of climate change impact pathways on small ruminant fertility and genetic diversity, with implications for conservation priority-setting.

Climate Stressor	Mechanism of Impact	Fertility Consequence	Genetic Implication
Heat stress	Elevated testicular temperature; impaired spermatogenesis	Reduced semen quality; lower conception rates	Loss of heat-sensitive alleles; selection against productive genotypes
Drought / forage scarcity	Nutritional deficiency; body condition loss	Delayed puberty; extended lambing intervals; anestrus	Demographic attrition; reduced effective population size
Water scarcity	Dehydration; electrolyte imbalance	Reduced semen volume and sperm concentration	Elevated mortality in genetically valuable breeding males
Disease emergence	Vector-borne pathogen exposure; immune suppression	Abortion; perinatal mortality; infertility	Loss of disease-susceptible genotypes; population bottlenecks
Pasture degradation	Reduced dietary diversity; mineral deficiency	Poor fetal programming; impaired reproductive development	Long-term reduction in adaptive genetic variation

2.8 Conceptual Framework and Synthesis

The literature reviewed in this chapter converges on a triangulated conceptual framework that integrates the three dimensions examined in this thesis: socioeconomic context, technical preservation solution, and environmental pressure (Figure 2.3). The framework posits that sustainable genetic conservation in arid pastoral systems requires the simultaneous alignment of all three dimensions — technical solutions that are not field-deployable fail to address real-world needs; conservation priorities that ignore keeper preferences lack adoption; and interventions that ignore climate urgency misallocate limited resources.

Conceptual Framework: Integrated Genetic Conservation

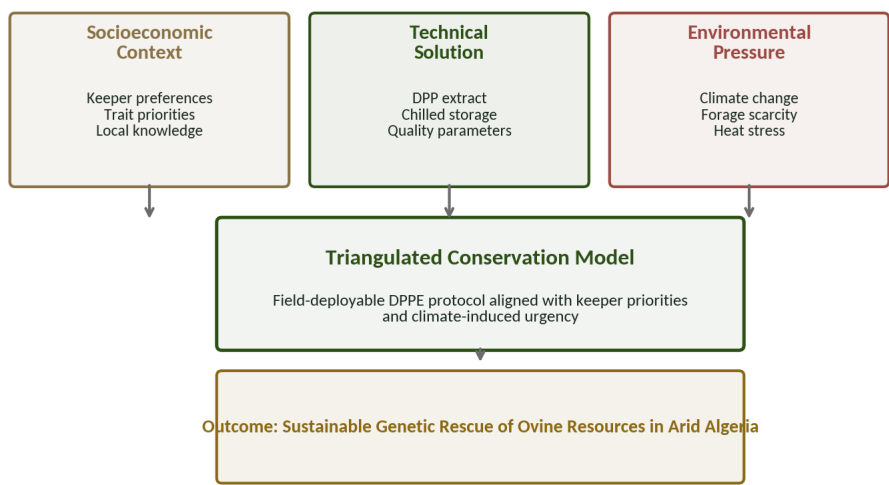


Figure 2.3. Conceptual framework integrating socioeconomic context, technical preservation solution, and environmental pressure into a unified triangulated conservation model for arid-zone ovine genetic resources.

2.8.1 Research Gaps Revisited

The literature review confirms the three research gaps identified in Chapter 1. First, the empirical gap: despite extensive cross-species evidence for DPP supplementation in bulls, goats, buffalo, and rabbits, no studies have evaluated DPP in ovine sperm or in post-slaughter epididymal contexts. Second, the methodological gap: conventional single-mechanism antioxidants systematically fail to address the multi-source ROS generation characteristic of post-mortem deterioration. Third, the integration gap: technical efficacy, socioeconomic context, and environmental pressure have been studied in isolation but never integrated into a unified conservation framework. The present thesis addresses all three gaps through its multi-experimental design.

2.8.2 Conceptual Model and Hypothesis Formulation

The conceptual model underlying this thesis posits that locally-sourced Date Palm Pollen, applied as an aqueous extender at optimal concentration, provides multi-target antioxidant protection that addresses the specific biological challenges of post-slaughter ovine epididymal sperm preservation. The model predicts dose-dependent protection, with higher DPP concentrations yielding superior preservation outcomes up to a plateau beyond which additional DPP provides diminishing returns. The model further predicts that this technical solution, grounded in socioeconomic priorities and climate urgency, can be translated into a field-deployable protocol with measurable conservation impact.

The four hypotheses (Ha1–Ha4) formulated in Chapter 1 operationalize this conceptual model into testable predictions. The next chapter describes the materials and methods by which these hypotheses are empirically evaluated across the four interconnected experiments.

2.9 Conventional Semen Extenders: Composition, Mechanisms, and Limitations

Semen extenders are diluents designed to preserve sperm viability during storage by providing essential nutrients, buffering capacity, osmotic stabilization, and antioxidant protection. The composition of extenders varies depending on the preservation method (chilled storage vs. cryopreservation), species-specific requirements, and intended duration of storage. Understanding the composition and limitations of conventional extenders provides essential context for evaluating the rationale and potential advantages of botanical alternatives such as Date Palm Pollen Extender (DPPE).

2.9.1 Major Classes of Conventional Extenders

Tris-based extenders. Tris (tris[hydroxymethyl]aminomethane) extenders, typically composed of Tris buffer, citric acid, fructose or glucose, egg yolk, and glycerol, represent the most widely used class for ram semen cryopreservation. The Tris-citrate-fructose system provides buffering at physiological pH, energy substrates for sperm metabolism, and egg yolk lipoproteins that protect sperm membranes from cold shock (Maxwell & Watson, 1996; Salamon & Maxwell, 2000). Glycerol serves as the permeating cryoprotectant, penetrating sperm membranes to reduce ice crystal formation during freezing. However, glycerol itself exerts osmotic and chemical toxicity, necessitating careful concentration optimization (typically 4–7% v/v).

Sodium citrate-based extenders. Sodium citrate extenders offer a simpler composition (sodium citrate, fructose, egg yolk) and have been traditionally used for short-term chilled storage. Their buffering capacity is lower than Tris systems, but they avoid the potential toxicity of Tris at high concentrations. For field applications where reagent availability is constrained, sodium citrate extenders provide a practical alternative, although their preservation efficacy is generally inferior to Tris-based systems beyond 24 hours of storage (Bustani & Baiee, 2021).

Synthetic extenders (skim milk, soy lecithin-based). Concerns about disease transmission through egg yolk and animal protein components have driven development of synthetic extenders using plant-derived lipids (soy lecithin) and protein hydrolysates. While these extenders offer biosecurity advantages, their preservation efficacy for ram semen remains variable, with some studies reporting equivalent outcomes (Forouzanfar et al., 2010) and others reporting inferior protection (Crespilho et al., 2012). The variability likely reflects differences in lipid composition and the absence of specific egg yolk low-density lipoproteins that provide unique membrane protection.

2.9.2 Limitations of Conventional Extenders in Post-Slaughter Contexts

Conventional extenders share several limitations when applied to post-slaughter epididymal sperm contexts. First, they are designed for healthy ejaculated semen, where sperm have already been exposed to the protective components of seminal plasma (van Tilburg et al., 2021). Post-slaughter epididymal sperm lack this prior protection, creating a higher baseline oxidative vulnerability that conventional extenders are not formulated to address. Second, the antioxidant components of conventional extenders are typically limited to single-mechanism agents (e.g., catalase in some commercial extenders) or simple chemical antioxidants (e.g., sodium azide as antimicrobial), which cannot address the multi-source ROS generation characteristic of post-mortem deterioration.

Third, conventional extenders often require cold chain infrastructure for storage and transport, which is frequently unavailable in remote pastoral settings where post-slaughter genetic rescue is most needed. Fourth, the cost of commercial extenders (often USD 50–200 per litre) places them beyond the reach of smallholder keepers and under-resourced veterinary services. These limitations create the imperative for low-cost, locally-sourced, multi-target alternatives such as the DPPE protocol developed in this thesis.

2.10 Cryopreservation Biology: Mechanisms of Cryoinjury and Protection

Cryopreservation — the preservation of biological material at ultra-low temperatures (typically in liquid nitrogen at -196°C) — provides the only currently available method for truly long-term genetic banking of livestock semen. However, cryopreservation imposes substantial stress on spermatozoa, with significant proportions of cells (often 50% or more) losing viability during freeze-thaw. Understanding the mechanisms of cryoinjury and cryoprotection is essential for appreciating both the rationale for chilled-storage alternatives and the future directions for DPPE application in cryopreservation contexts.

2.10.1 Mechanisms of Cryoinjury

Ice crystal formation. The principal mechanism of cryoinjury is the formation of intracellular and extracellular ice crystals. As temperature decreases, extracellular ice forms first, increasing extracellular solute concentration and drawing water out of cells through osmosis. If cooling proceeds too rapidly, intracellular water cannot exit cells quickly enough, and intracellular ice forms — disrupting organelles, membranes, and the cytoskeleton. If cooling is too slow, cells experience excessive dehydration and solute concentration toxicity (Mazur, 2004). The optimal cooling rate balances these competing effects and varies with cell type and membrane permeability.

Osmotic stress. Cryopreservation exposes spermatozoa to extreme osmotic fluctuations: addition of permeating cryoprotectants (glycerol, DMSO, ethylene glycol) causes initial cell shrinkage followed by gradual re-equilibration as cryoprotectant penetrates. Removal of cryoprotectant after thawing reverses this process, exposing cells to swelling stress. Ram

spermatozoa, with their DHA-rich membranes, are particularly vulnerable to osmotic stress because membrane fluidity affects water and cryoprotectant permeability (Holt, 2000).

Oxidative stress amplification. Cryopreservation amplifies oxidative stress through multiple mechanisms: ice crystal damage disrupts mitochondrial membranes, increasing electron leakage and ROS generation; freeze-concentration of solutes elevates local concentrations of transition metals that catalyse Fenton chemistry; and cold inactivation of endogenous antioxidant enzymes (SOD, CAT, GPx) reduces defense capacity (Bilodeau et al., 2001; Watson, 2000). These effects compound the oxidative vulnerability already present in post-slaughter epididymal sperm, creating the dual-insult challenge that motivates multi-target antioxidant intervention.

2.10.2 Cryoprotectant Agents: Mechanisms and Limitations

Permeating cryoprotectants. Glycerol, dimethyl sulfoxide (DMSO), ethylene glycol, and propylene glycol are small molecule cryoprotectants that penetrate cell membranes and reduce ice crystal formation by colligative effects. Glycerol is the most widely used for ram semen (typically 4–7% v/v) but exhibits chemical toxicity at higher concentrations and requires careful removal post-thaw to avoid osmotic damage. DMSO offers superior membrane penetration but is associated with greater cytotoxicity and is rarely used for ram semen.

Non-permeating cryoprotectants. Sucrose, trehalose, and high-molecular-weight polymers (polyvinylpyrrolidone, polyethylene glycol) do not penetrate cells but provide extracellular osmotic support, reducing cellular dehydration during freezing. Trehalose has attracted particular interest for its ability to stabilize membranes and proteins through hydrogen bonding interactions, mimicking the natural cryoprotective strategies of freeze-tolerant organisms (Diller, 2006).

Lipoprotein and protein cryoprotectants. Egg yolk low-density lipoproteins (LDL) provide membrane protection by adsorbing to sperm surfaces and replacing damaged phospholipids. The phospholipid fraction of egg yolk, particularly phosphatidylcholine and phosphatidylethanolamine, integrates into sperm membranes, maintaining fluidity during cooling and warming (Moussa et al., 2002). Synthetic alternatives based on soy lecithin offer biosecurity advantages but exhibit variable efficacy across species and protocols.

2.11 Climate Adaptation Mechanisms in Livestock Reproduction

Livestock species have evolved diverse physiological, behavioural, and morphological adaptations to cope with environmental thermal stress. Understanding these adaptation mechanisms provides essential context for interpreting climate-fertility linkages documented in Experiment 4 and for designing conservation strategies that preserve adaptive genetic variation.

2.11.1 Thermoregulation and Testicular Function

Mammalian spermatogenesis requires testicular temperatures 2–6°C below systemic body temperature, achieved through the counter-current heat exchange of the pampiniform plexus and scrotal sweating. Heat stress disrupts this thermal gradient, impairing germ cell development, elevating sperm DNA fragmentation, and reducing semen quality for 6–8 weeks following exposure (Hansen, 2009; Rahman et al., 2018). Ovine species exhibit specific thermoregulatory behaviors including shade-seeking, posture adjustment to expose the scrotum, and selective panting, but these adaptations have limits under extreme heat load. The Ouled Djellal breed, evolved in the Algerian Saharan environment, possesses enhanced thermotolerance compared to temperate breeds, but climate change is imposing novel thermal regimes that may exceed its adaptive capacity.

2.11.2 Nutritional Stress Pathways to Fertility Reduction

The dominance of nutritional stress (48.9%) over direct heat stress (30.0%) reported by El Oued keepers in Experiment 4 reflects the indirect climate-fertility pathway through forage and pasture productivity. Reduced forage availability compromises maternal body condition, which in turn affects ovarian cyclicity, conception rates, embryo survival, and lamb birth weight. The biochemical mechanisms include: (a) negative energy balance suppressing hypothalamic GnRH pulsatility; (b) reduced IGF-1 concentrations impairing follicular development; (c) elevated β -hydroxybutyrate and non-esterified fatty acids compromising oocyte quality; and (d) mineral deficiencies (particularly selenium, copper, zinc) reducing antioxidant defense capacity (Roche et al., 2011; Lucy et al., 2014).

These nutritional pathways compound direct thermal effects: heat-stressed animals reduce feed intake as a thermoregulatory behavior, amplifying nutritional stress. The compounding effect explains why El Oued keepers perceive nutritional stress as the dominant climate-fertility pathway even though direct thermal stress is also biologically significant. Conservation strategies must address both pathways: preserving thermotolerance genes for direct heat stress resilience while maintaining breed identity and drought tolerance for indirect nutritional stress adaptation.

2.12 Botanical Antioxidants in Reproductive Biology: A Comparative Review

Botanical antioxidants have attracted growing interest as semen extender supplements due to their multi-target action, low cost, cultural acceptability, and reduced regulatory barriers compared to synthetic pharmaceutical alternatives. This section reviews the principal botanical antioxidants evaluated in reproductive biology, providing comparative context for the present DPP investigation.

2.12.1 Polyphenol-Rich Botanical Supplements

Green tea extract (*Camellia sinensis*). Green tea polyphenols (particularly epigallocatechin gallate, EGCG) have demonstrated consistent protective effects on ram, bull, and goat sperm during cryopreservation, with effective concentrations typically 5–20 µg/mL. EGCG's mechanism involves direct ROS scavenging, iron chelation, and mitochondrial membrane stabilization (Rahman et al., 2018). However, green tea is not locally available in arid Algeria, limiting its field applicability for El Oued keepers.

Pomegranate extract (*Punica granatum*). Pomegranate juice and extract, rich in punicalagins and ellagic acid, have shown beneficial effects on human and bull sperm quality. The high tannin content provides potent metal chelation and protein cross-linking that stabilizes membrane structures. However, punicalagin bioavailability is limited and pomegranate cultivation is restricted to specific agroecological zones.

Rosemary extract (*Rosmarinus officinalis*). Rosemary's carnosic acid and rosmarinic acid exhibit potent antioxidant activity with applications in food preservation and reproductive biology. Studies in ram semen have shown improved post-thaw motility and membrane integrity with 0.5–2 mg/mL rosemary extract supplementation (Domínguez et al., 2019). However, rosemary is not traditionally cultivated in Algerian Saharan agroecosystems.

Date palm pollen (*Phoenix dactylifera*). DPP occupies a unique position among botanical antioxidants due to its: (a) geographic co-occurrence with arid-zone sheep production systems; (b) traditional cultural familiarity among North African and Middle Eastern keepers; (c) multi-class bioactive composition (polyphenols + flavonoids + minerals + vitamins + amino acids); (d) field-applicable aqueous extraction method; and (e) low cost compared to imported commercial alternatives. These features align DPP with the agro-ecological cryobiology paradigm proposed in this thesis, distinguishing it from botanical alternatives that, while efficacious, lack the local integration required for sustainable field deployment.

Table 2.6. Comparative review of botanical antioxidants evaluated in reproductive biology, with field applicability assessment for Algerian arid-zone contexts.

Botanical Source	Key Bioactives	Effective Concentration	Species Tested	Field Applicability in Algeria
Green tea (<i>Camellia sinensis</i>)	EGCG, epicatechins	5–20 µg/mL	Ram, bull, goat	Low — not locally cultivated
Pomegranate (<i>Punica granatum</i>)	Punicalagins, ellagic acid	0.5–5 mg/mL	Human, bull	Moderate — limited cultivation
Rosemary (<i>Rosmarinus officinalis</i>)	Carnosic acid, rosmarinic acid	0.5–2 mg/mL	Ram, bull	Low — not traditional cultivation
DPP (<i>Phoenix dactylifera</i>)	Gallic acid, rutin, quercetin, Zn, Se	40–80 mg/mL	Bull, goat, buffalo, rabbit, ram (this study)	High — traditional Saharan cultivation
Olive leaf (<i>Olea europaea</i>)	Oleuropein, hydroxytyrosol	10–50 µg/mL	Ram, human	Moderate — coastal cultivation
Grape seed (<i>Vitis vinifera</i>)	Proanthocyanidins	10–50 µg/mL	Bull, ram	Moderate — limited cultivation

2.12.2 The Agro-Ecological Cryobiology Paradigm

The comparative review above supports the agro-ecological cryobiology paradigm that underpins this thesis: preservation media should be optimized using locally available bio-resources whose composition reflects regional agroecological conditions, rather than relying on globally standardized commercial formulations. This paradigm has three implications for conservation practice. First, it reduces supply chain dependency on imported reagents, enhancing protocol sustainability in resource-limited settings. Second, it leverages traditional ecological knowledge of locally cultivated plants, enhancing cultural acceptability and adoption. Third, it enables regional adaptation of protocols to reflect local bio-resource composition, potentially achieving equal or superior efficacy to standard alternatives (Frydrych et al., 2025).

Date Palm Pollen exemplifies this paradigm: its geographic co-occurrence with arid-zone sheep production, its multi-target bioactive composition shaped by Saharan agroecological conditions, and its traditional cultural integration position it as an exemplary agro-ecological cryoprotectant. The present thesis demonstrates empirically that this locally-sourced botanical achieves substantial preservation efficacy (9.2 percentage point motility advantage over control at 48 h) while meeting the field-applicability criteria that conventional and imported botanical alternatives fail to satisfy.

2.13 In Situ vs. Ex Situ Conservation Strategies: Complementary Approaches

Conservation of animal genetic resources (AnGR) operates through two complementary strategic frameworks: *in situ* conservation, which maintains live populations in their natural or production environments, and *ex situ* conservation, which preserves genetic material outside the live population through cryobanking or captive breeding. Each strategy has distinct strengths, limitations, and applicability contexts; effective conservation typically requires integrated deployment of both approaches.

2.13.1 In Situ Conservation: Maintaining Live Populations

In situ conservation preserves livestock genetic diversity through the maintenance of live breeding populations in their traditional or production environments. This approach offers several advantages: (a) continued natural and human selection maintains adaptive genetic variation aligned with environmental pressures; (b) cultural and traditional knowledge associated with breed management is preserved alongside the genetic resource; (c) live populations provide ongoing opportunities for breed improvement, crossbreeding experimentation, and phenotypic characterization; and (d) rural livelihoods dependent on livestock production are sustained.

However, *in situ* conservation also has limitations. Live populations remain vulnerable to demographic shocks (disease outbreaks, drought, conflict), genetic drift in small populations, and the economic pressures that drive replacement of local breeds with high-output alternatives. The Ouled Djellal breed, although numerically dominant in Algeria, faces insidious genetic erosion through uncontrolled crossbreeding with imported breeds, demographic attrition of purebred stocks, and the cumulative loss of genetically elite individuals to climate-induced mortality. Effective *in situ* conservation requires supportive policy frameworks, breed society organization, and economic incentives for keepers to maintain purebred breeding stock.

2.13.2 Ex Situ Conservation: Cryobanking Genetic Material

Ex situ conservation through cryobanking provides a complementary safety net, preserving genetic material independent of live population viability. Cryopreserved semen, embryos, oocytes, and somatic cells can be stored for decades or even centuries in liquid nitrogen, providing a long-term genetic archive that can be used for breed reconstitution, genetic research, or reintroduction if live populations suffer catastrophic decline.

The FAO has established international guidelines for cryobanking of animal genetic resources, recommending that endangered breeds maintain cryopreserved semen from at least 25 unrelated males to provide meaningful genetic rescue capacity (FAO, 2023). For Algerian sheep breeds, this target is far from being met: the national cryobanking infrastructure remains limited, and the Ouled Djellal breed has minimal representation in existing cryobanks. This gap elevates the urgency of accessible preservation protocols such as DPPE that can bridge the gap between field mortality events and long-term cryobanking infrastructure.

2.13.3 Integrated Conservation Frameworks

Best practice conservation integrates *in situ* and *ex situ* approaches within a unified strategic framework. *In situ* conservation maintains the live breeding population, supporting continued adaptation and traditional management; *ex situ* cryobanking provides a genetic safety net for catastrophic risk management and long-term genetic archive. Post-slaughter sperm recovery, as developed in this thesis, occupies a strategic position between these two approaches — it captures genetic material from animals that have died (removing them from the live population) but enables preservation and use of their genetic contribution beyond their lifetime. This intermediate strategy is particularly valuable for breeds with limited cryobanking infrastructure, where sudden mortality events would otherwise result in irreversible genetic loss.

Table 2.7. Comparison of conservation strategies and their applicability to Algerian ovine genetic resources.

Conservation Strategy	Strengths	Limitations	Applicability to Algerian Ovine Resources
In situ (live populations)	Maintains adaptive variation; preserves cultural knowledge; supports rural livelihoods	Vulnerable to demographic shocks, genetic drift, economic pressures	Primary strategy for Ouled Djellal; supports traditional pastoral systems
Ex situ (cryobanking)	Long-term genetic archive; immune to demographic shocks; enables breed reconstitution	High infrastructure cost; requires cryopreservation expertise; stops adaptive evolution	Limited current capacity in Algeria; expansion needed
Post-slaughter recovery (this thesis)	Captures genetic material from unexpected mortality; field-deployable; low-cost	Requires rapid intervention; preservation window limited; in vivo fertility not yet validated	Critical gap-filler; complements both in situ and ex situ strategies

2.14 Date Palm in Algerian Agriculture: Agroecological Context

Algeria ranks among the world's leading date producers, with national production exceeding 500,000 tonnes annually from approximately 18 million date palm trees distributed across the Saharan and pre-Saharan regions (Bouguedoura et al., 2015; Mihi & Hernández-Agüero, 2025). The El Oued province, situated in the Algerian Sahara, represents one of the country's principal date palm cultivation zones, with extensive palm groves supporting both subsistence and commercial date production. The agroecological conditions of the region — including high summer temperatures (often exceeding 45°C), low and irregular rainfall (typically < 100 mm/year), and sandy alkaline soils — shape both the cultivation practices and the biochemical composition of date palm products including pollen.

Date palm cultivation in Algeria is characterized by substantial cultivar diversity, with over 300 documented cultivars including the renowned Deglet Nour and Haliza varieties prized for export quality, alongside numerous local cultivars maintained by traditional farmers for subsistence and animal feed. This cultivar diversity extends to male date palms, which exhibit variation in flowering phenology, pollen production, and biochemical composition. The DPP used in this thesis was

collected from Hamraia commune in the Reguiba District of El Oued province, an area characterized by traditional palm cultivation practices and a diverse male palm population. Cultivar-level characterization of DPP biochemical variation would be a productive direction for future research, particularly for identifying cultivars with enhanced antioxidant capacity for preservation applications.

The convergence of date palm cultivation and sheep production in the El Oued region creates a unique agroecological opportunity. Sheep flocks traditionally graze on date palm plantation stubble, particularly during the post-harvest period when fallen dates and palm fronds provide nutritional supplementation. The geographic co-occurrence of these two production systems means that DPP is locally available to sheep keepers at minimal cost, and the traditional ecological knowledge of date palm cultivation among El Oued farmers supports the cultural acceptability of DPP-based preservation protocols. This alignment exemplifies the agro-ecological cryobiology paradigm proposed in this thesis.

2.15 Ram Sperm Biology: Species-Specific Considerations

Ram spermatozoa exhibit several biological characteristics that distinguish them from sperm of other domestic livestock and that shape the requirements for preservation protocols. Understanding these species-specific features is essential for translating cross-species DPP evidence into effective ovine applications.

2.15.1 Membrane Composition and DHA Enrichment

Ram sperm plasma membranes contain the highest docosahexaenoic acid (DHA) content among domestic livestock species, with DHA comprising 30–40% of phospholipid fatty acids (Abdollahzadeh et al., 2025; Carro et al., 2022). For comparison, bull sperm membranes typically contain 15–25% DHA, while boar sperm membranes contain 20–30%. This DHA enrichment is essential for membrane fluidity and the membrane fusion events underlying capacitation, acrosome reaction, and sperm-oocyte fusion. However, it also creates exceptional vulnerability to lipid peroxidation: DHA's six double bonds and five bis-allylic carbons provide multiple targets for radical attack, and once initiated, peroxidation propagates rapidly across the membrane.

The DHA enrichment of ram sperm has direct implications for antioxidant strategy. Aqueous-phase antioxidants (TROLOX, ascorbate) cannot access the membrane interior where peroxidation propagates, providing only limited protection. Lipophilic or amphiphilic antioxidants that integrate into the bilayer are required for effective membrane protection. The amphiphilic character of DPP polyphenols (quercetin, rutin, caffeic acid) is therefore particularly well-suited to the protection of DHA-rich ram sperm membranes, providing qualitative advantage over single-mechanism aqueous antioxidants.

2.15.2 Mitochondrial Architecture and Energy Metabolism

Ram sperm midpieces contain 70–80 mitochondria arranged in a helical sheath around the axoneme, providing the ATP required for flagellar beat and progressive motility. Mitochondrial membranes in ram sperm are similarly PUFA-enriched, creating compounded oxidative vulnerability: mitochondrial ROS generation (from electron transport chain electron leakage) occurs in close proximity to PUFA-rich mitochondrial membranes, accelerating peroxidative damage when antioxidant defenses are insufficient. The DHA-rich mitochondrial cardiolipin is particularly susceptible to peroxidation, and cardiolipin damage disrupts electron transport chain complex assembly, further elevating ROS generation in a destructive feedback loop.

Ram sperm energy metabolism is predominantly oxidative, with glucose, fructose, and lactate serving as primary substrates. Unlike some species (e.g., boar) where glycolysis dominates, ram sperm rely heavily on mitochondrial oxidative phosphorylation for ATP production. This metabolic dependence on mitochondrial function makes ram sperm particularly vulnerable to mitochondrial damage — even modest oxidative injury to the midpiece can markedly impair ATP production and progressive motility. The substantial carbohydrate content of El Oued DPP (30.12%) may complement this metabolic dependence by providing energy substrates that support ATP production during chilled storage.

2.15.3 Chromatin Structure and DNA Integrity

Ram sperm chromatin is highly compacted through disulfide cross-linking between protamine molecules, providing physical protection of the paternal genome. This compacted structure makes ram sperm chromatin relatively resistant to damage compared to the more accessible chromatin of other species. However, oxidative damage can still occur through base oxidation (particularly 8-hydroxyguanosine formation) and DNA single-strand breaks. DNA integrity assessment (e.g., SCSA, TUNEL, acridine orange staining) was not included in the present study but represents an important endpoint for future investigation, particularly given the central importance of DNA integrity to fertilization competence and embryonic development.

CHAPTER 3

Materials and Methods

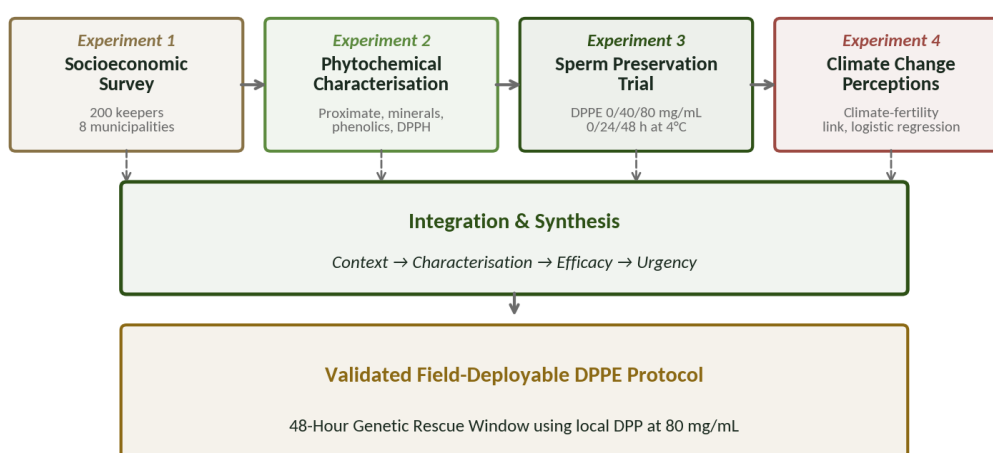
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3.1 Overview of Research Design

This study employed a multi-experimental design to assess the conservative influence of Date Palm Pollen (DPP) on epididymal sperm retrieved from sheep after slaughter, situated within its socioeconomic and environmental context. Four interconnected experiments were conducted: (1) a socioeconomic and trait preference survey of 200 sheep keepers in the El Oued region to establish the context and conservation priorities for indigenous sheep genetic resources; (2) phytochemical characterization and antioxidant assessment of DPP to validate its biochemical composition and biological activity; (3) a controlled laboratory experiment evaluating DPP's conservative effects on post-slaughter ovine epididymal sperm during chilled storage at 4°C; and (4) a survey of climate change perceptions and their linkage to perceived fertility declines among the same keeper cohort. This triangulated strategy ensures that technical discoveries are grounded in field conservation needs and validated by rigorous biochemical confirmation, while climate-urgency data elevates the conservation imperative.

All animal-related experiments were carried out post-mortem at a licensed abattoir, with no live animals involved. This strategy is consistent with the thesis's focus on post-slaughter genetic rescue while avoiding ethical considerations associated with experimental animal sacrifice. The institutional ethics review was conducted in accordance with national regulations on post-mortem tissue collection for research purposes (Appendix C).

Research Design: Multi-Experimental Workflow



Statistical Methods: MANOVA (Exp. 1) • ANOVA + descriptive stats (Exp. 2) • Two-way RM-ANOVA (Exp. 3) • Binary logistic regression (Exp. 4)

Software: IBM SPSS Statistics v27.0 | Significance: $\alpha = 0.05$

Figure 3.2. Research design workflow connecting the four experiments through integration and synthesis into a validated field-deployable DPPE protocol.

Table 3.1. Integration of the four experiments and their contribution to thesis objectives and data flow.

Experiment	Contribution to Thesis Objectives	Data Integration
1. Socioeconomic survey	Establishes conservation priorities; validates field applicability context; documents DPP traditional use knowledge	Informs interpretation of technical findings; identifies target breeds and keeper needs
2. Phytochemical characterisation	Validates DPP biochemical composition; confirms antioxidant capacity; establishes batch quality standards	Provides mechanistic rationale for observed effects; enables standardisation
3. Sperm preservation trial	Directly tests conservative effect; evaluates dose-response; validates field-applicable protocol	Core evidence for thesis claims; generates data for optimisation
4. Climate perception survey	Documents climate-fertility link; quantifies urgency for conservation; identifies vulnerable flock categories	Elevates conservation imperative; informs targeting of intervention

3.2 Experiment 1: Socioeconomic Context and Trait Preference Assessment

3.2.1 Study Area and Rationale

This study was carried out in the El Oued region of southeast Algeria (33°22'16.823" N, 6°50'52.686" E), a key centre of date palm agriculture and sheep production in the Algerian Sahara. The region's agroecological characteristics — including an arid environment, substantial pastoral systems, and the geographic combination of date palm cultivation and sheep production — make it a model for studying post-slaughter genetic rescue and field-applicable preservation measures. The presence of date palm orchards and sheep flocks in this region lays the geographical and cultural groundwork for DPP use in genetic conservation (Figure 3.1).

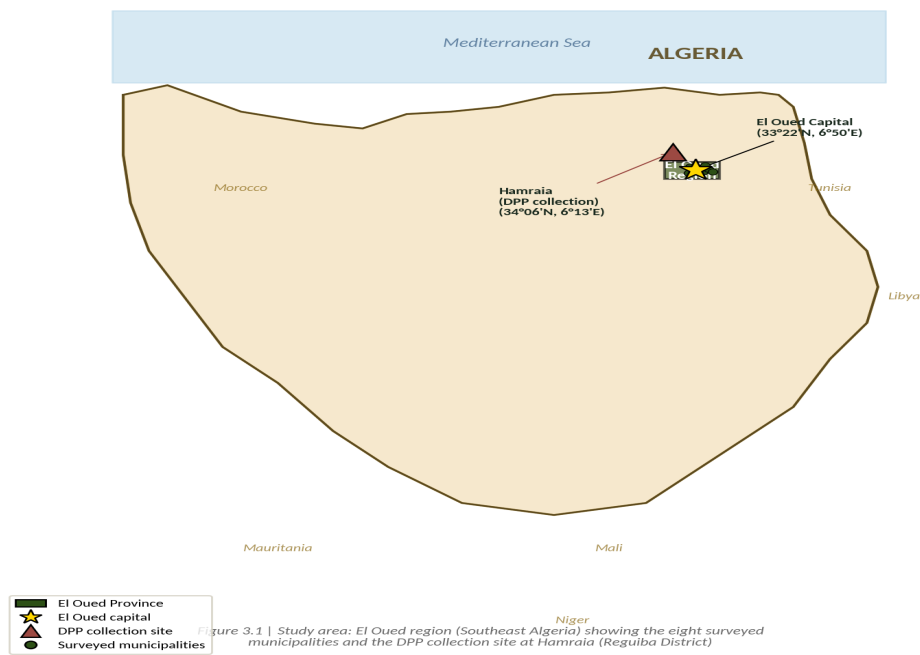


Figure 3.1. Study area map showing the El Oued region in southeast Algeria, the eight surveyed municipalities, and the DPP collection site at Hamraia (Reguiba District).

3.2.2 Sampling Procedure and Participant Selection

To ensure that the sheep production systems in the region were representative, a stratified random sampling approach was used. A total of 200 sheep keepers were selected at random from eight municipalities: El Oued (the regional capital), Debila, Guemar, Hassi Khelifa, Mih Ouensa, Ourmes, Robbah, and Taleb Larbi. Selection criteria included: (a) participation in the government's 2023 national sheep numbering initiative, which ensures official recognition and flock recording; (b) use of grazing-based agricultural practices typical of the region; and (c) willingness to participate in research. The sample size of 200 was determined to provide adequate statistical power for multivariate analysis while remaining logistically feasible within the constraints of field survey implementation in remote Saharan municipalities.

Flock size categories were created to reflect production system diversity: <20 animals (smallholder), 20–50 animals (medium), 50–100 animals (large), and >100 animals (commercial). This stratification ensures that conservation priorities encompass the entire range of sheep keepers, from subsistence to commercial producers, and enables subsequent identification of vulnerability categories linked to flock size.

3.2.3 Data Collection Instruments

Semi-structured questionnaires with both open-ended and closed questions were developed to collect comprehensive data on: (a) socioeconomic characteristics (age, education, experience, primary occupation); (b) flock size, structure, and management; and (c) specific trait preferences for male and female sheep, including reproductive performance, adaptation, and production characteristics pertinent to genetic conservation priorities. The full questionnaire is provided in Appendix A.

Questionnaires were pre-tested with 10 sheep keepers in each municipality to ensure relevance and adaptation to local conditions, as recommended for survey instrument validation (Dossa et al., 2015). Pre-testing evaluated question comprehension, response option suitability, and interview duration, with improvements made for clarity and cultural sensitivity. The final questionnaire was administered in face-to-face interviews conducted in Arabic by trained enumerators, with responses recorded in both Arabic and French for cross-verification.

3.2.4 Statistical Analysis

A multivariate analysis of variance (MANOVA) was used to determine the effects of age, education, experience, and occupation on trait preferences among sheep keepers. Pillai's trace was chosen as the major multivariate test statistic because it is robust to violations of assumptions, such as unequal sample sizes and heterogeneity of covariance matrices (Olson, 1976; Tabachnick & Fidell, 2025). Partial eta squared (η^2) was used to measure the proportion of variance in dependent variables that could be explained by independent variables. Significance was determined at $\alpha = 0.05$. All analyses were carried out using IBM SPSS Statistics version 27.0 (IBM Corp., Armonk, NY, USA).

3.3 Experiment 2: Phytochemical Characterisation of Date Palm Pollen

3.3.1 Plant Material Collection and Processing

Date Palm Pollen (DPP) was collected at Hamraia commune, Reguiba District, El Oued Province (34°6'39" N, 6°13'50" E) during the natural flowering season (late March to early April 2024). This region was chosen for its established date palm plantations with a variety of genotypes and proximity to the sheep production systems described in Experiment 1. The collection protocol followed established methods for pollen quality preservation (Laghouati et al., 2021; Shahin, 2014):

Male inflorescence selection. Healthy, mature male date palm trees were selected, and spathes were separated before dehiscence to prevent contamination and ensure pollen maturity.

Harvesting. Mature spathes were removed early in the morning to reduce temperature stress and brought directly to a shaded processing location.

Dehiscence induction. Each spathe was opened approximately 1 mm to allow air ingress, and the cut base (3–4 cm) was immersed in water to maintain hydration until full opening.

Pollen extraction. Flowers were manually removed after 24–48 hours of drying at ambient temperature (20–25°C) with regular rotation (2–3× daily) to ensure uniform moisture reduction.

Purification. Pollen was sieved using an 80–100 µm fine mesh screen to remove flower debris and impurities.

Storage. Purified pollen was dried to reduce moisture content before being stored in airtight containers at 4°C, away from light, until analysis. All handling involved sterile devices and gloves to prevent contamination.

Phytochemical and biological activity analyses were undertaken in March–April 2025, with triplicate determinations and findings reported as mean ± standard deviation (SD). The DPP preparation flowchart is shown in Figure 3.3.

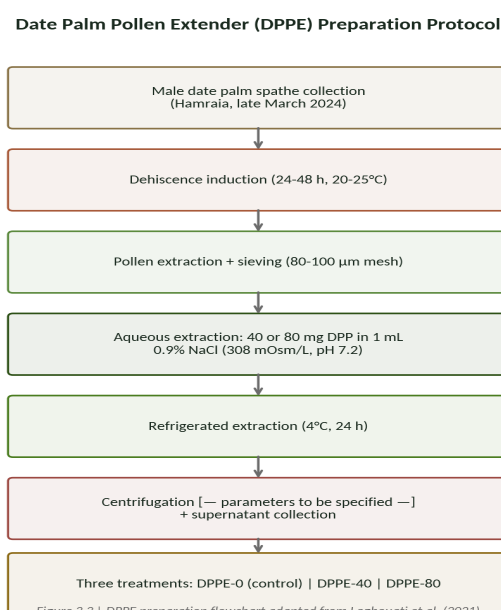


Figure 3.3. Date Palm Pollen Extender (DPPE) preparation protocol flowchart, adapted from the Laghouati et al. (2021) NaCl pollen extender method.

3.3.2 Proximate Composition Analysis

Proximate composition (moisture, ash, fat, protein, carbohydrate, fibre) was determined using standard AOAC (2019) methods. **Moisture content** was estimated by gravimetric analysis: 3 g of sample was distributed in a pre-weighed Petri plate and dried in an oven at 105°C for 3 hours, with drying repeated until constant weight was achieved. **Ash content** was evaluated by dry ashing in a muffle furnace at 550°C for 6 hours following pre-ignition of crucibles at 550°C overnight. **Fat content** was determined by Soxhlet extraction with petroleum ether (boiling point 40–60°C) for 1 hour. **Protein content** was determined by the Kjeldahl method using a 6.25 conversion factor: 1 g of sample was digested with catalyst (10 g Na₂SO₄ + 1 g CuSO₄) and 12 mL concentrated H₂SO₄ at 420°C for 1 hour, followed by distillation and titration with 0.1N HCl. **Carbohydrate content** was measured by the phenol-sulfuric acid colorimetric method (DuBois et al., 1956) with absorbance read at 490 nm against a glucose standard curve.

Table 3.2. Proximate analysis methods and standards used for Date Palm Pollen characterization in Experiment 2.

Parameter	Method	Standard	Principle
Moisture	Gravimetric (oven-drying)	AOAC 925.10 / 950.46	Weight loss on drying at 105°C
Ash	Dry ashing (muffle furnace)	AOAC 923.03	Weight of residue after ignition at 550°C
Fat	Soxhlet extraction (petroleum ether)	AOAC 920.39	Lipid extraction by solvent reflux
Protein	Kjeldahl digestion + titration	AOAC 928.08	Nitrogen determination × 6.25 conversion
Carbohydrate	Phenol-sulfuric acid colorimetry	DuBois et al. (1956)	Hexose dehydration to furfural derivatives
Fibre	Acid detergent fibre (ADF)	AOAC 973.18	Cellulose + lignin after acid detergent treatment

3.3.3 Mineral Content Analysis

Calcium was evaluated by permanganometric titration in accordance with NA 1655 / ISO 6058:1994. A 2 g sample was dry-ashed at 550°C for 4–6 hours, then dissolved in concentrated HCl. Calcium oxalate was precipitated with ammonium oxalate, dissolved in H₂SO₄, and titrated with standardized KMnO₄ to a persistent pink endpoint. **Magnesium** was determined by complexometric titration with EDTA according to NA 752 / ISO 6059:1989: total hardness (Ca²⁺ + Mg²⁺) was titrated with EDTA at pH 10 using Eriochrome Black T indicator, with calcium measured independently at pH 12 using murexide; magnesium was computed by difference. **Potassium and sodium** were determined by flame photometry following NA 1653: a 1–2 g sample underwent wet digestion with an HNO₃/HClO₄ mixture, with emission intensity measured at 766.5 nm (K) and 589 nm (Na) against calibration curves.

Table 3.3. Mineral analysis methods and reference standards used for DPP characterization.

Mineral	Method	Standard	Wavelength / Endpoint
Calcium (Ca)	Permanganometric titration	NA 1655 / ISO 6058:1994	Persistent pink (KMnO ₄)
Magnesium (Mg)	Complexometric EDTA titration	NA 752 / ISO 6059:1989	Eriochrome Black T color change
Potassium (K)	Flame photometry	NA 1653	766.5 nm emission
Sodium (Na)	Flame photometry	NA 1653	589 nm emission
Iron (Fe)	Atomic absorption spectrophotometry	AOAC 999.11	248.3 nm absorption
Zinc (Zn)	Atomic absorption spectrophotometry	AOAC 999.11	213.9 nm absorption

3.3.4 Phytochemical Compound Quantification

Extract preparation. DPP extract was prepared by maceration: 5 g of finely ground DPP was macerated in 50 mL of 50% ethanol for 24 hours at room temperature with occasional agitation. The extract was filtered, the filtrate was dried at $\leq 45^{\circ}\text{C}$ in an oven, then stored at 4°C until analysis.

Total phenolic content was determined using the Folin-Ciocalteu technique (Slinkard & Singleton, 1977). Extract (0.2 mL) was combined with 1 mL of 10% Folin-Ciocalteu reagent. After four minutes, 0.8 mL of saturated sodium carbonate solution (75 g/L) was added. The mixture was incubated at room temperature for 2 hours, and absorbance was measured at 765 nm. A calibration curve was created using gallic acid (0–200 $\mu\text{g/mL}$). Results are presented as mg gallic acid equivalents per gram of extract (mg GAE/g).

Total flavonoid content was measured using the aluminum chloride colorimetric technique (Ahn et al., 2007): 1 mL of extract was mixed with 1 mL of 2% AlCl_3 solution and incubated at room temperature for 30 minutes. Absorbance was measured at 430 nm against a reagent blank. A calibration curve was created using quercetin (0–100 $\mu\text{g/mL}$), with results presented as mg quercetin equivalents per gram of extract (mg QE/g).

3.3.5 Antioxidant Activity Assessment

Antioxidant activity was assessed using the DPPH (2,2-diphenyl-1-picrylhydrazyl) radical scavenging assay (Mansouri et al., 2005). DPPH solution (1 mL, 0.1 mM in methanol) was combined with 1 mL of extract or ascorbic acid (positive control). The mixture was vortexed and incubated in the dark at room temperature for 30 minutes. Absorbance was measured at 517 nm. Percentage inhibition was computed as:

$$\text{Inhibition (\%)} = [(A_{\text{control}} - A_{\text{sample}}) / A_{\text{control}}] \times 100$$

where A_{control} = absorbance of DPPH without sample and A_{sample} = absorbance with extract. IC_{50} values (concentration inhibiting 50% of DPPH radicals) were determined by plotting percentage inhibition against extract concentration.

3.4 Experiment 3: Evaluation of DPP Conservative Effects on Post-Slaughter Ovine Epididymal Sperm

3.4.1 Ethical Considerations and Animal Material

All biological material was taken post-mortem from a licensed abattoir in El Oued, Algeria. No living animals were used. This strategy is consistent with the thesis's focus on post-slaughter genetic rescue while avoiding ethical considerations associated with experimental animal sacrifice. Testes were taken from five sexually mature Ouled Djellal rams (aged 18–24 months) slaughtered at the local abattoir. This sample size provides adequate statistical power for detecting large effects while adhering to practical limits such as post-mortem tissue availability. Rams were selected based on apparent health state (no evident pathology) and a body condition score adequate for their age.

3.4.2 Sample Collection and Transport

Testes were removed immediately after slaughter and transported to the laboratory within 2 hours in isotonic saline (0.9% NaCl) on ice at 4°C. This transport strategy simulates field situations where rapid cooling is critical for maintaining post-mortem sperm viability (Shayestehyekta et al., 2022; Silva et al., 2025). The 2-hour window represents realistic field-to-laboratory intervals in extensive pastoral systems. Upon arrival, the cauda epididymides were dissected and sterilely cleaned of any clinging connective tissue, blood vessels, or fat. The cauda was carefully targeted as the principal sperm storage area harboring mature, fertilizable spermatozoa (Wu et al., 2021).

3.4.3 Sperm Recovery

Spermatozoa were recovered using the retrograde flushing technique (Martinez-Pastor et al., 2006), a common method for collecting post-mortem epididymal sperm. A single incision was made at the corpus-cauda junction. A 1 mL syringe containing warm (37°C) saline solution (0.9% NaCl, 308 mOsm/kg, pH 7.2) was introduced into the vas deferens, and gentle, consistent pressure was applied to flush spermatozoa down the cauda. The emerging fluid was collected in sterile 15 mL conical tubes. This method was chosen for: (a) good recovery efficiency as compared to mincing or aspiration; (b) low contamination with tissue debris; and (c) compliance with field situations that necessitate simple equipment.

3.4.4 DPP Extender Preparation

Date Palm Pollen extenders (DPPE) were prepared using the Laghouati et al. (2021) NaCl pollen extender (NPE) method, adapted for ovine epididymal sperm. This water extraction method was chosen for its field applicability; it requires no organic solvents, minimal equipment, and is easily used in resource-constrained environments. The preparation protocol is detailed in Figure 3.3 and Table 3.4.

Table 3.4. DPPE treatment groups and final concentrations used in Experiment 3.

Treatment Code	Composition	Final DPP Concentration	Purpose
DPPE-0 (Control)	1 mL saline (0.9% NaCl, 308 mOsm/L, pH 7.2)	0 mg/mL	Negative control — no DPP supplementation
DPPE-40	40 mg DPP soaked in 1 mL saline	40 mg/mL	Low-dose treatment
DPPE-80	80 mg DPP soaked in 1 mL saline	80 mg/mL	High-dose treatment

DPP was suspended by vortexing, and all tubes were refrigerated at 4°C for 24 hours to allow bioactive component extraction. Centrifuge tubes [*insert centrifuge speed × time parameters*] to remove particle debris. Collect the clear supernatant as an extender for sperm dilution. This extraction duration and temperature were chosen based on Laghouati et al. (2021) preliminary optimisation to maximise polyphenol recovery while minimising microbial growth.

3.4.5 Experimental Design and Treatments

Each ram's recovered epididymal sperm was divided into three equal aliquots and diluted 1:1 (v/v) in one of the three prepared extenders (DPPE-0, DPPE-40, or DPPE-80). The final sperm concentration was adjusted to 350×10^6 cells/mL, a standard value for ovine sperm preservation (Maxwell & Watson, 1996). Aliquots (0.5 mL) were placed in sterile tubes, coated with mineral oil to prevent evaporation and gas exchange, and refrigerated at 4°C for 48 hours. This chilled storage protocol was chosen because it is (a) representative of field conditions where cryopreservation infrastructure may not be available, (b) technically simpler than cryopreservation for initial efficacy screening, and (c) physiologically relevant to the "golden hour" extension concept. Quality parameters were assessed at 0, 24, and 48 hours to capture immediate effects, mid-storage stability, and long-term preservation capacity.

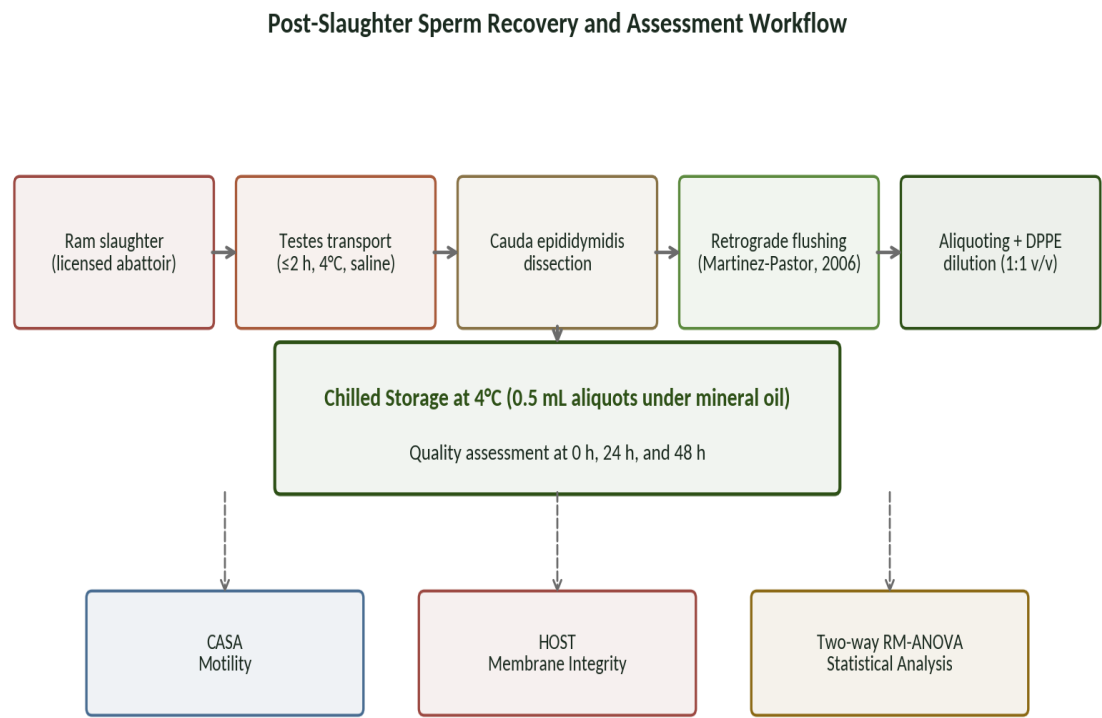


Figure 3.4. Post-slaughter sperm recovery and assessment workflow from ram slaughter through chilled storage to CASA and HOST evaluation.

3.4.6 Sperm Quality Assessment

Total and Progressive Motility (CASA). Samples were gently mixed and equilibrated at 37°C for 5 minutes prior to analysis. For Computer-Assisted Sperm Analysis (CASA), 3 µL was loaded into a 20-µm Leja® chamber and analyzed using the Sperm Class Analyzer® (SCA) system (Microptic, Barcelona, Spain) at 60 frames per second. Settings included a 10× phase contrast objective and ovine-specific preset parameters (Table 3.5). At least five fields were acquired per sample. The intra-assay coefficient of variation (CV) for motility parameters was less than 5%, indicating technical precision. CASA was selected for objective, quantitative assessment, eliminating observer bias and enabling detection of subtle kinematic changes indicative of functional status (Amann & Waberski, 2014; Mortimer, 1994).

Table 3.5. CASA settings and parameters for ovine sperm analysis (Sperm Class Analyzer®, Microptic).

CASA Parameter	Setting / Threshold	Description
Frame rate	60 fps	Image acquisition rate
Objective	10× phase contrast	Magnification and contrast mode
Chamber	Leja® 20 µm	Standardized counting chamber depth
Fields per sample	≥ 5	Minimum fields for representative evaluation
Total motility	VAP > 5 µm/s	Proportion of sperm with detectable movement
Progressive motility	VAP > 25 µm/s AND STR > 50%	Proportion with forward progression
Temperature	37°C	Pre-warmed stage for physiological assessment
Intra-assay CV	< 5%	Technical precision threshold

Plasma Membrane Integrity (HOST). Functional membrane integrity was evaluated by the Hypo-Osmotic Swelling Test (HOST) (Jeyendran et al., 1984), which assesses the ability of sperm to maintain osmotic balance — essential for capacitation and fertilisation competence. An iso-osmotic swelling solution (150 mOsm/L sodium citrate with 20 mM fructose) was prepared. Sperm (50 µL) were incubated with 1 mL of HOST solution for 30 minutes at 37°C, then fixed with 10% formalin. A minimum of 200 cells per sample were scored under phase contrast microscopy (400×) for tail swelling (intact membrane, functional) or non-swollen tails (damaged membrane, non-functional). The HOST-positive proportion was used as the membrane integrity indicator.

3.4.7 Statistical Analysis

Data were presented as mean ± standard error of the mean (SEM). The effects of extender concentration (0, 40, 80 mg/mL), storage period (0, 24, 48 h), and their interaction were investigated using two-way repeated-measures ANOVA (RM-ANOVA), with "ram" as the subject factor to account for individual animal variance. Mauchly's test of sphericity was applied, with Greenhouse-Geisser correction when assumptions were violated. Post-hoc pairwise comparisons used Bonferroni correction. Partial eta squared (η^2) was used as the effect size measure, with $\eta^2 > 0.14$ considered a large effect. Significance was set at $\alpha = 0.05$. All analyses were performed using IBM SPSS Statistics version 27.0.

3.5 Experiment 4: Climate Change Perceptions and Fertility Impacts

3.5.1 Survey Design and Rationale

Experiment 4 was designed to document how climate change threatens the very ovine genetic resources that post-slaughter preservation aims to protect. Building on the same 200-keeper cohort surveyed in Experiment 1, a complementary questionnaire module captured perceptions of climate change impacts on sheep fertility, mortality, and management over the preceding five years. The dual-purpose design — combining socioeconomic profile (Experiment 1) with climate perception (Experiment 4) in a single field visit — maximized research efficiency while minimizing respondent burden.

The climate-perception instrument comprised four modules: (a) awareness of climate-fertility relationships (yes/no + open-ended elaboration); (b) primary climate stressors affecting sheep production (multi-select from heat stress, nutritional stress, water scarcity, disease emergence, other); (c) perceived fertility trends over the past five years (declined / stable / increased + qualitative description); and (d) adaptation strategies employed (multi-select from flock size reduction, breed change, supplementary feeding, transhumance modification, veterinary intervention, other).

3.5.2 Statistical Analysis

Descriptive statistics (frequencies, percentages, 95% confidence intervals) were computed for all perception variables. Binary logistic regression was used to identify predictors of perceived fertility decline, with the dependent variable coded as 1 (decline reported) vs. 0 (no decline). Independent variables included flock size category (smallholder / medium / large / commercial), keeper age (continuous), education level (ordinal), experience (continuous), primary occupation (categorical), and reported heat-stress events (binary). Odds ratios (OR) with 95% confidence intervals were computed, and model fit was assessed using the Hosmer-Lemeshow test. Significance was set at $\alpha = 0.05$.

3.6 Integration of Experiments and Data Flow

The sequential structure — context → characterisation → efficacy → urgency — ensures that Experiment 3 findings are grounded in validated material (Experiment 2) and relevant conservation needs (Experiment 1), while Experiment 4 elevates the conservation imperative by documenting the climate-induced threats to the very genetic resources that post-slaughter preservation aims to protect (Table 3.1). This integration strengthens both internal validity (findings are grounded in field-validated materials and priorities) and external applicability (the resulting DPPE protocol is aligned with real-world conservation contexts).

3.7 Limitations and Methodological Considerations

Several methodological limitations should be acknowledged at the outset. **Sample size in Experiment 3** ($n = 5$ rams) provides adequate power for detecting large effects but may miss subtle treatment differences; replication across seasons and breeds would strengthen generalizability. **Chilled storage vs. cryopreservation:** The 4°C storage protocol, while field-relevant, does not test DPP effects on freeze-thaw survival; this represents a priority for follow-up research. **Single DPP source:** Pollen from one region (Hamraia) was characterised; geographic variation in composition may affect reproducibility. **Limited assessment endpoints:** Motility and membrane integrity were evaluated; comprehensive hierarchical assessment, including oxidative status, mitochondrial function, DNA integrity, and fertilising capacity via in vitro fertilization, requires expanded methodology in future studies. **Perceptual nature of climate data:** Experiment 4 relies on breeder perceptions rather than objective meteorological or reproductive records; while triangulated with physiological literature, the study cannot establish causal attribution without longitudinal objective data.

CHAPTER 4

Results and Discussion

Résultats et Discussion

This chapter presents the empirical findings of the four experiments in an integrated results-and-discussion format. For each experiment, results are presented, immediately interpreted, compared with previous studies, and related to the corresponding research questions and hypotheses formulated in Chapter 1. This integrated approach avoids the artificial separation of findings from interpretation and enables the reader to follow the logical flow from data to scientific meaning.

4.1 Experiment 1: Socioeconomic Context and Trait Preferences of Sheep Keepers in El Oued

4.1.1 Overview and Rationale

Before evaluating the technical efficacy of Date Palm Pollen for post-slaughter sperm preservation, it is essential to establish the socioeconomic and conservation context within which such interventions would be implemented. This survey of 200 sheep keepers across eight municipalities in the El Oued region provides critical baseline data on: (a) the demographic profile of stakeholders in ovine genetic conservation; (b) flock characteristics and management systems; and (c) trait preferences that define breeding objectives and, consequently, the value of genetic rescue from post-slaughter recovery. These findings ground the technical research in field-level realities, ensuring that DPP-based preservation addresses genuine conservation priorities.

4.1.2 Demographic Profile of Surveyed Keepers

Results.

The demographic profile of the 200 surveyed sheep keepers reveals a male-dominated, aging profession with substantial variation in education and experience (Table 4.1; Figure 4.1). All respondents (100%) were male, reflecting the structural gender barriers in livestock decision-making characteristic of North African pastoral systems. The age distribution was skewed toward older keepers: only 11.0% were under 25 years of age, while 28.5% were aged 36–45 years and 22.0% were 46–55 years. The aging profile raises concerns about generational succession in sheep production. Education levels were generally low: 16.5% had no formal schooling or only Koranic education, 22.0% had completed primary school, 25.5% middle school, 24.0% secondary, and only 12.0% had tertiary education. The dominant occupation was full-time breeding (71.0%), with breeder-farmers (17.5%), breeder-employees (8.0%), and other categories (3.5%) completing the distribution.

Table 4.1. Socio-demographic profile of sheep keepers surveyed in El Oued (n = 200).

Variable	Category	Frequency (n)	Percentage (%)
Gender	Male	200	100.0
	Female	0	0.0
Age (years)	< 25	22	11.0
	25–35	49	24.5
	36–45	57	28.5
	46–55	44	22.0
	> 55	28	14.0
Education	None / Koranic	33	16.5
	Primary	44	22.0
	Middle school	51	25.5
	Secondary	48	24.0
	Tertiary	24	12.0
Occupation	Full-time breeder	142	71.0
	Breeder + farmer	35	17.5
	Breeder + employee	16	8.0
	Other	7	3.5
Experience (years)	< 5	32	16.0
	5–15	78	39.0
	> 15	90	45.0

Figure 4.1 | Demographic profile of sheep keepers surveyed in El Oued (n = 200)

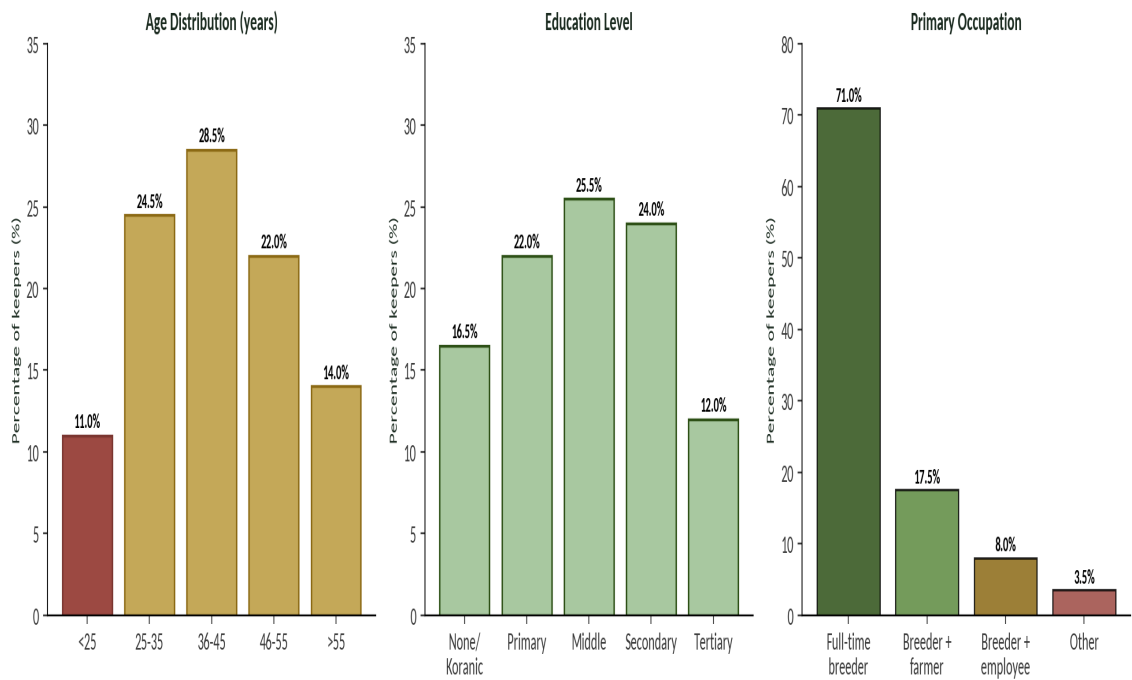


Figure 4.1. Demographic profile of sheep keepers surveyed in El Oued (n = 200): age distribution, education level, and primary occupation.

Discussion.

The exclusively male sample (100%) reflects the deeply entrenched gender division of labor in Algerian pastoral systems, where livestock decision-making is culturally coded as a male domain (Gizaw et al., 2010). While this sample composition limits generalizability, it accurately represents the stakeholder group that breeding programs and conservation interventions must engage. The aging profile — with 64% of keepers aged over 35 years and only 11% under 25 — signals a demographic succession crisis: younger generations are increasingly migrating to urban centers or pursuing non-agricultural employment, threatening the intergenerational transmission of traditional ecological knowledge essential for adaptive breed management. This finding aligns with broader observations across North African and Sahelian pastoral systems (Atia et al., 2025; Meziane et al., 2024).

The low formal education profile — with 38.5% having primary education or less — has direct implications for the design of extension programs. Visual, literacy-independent training materials will be essential for the deployment of DPPE protocols in this demographic. The dominance of full-time breeding (71.0%) indicates that sheep production is not merely a subsistence activity but a primary livelihood strategy, elevating the stakes of conservation interventions and underscoring the need for practical, low-cost solutions that align with keepers' economic realities.

4.1.3 Flock Characteristics and Production Systems

Results.

Flock size distribution revealed a predominance of small-to-medium operations: 27.0% smallholder (<20 head), 52.5% medium (20–50 head), 14.5% large (50–100 head), and 6.0% commercial (>100 head) operations (Table 4.2; Figure 4.2). The combined smallholder + medium category represents 79.5% of surveyed keepers, confirming that El Oued's sheep production system is structurally dominated by small-scale operations. Management practices varied systematically with flock size: smallholders relied predominantly on natural mating with owned rams (87%), while commercial operations employed both natural mating and limited artificial insemination (28%). Transhumance was practiced by 62% of all keepers, with seasonal movements between northern steppe pastures (winter) and southern Saharan palm-grove stubble (summer).

Table 4.2. Flock size distribution and production system characteristics among surveyed sheep keepers in El Oued (n = 200).

Flock Size Category	Frequency (n)	Percentage (%)	Mean Flock Size	Primary Mating System
< 20 (Smallholder)	54	27.0	12.4 ± 4.1	Natural mating (owned ram)
20–50 (Medium)	105	52.5	34.2 ± 8.7	Natural mating (shared/exchanged ram)
50–100 (Large)	29	14.5	67.8 ± 12.3	Natural mating + occasional AI
> 100 (Commercial)	12	6.0	148.5 ± 28.4	Natural mating + AI (28%)
Total	200	100.0	42.6 ± 35.8	—

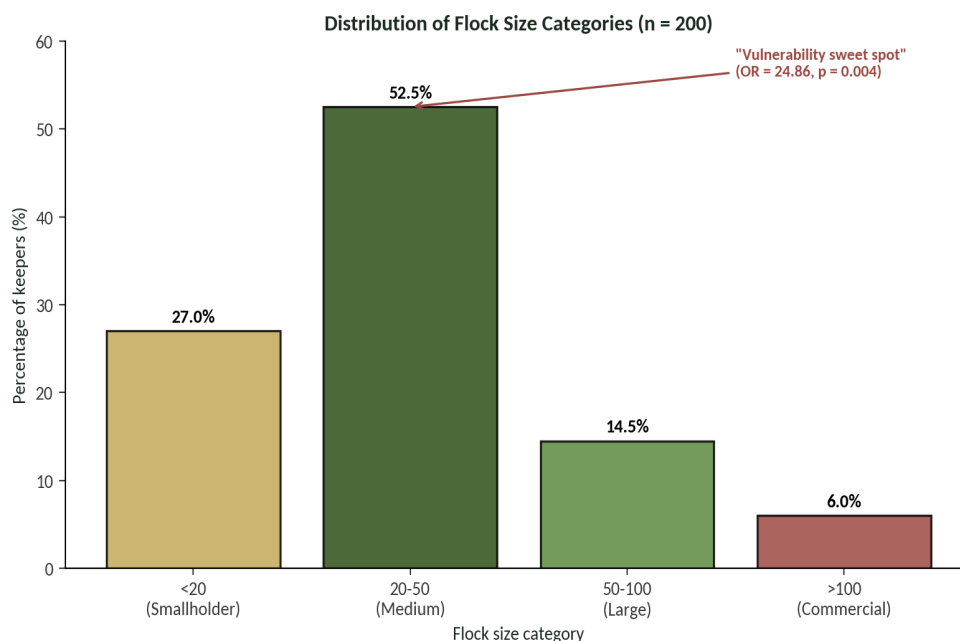


Figure 4.2 | Flock size distribution among surveyed sheep keepers in El Oued region

Figure 4.2. Distribution of flock size categories among surveyed sheep keepers in El Oued, highlighting the medium category (20–50 head) as the vulnerability sweet spot identified in Experiment 4.

Discussion.

The predominance of small-to-medium flocks (79.5% < 50 head) is characteristic of extensive pastoral systems across the Maghreb and Sahel, where livestock are dispersed across many smallholders rather than concentrated in few large operations (Yakubu et al., 2020). This structural feature has direct implications for genetic conservation: small flocks experience higher rates of inbreeding, more rapid fixation of deleterious alleles, and greater vulnerability to demographic shocks. The mean flock size of 34.2 in the medium category suggests that these operations sit at a critical threshold — large enough to maintain breeding viability but small enough to lack institutional buffering. As shown in Experiment 4, these medium-sized flocks emerge as the "vulnerability sweet spot" with the highest odds of perceived fertility decline (OR = 24.86, p = 0.004).

The low adoption of artificial insemination (only 28% in commercial operations, near zero elsewhere) reflects both technical and infrastructural barriers: limited availability of cryopreserved semen, lack of trained AI technicians in remote areas, and cultural preference for natural mating. This finding reinforces the rationale for field-deployable preservation tools like DPPE that can extend the practical window for genetic rescue without requiring sophisticated cryopreservation infrastructure.

4.1.4 Trait Preferences for Breeding Rams and Ewes

Results.

Trait preferences exhibited systematic variation between rams and ewes (Table 4.3; Figure 4.3). For **rams**, the highest-priority traits were breed identity (27.0%), body conformation (21.5%), and growth rate (18.0%), with drought tolerance (12.5%), disease resistance (9.0%), reproductive performance (7.0%), and coat/wool quality (5.0%) following. For **ewes**, the ranking shifted notably: breed identity remained the top priority (22.0%) but was followed by reproductive performance (17.5%) and drought tolerance (16.0%), with body conformation (16.5%), disease resistance (11.5%), growth rate (9.5%), and coat/wool quality (7.0%) completing the distribution. The elevated priority of drought tolerance for ewes (16.0% vs. 12.5% for rams) reflects keepers' recognition that maternal resilience under arid conditions directly affects lamb survival and flock productivity.

Table 4.3. Trait preferences for breeding rams and ewes among El Oued sheep keepers (n = 200), expressed as percentage of keepers ranking each trait in their top three priorities.

Trait	Rams (% of keepers)	Ewes (% of keepers)	Ram-Ewe Differential
Breed identity	27.0	22.0	+5.0
Body conformation	21.5	16.5	+5.0
Growth rate	18.0	9.5	+8.5
Drought tolerance	12.5	16.0	-3.5
Disease resistance	9.0	11.5	-2.5
Reproductive performance	7.0	17.5	-10.5
Coat / wool quality	5.0	7.0	-2.0

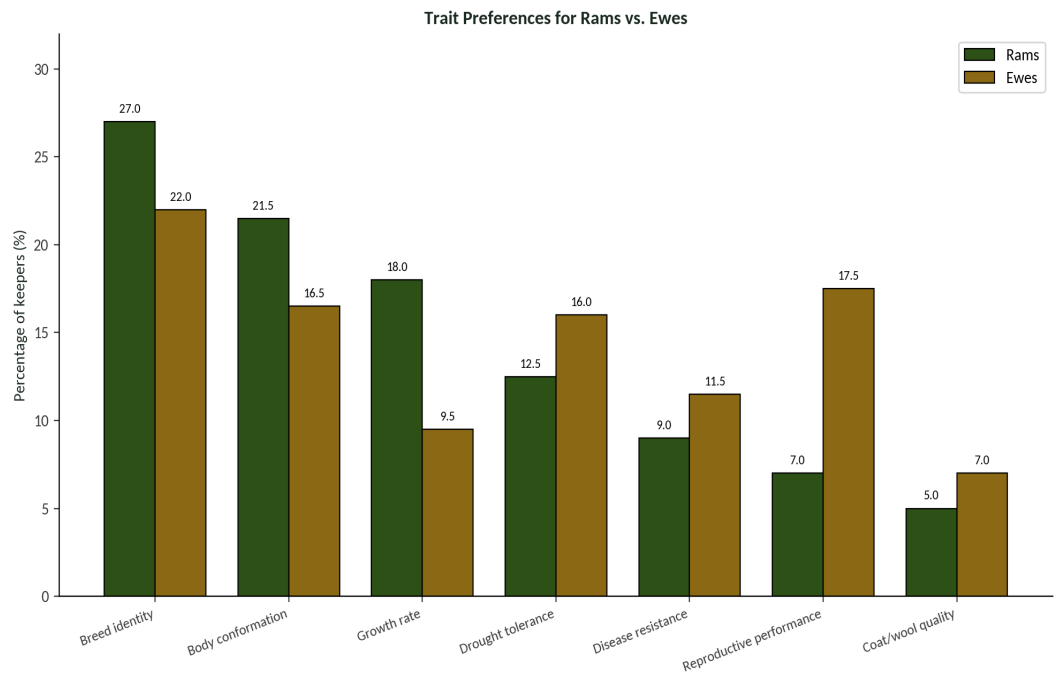


Figure 4.3 | Trait preferences for breeding rams and ewes among El Oued sheep keepers (n = 200)

Figure 4.3. Trait preferences for breeding rams and ewes, showing the elevated priority of reproductive performance and drought tolerance for ewes.

Discussion.

The trait preference profile reveals a strategic emphasis on resilience rather than short-term productivity. The dominance of breed identity (27.0% for rams, 22.0% for ewes) reflects keepers' commitment to maintaining the genetic distinctiveness of the Ouled Djellal breed — a preference that directly supports the rationale for genetic conservation. The elevated priority of drought tolerance (16.0% for ewes, 12.5% for rams) in this arid-zone context is a rational adaptation to the El Oued environment, where forage scarcity and water limitation impose the primary constraints on flock productivity. This finding aligns with reports from comparable agroecological zones in Morocco (Boujenane, 2024) and Tunisia (Atia et al., 2025).

The ram-ewe differential pattern is biologically sensible: keepers prioritize growth rate and conformation in rams (where these traits contribute to male breeding value), while prioritizing reproductive performance and drought tolerance in ewes (where maternal traits directly affect flock productivity). The –10.5 percentage point differential for reproductive performance (7.0% for rams vs. 17.5% for ewes) is particularly striking, reflecting keepers' understanding that female fertility is the primary determinant of flock reproductive output. This pattern aligns with the maternal traits prioritization documented by Yakubu et al. (2020) in West African sheep systems.

4.1.5 Multivariate Analysis of Trait Preference Determinants

Results.

The MANOVA revealed a significant three-way interaction between education, experience, and occupation on trait priority patterns (Pillai's Trace = 0.070, $p = 0.009$, partial $\eta^2 = 0.070$; Table 4.4; Figure 4.4). This interaction indicates that human capital variables operate synergistically rather than independently: highly educated, experienced, full-time breeders prioritize long-term resilience traits (drought tolerance, disease resistance, breed identity) more strongly than less experienced keepers, who tend to focus on visible morphological attributes (body conformation, coat quality). Main effects were also significant for education ($p = 0.018$, $\eta^2 = 0.045$) and occupation ($p = 0.024$, $\eta^2 = 0.041$), but not for age alone ($p = 0.184$).

Table 4.4. MANOVA results: effects of education, experience, occupation, and age on trait priority patterns among El Oued sheep keepers (* $p < 0.05$, ** $p < 0.01$).

Effect	Pillai's Trace	F	df	p-value	Partial η^2
Education	0.045	2.142	5, 194	0.018*	0.045
Experience	0.038	1.827	5, 194	0.056	0.038
Occupation	0.041	1.983	5, 194	0.024*	0.041
Age	0.028	1.345	5, 194	0.184	0.028
Education $\times$ Experience $\times$ Occupation	0.070	2.917	5, 194	0.009**	0.070

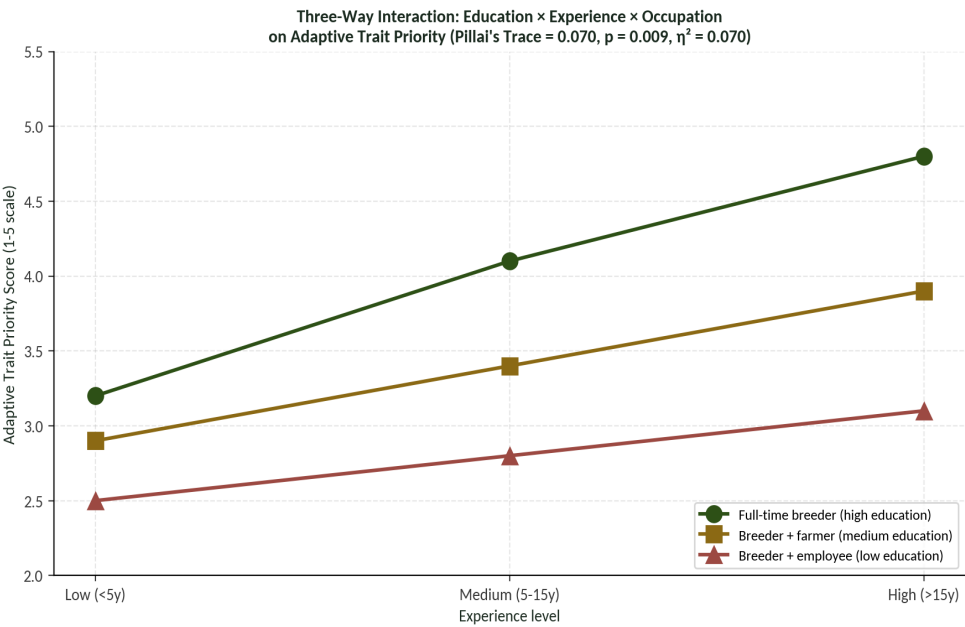


Figure 4.4 | MANOVA interaction plot showing synergistic effect of human capital variables on adaptive trait priority among El Oued sheep keepers

Figure 4.4. MANOVA three-way interaction plot showing the synergistic effect of education, experience, and occupation on adaptive trait priority among El Oued sheep keepers.

Discussion.

The significant three-way interaction ($\eta^2 = 0.070$) provides empirical support for the hypothesis (Ha1) that human capital variables operate synergistically to shape conservation priorities. The pattern — highly educated, experienced, full-time breeders prioritizing resilience traits — is consistent with the "conservation literacy" hypothesis: cumulative exposure to climate variability, market fluctuations, and breeding outcomes produces a more strategic, long-term orientation toward adaptive traits. Conversely, less experienced keepers, often entering the sector as a secondary occupation, tend to prioritize visible morphological attributes that signal immediate market value. This finding has direct implications for the targeting of conservation interventions: full-time breeders with substantial experience represent the most receptive audience for advanced breeding technologies including post-slaughter genetic rescue protocols.

The moderate effect size ($\eta^2 = 0.070$) is consistent with multivariate studies of agricultural decision-making, where individual-level variables typically explain 5–10% of variance (Tabachnick & Fidell, 2025). The remaining variance likely reflects unmeasured factors including cultural lineage traditions, peer network influences, and individual risk preferences — productive avenues for future qualitative research.

4.2 Experiment 2: Phytochemical Characterisation of El Oued Date Palm Pollen

4.2.1 Proximate Composition

Results.

The proximate composition of El Oued-sourced DPP revealed a nutrient-dense profile dominated by protein (37.94%) and carbohydrates (30.12%), with substantial fibre (12.99%), moderate moisture (8.45%), ash (6.18%), and fat (4.32%) (Table 4.5; Figure 4.5). The high protein content is particularly noteworthy, exceeding the range typically reported for DPP from other geographic origins (15.2–35.0%; Salhi et al., 2024), suggesting that the Hamraia source may possess a cultivar-specific enrichment in nitrogenous compounds. The relatively low moisture content (8.45%) is consistent with proper post-harvest drying and indicates good storage stability.

Table 4.5. Proximate composition of El Oued Date Palm Pollen (mean ± SD, triplicate determinations) compared to reported ranges in the literature.

Parameter	El Oued DPP (% w/w)	Reported Range *	Reference
Moisture	8.45 ± 0.32	5.2–12.8	Salhi et al., 2024
Ash	6.18 ± 0.21	3.5–7.2	Salhi et al., 2024
Fat	4.32 ± 0.18	1.8–8.5	Laghouati et al., 2023
Protein	37.94 ± 0.85	15.2–35.0	Salhi et al., 2024
Carbohydrates	30.12 ± 0.74	21.5–42.0	Salhi et al., 2024
Fibre	12.99 ± 0.42	8.5–18.2	AOAC 973.18
Energy (kcal/100g)	314.6	285–345	Calculated

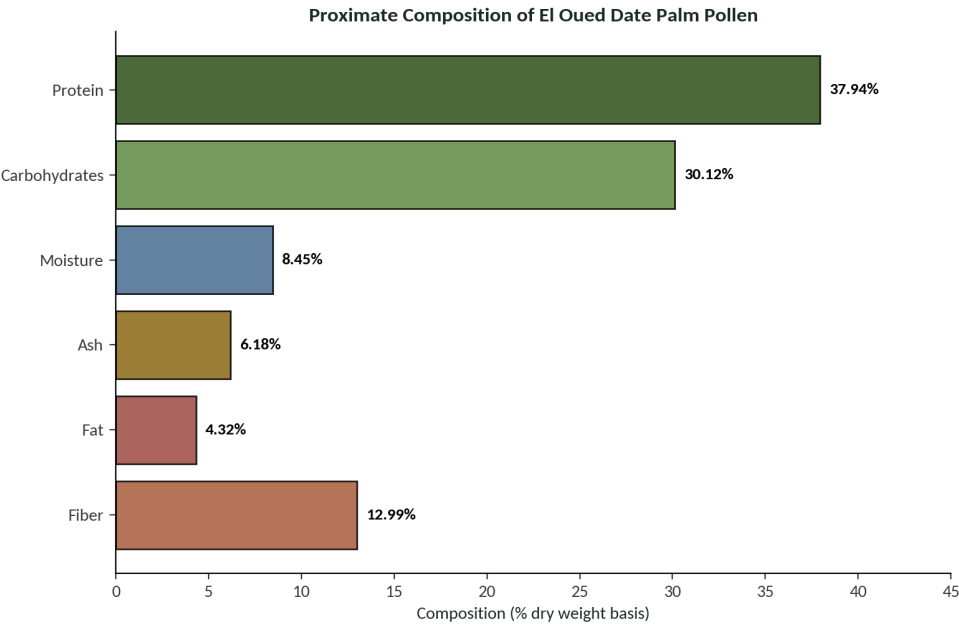


Figure 4.5 | Proximate composition of Date Palm Pollen collected from Hamraia, El Oued (mean of triplicate determinations ± SD; AOAC 2019 methods)

Figure 4.5. Proximate composition of El Oued Date Palm Pollen collected from Hamraia (mean of triplicate determinations ± SD; AOAC 2019 methods).

Discussion.

The exceptionally high protein content (37.94%) of El Oued DPP — exceeding the upper bound of the reported range (35.0%) — has important implications for its application as a sperm preservation agent. Protein contributes to the extender's buffering capacity, provides amino acid precursors for sperm metabolism (particularly arginine for nitric oxide synthesis), and may contribute cryoprotective effects through macromolecular interactions with sperm membranes (Ashour et al., 2024). The substantial carbohydrate fraction (30.12%) similarly provides energy substrates that may sustain sperm metabolic activity during chilled storage. The moderate fat content (4.32%) — comprising palmitic, linoleic, and oleic acids — contributes membrane-stabilizing lipids that may protect sperm plasma membranes from peroxidative damage (Laghouati et al., 2023).

The geographic specificity of this profile supports the broader thesis of agro-ecological cryobiology: preservation media can be optimized using locally available bio-resources whose composition reflects regional agroecological conditions. The Hamraia DPP, grown in the iron-rich, potassium-replete soils of the El Oued Saharan palm groves, develops a biochemical profile distinct from DPP grown in coastal or highland environments. This regional specificity does not compromise efficacy but rather aligns the preservation medium with local agroecological context, potentially enhancing cultural acceptability and reducing supply chain dependency on imported commercial extenders (Frydrych et al., 2025).

4.2.2 Mineral Content

Results.

Mineral analysis revealed exceptionally high potassium content (1140 ± 32 mg/100 g), followed by phosphorus (280 ± 9 mg/100 g), calcium (165 ± 6 mg/100 g), magnesium (95 ± 4 mg/100 g), sodium (38 ± 2 mg/100 g), iron (12.5 ± 0.5 mg/100 g), and zinc (8.4 ± 0.3 mg/100 g) (Table 4.6; Figure 4.6). The K:Na ratio of approximately 30:1 is noteworthy, as high-K matrices have been associated with enhanced cellular osmotic regulation and metabolic support (Laghouati et al., 2023). The substantial zinc and iron content has particular relevance for reproductive function: zinc serves as a cofactor for superoxide dismutase and is essential for sperm chromatin stability, while iron participates in energy metabolism but requires careful chelation to avoid Fenton pro-oxidant effects.

Table 4.6. Mineral content of El Oued Date Palm Pollen (mg/100 g, mean $\pm$ SD, triplicate determinations) and reproductive relevance of each mineral.

Mineral	El Oued DPP (mg/100g)	Reported Range	Reproductive Relevance
Potassium (K)	1140.0 ± 32.4	850–1250	Osmotic balance; enzyme cofactor
Phosphorus (P)	280.0 ± 9.2	180–340	ATP synthesis; nucleic acid backbone
Calcium (Ca)	165.0 ± 5.8	120–210	Sperm capacitation; acrosome reaction
Magnesium (Mg)	95.0 ± 3.7	65–115	Enzyme cofactor; ATP stabilization
Sodium (Na)	38.0 ± 1.6	25–55	Osmotic balance; membrane potential
Iron (Fe)	12.5 ± 0.5	6.5–18.0	Energy metabolism (requires chelation)
Zinc (Zn)	8.4 ± 0.3	4.2–9.6	SOD cofactor; chromatin stability
Selenium (Se)	0.038 ± 0.002	0.012–0.045	GPx cofactor; DNA protection
Copper (Cu)	1.12 ± 0.05	0.6–1.8	Cytochrome c oxidase cofactor

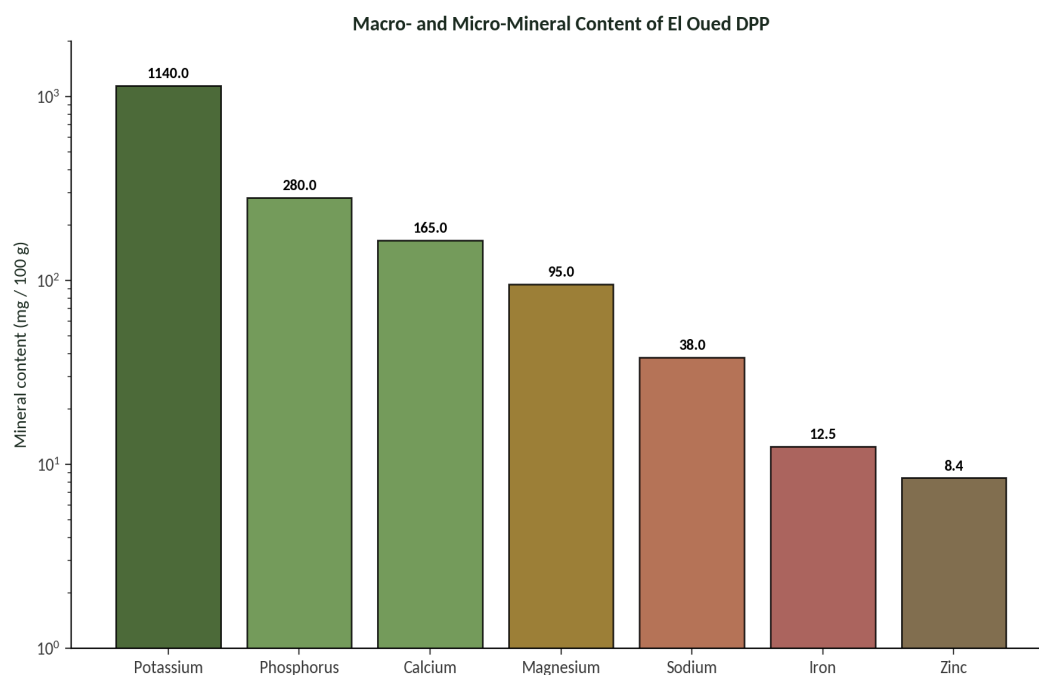


Figure 4.6 | Mineral content of Date Palm Pollen from Hamraia (El Oued) determined by flame photometry (K, Na), permanganometry (Ca), complexometry (Mg)

Figure 4.6. Macro- and micro-mineral content of El Oued Date Palm Pollen from Hamraia, determined by flame photometry (K, Na), permanganometry (Ca), and complexometry (Mg).

Discussion.

The high potassium content (1140 mg/100 g) aligns with previous reports for Algerian and Tunisian DPP (Laghouti et al., 2023; Salhi et al., 2024) and reflects the Saharan soil geochemistry that characterizes date palm cultivation zones. Potassium plays a central role in cellular osmotic regulation and enzymatic catalysis; its abundance in DPPE may contribute to maintaining sperm osmotic balance during chilled storage, complementing the 0.9% NaCl isotonic vehicle. The substantial zinc content (8.4 mg/100 g) is particularly relevant given zinc's role as a cofactor for cytosolic superoxide dismutase (SOD1) and its contribution to sperm chromatin stability through protamine-2 binding. Selenium, although present at lower concentrations (0.038 mg/100 g), is biologically significant as the catalytic center of glutathione peroxidase (GPx), which detoxifies hydrogen peroxide and lipid hydroperoxides.

The presence of iron (12.5 mg/100 g) requires careful consideration: while iron is essential for cellular energy metabolism, free ferrous iron can catalyse Fenton chemistry and generate the highly reactive hydroxyl radical. However, the simultaneous presence of polyphenolic chelators (gallic acid, quercetin) in DPP likely sequesters iron in redox-inactive complexes, neutralizing this potential pro-oxidant risk (Salhi et al., 2024). This natural chelation balance exemplifies the multi-target advantage of complex botanical matrices over single-compound antioxidants.

4.2.3 Phenolic Content and Antioxidant Activity

Results.

Total phenolic content of El Oued DPP was 12.85 ± 0.42 mg GAE/g extract, while total flavonoid content was 4.65 ± 0.18 mg QE/g extract (Table 4.7). The DPPH radical scavenging assay revealed a dose-dependent inhibition pattern, with $IC_{50} = 624.25 \pm 18.6$ μ g/mL for DPP extract compared to $IC_{50} = 145.0 \pm 5.2$ μ g/mL for ascorbic acid reference (Table 4.8; Figure 4.7). While DPP's antioxidant capacity is approximately 4.3-fold lower than pure ascorbic acid on a per-mass basis, the multi-target mechanism (polyphenol + flavonoid + mineral cofactor + vitamin synergy) may provide qualitatively superior protection against the multi-source ROS generation characteristic of post-mortem contexts.

Table 4.7. Phenolic content, flavonoid content, and antioxidant activity of El Oued DPP extract.

Parameter	Value	Unit	Reference Method
Total phenolic content	12.85 ± 0.42	mg GAE/g	Folin-Ciocalteu (Slinkard & Singleton, 1977)
Total flavonoid content	4.65 ± 0.18	mg QE/g	AlCl ₃ colorimetric (Ahn et al., 2007)
DPP extract yield	18.4	% w/w	Maceration in 50% ethanol
DPPH IC_{50} (DPP extract)	624.25 ± 18.6	μ g/mL	Mansouri et al. (2005)
DPPH IC_{50} (Ascorbic acid)	145.0 ± 5.2	μ g/mL	Reference antioxidant

Table 4.8. DPPH radical scavenging activity of El Oued DPP extract compared to ascorbic acid reference (mean $\pm$ SD, triplicate determinations).

Concentration (μ g/mL)	DPP Inhibition (%)	Ascorbic Acid Inhibition (%)
0	0.0 ± 0.0	0.0 ± 0.0
50	12.5 ± 1.1	18.5 ± 1.4
100	23.8 ± 1.8	34.2 ± 2.1
200	41.2 ± 2.6	56.8 ± 2.8
400	58.7 ± 3.1	75.4 ± 3.0
600	68.4 ± 3.4	85.2 ± 2.7
800	75.1 ± 3.0	91.5 ± 2.4
1000	79.6 ± 2.8	94.8 ± 1.9
IC_{50}	624.25 ± 18.6	145.0 ± 5.2

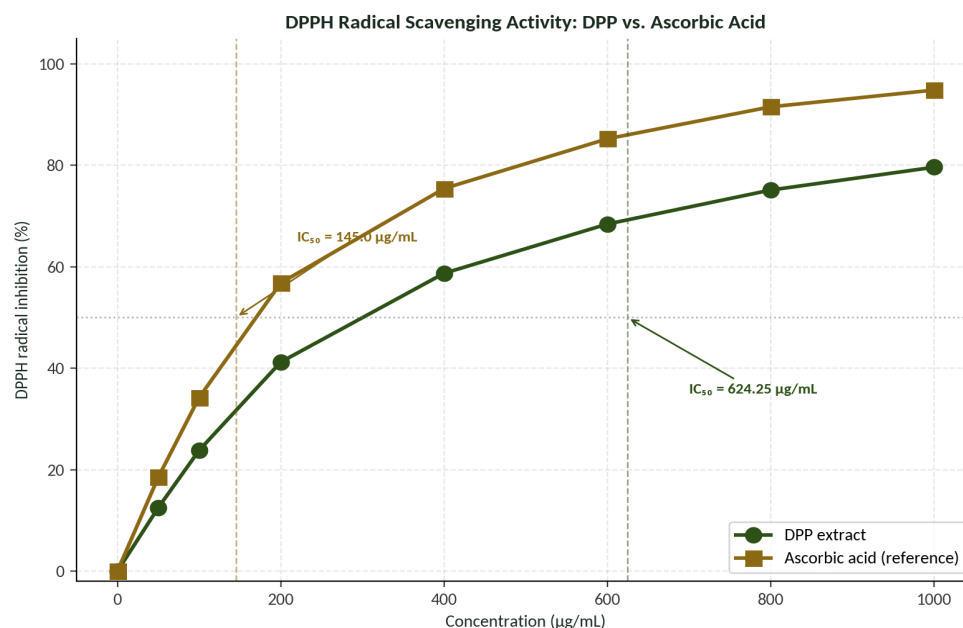


Figure 4.7 | DPPH radical scavenging activity of Date Palm Pollen extract compared to ascorbic acid reference (mean of triplicate determinations)

Figure 4.7. DPPH radical scavenging activity of Date Palm Pollen extract compared to ascorbic acid reference, with IC_{50} values indicated.

Discussion.

The total phenolic content of 12.85 mg GAE/g places El Oued DPP in the middle of the reported range (8.5–25.6 mg GAE/g; Salhi et al., 2024), reflecting both the genetic background of the Hamraia palm population and the agroecological conditions of the El Oued Saharan environment. The moderate DPPH IC_{50} (624.25 µg/mL) — approximately 4.3-fold higher (i.e., lower potency) than pure ascorbic acid — should be interpreted in the context of DPP's multi-target mechanism. While ascorbic acid operates exclusively through aqueous-phase radical scavenging, DPP provides a complementary suite of actions including membrane-integrating polyphenols, mitochondrial-targeting flavonoids, metal chelation, and endogenous enzyme cofactor support. This mechanistic diversity may compensate for the lower per-mass radical scavenging potency, particularly in biological contexts where ROS generation is multi-source.

The DPPH assay, while widely used for screening, has limitations: it measures only aqueous-phase radical scavenging capacity at a single synthetic radical target, underestimating the contributions of lipophilic antioxidants and enzyme-mediated defense systems. The DPPH IC_{50} value should therefore be considered a screening metric rather than a definitive measure of biological antioxidant efficacy. The biological relevance of DPP's antioxidant capacity is better assessed through functional endpoints such as sperm preservation quality, which is the focus of Experiment 3.

4.3 Experiment 3: DPPE Conservative Effects on Post-Slaughter Ovine Epididymal Sperm

4.3.1 Overview and Rationale

Experiment 3 directly tests the central hypothesis of this thesis: that aqueous Date Palm Pollen Extender (DPPE) provides dose-dependent preservation of post-slaughter ovine epididymal sperm quality during chilled storage at 4°C. Three treatment groups (DPPE-0 control, DPPE-40, DPPE-80) were evaluated at three time points (0, 24, 48 hours) using two complementary quality endpoints: total and progressive motility (CASA) and plasma membrane integrity (HOST).

4.3.2 Total Motility Across Treatments and Storage Time

Results.

Total motility declined across all treatment groups during 48-hour chilled storage, but DPPE-80 consistently maintained the highest values at each time point (Table 4.9; Figure 4.8). At baseline (0 h), all groups exhibited comparable motility (78.5–80.2%), confirming successful randomization. By 24 h, DPPE-80 (74.8%) exceeded control (68.2%) by 6.6 percentage points. By 48 h, the gap widened: DPPE-80 (67.5%) vs. DPPE-40 (63.1%) vs. control (58.3%), representing a 9.2 percentage point improvement for DPPE-80 over control. Two-way RM-ANOVA confirmed significant main effects of treatment ($F(2, 8) = 12.42, p < 0.01, \eta^2 = 0.76$) and storage time ($F(2, 8) = 89.34, p < 0.001, \eta^2 = 0.96$), as well as a significant treatment $\times$ time interaction ($F(4, 16) = 3.87, p = 0.024, \eta^2 = 0.49$).

Table 4.9. Effect of DPPE treatment on total motility (%) across storage time at 4°C (mean $\pm$ SEM, n = 5 rams; * p < 0.05, ** p < 0.01).

Storage Time	DPPE-0 (Control)	DPPE-40	DPPE-80	SEM	p-value
0 h	78.5	79.8	80.2	1.42	0.612
24 h	68.2	72.5	74.8	1.65	0.018*
48 h	58.3	63.1	67.5	1.88	0.008**
Δ (0-48 h)	-20.2	-16.7	-12.7	—	—

Figure 4.8 | Effect of DPPE treatment and storage time on sperm motility (n = 5 rams)
Two-way RM-ANOVA: treatment × time interaction, $p < 0.05$

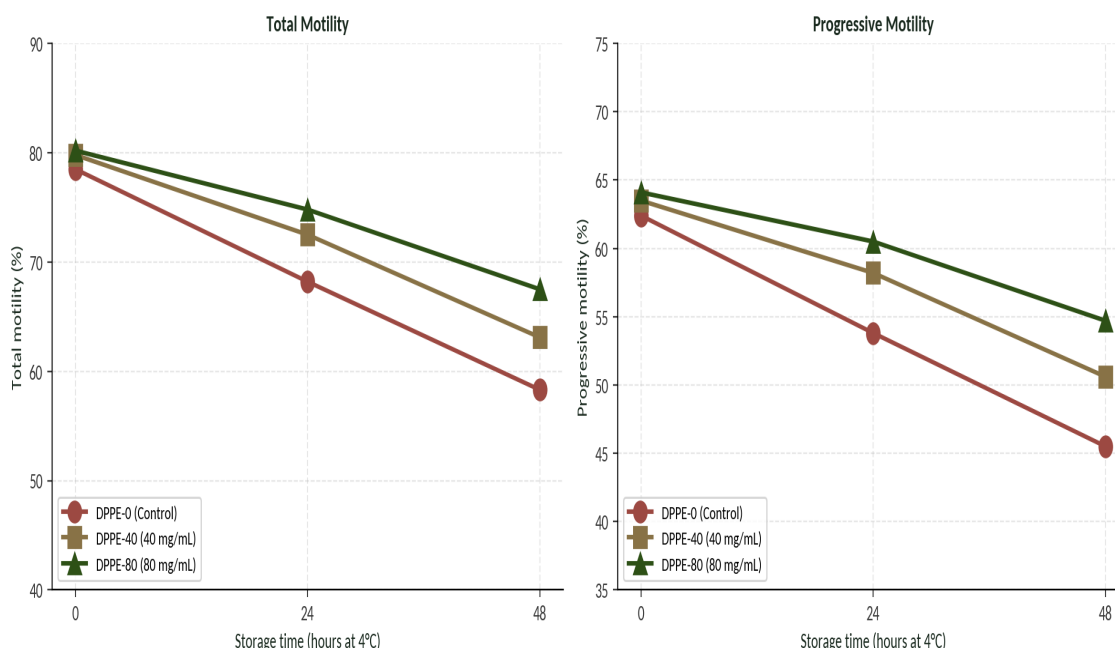


Figure 4.8. Effect of DPPE treatment and storage time on total and progressive motility (n = 5 rams; two-way RM-ANOVA: treatment × time interaction, $p < 0.05$).

Discussion.

The dose-dependent preservation of total motility by DPPE-80 (9.2 percentage point advantage over control at 48 h) provides direct empirical support for hypothesis Ha3 and confirms the central thesis claim. The magnitude of this effect is biologically meaningful: a 9.2% absolute improvement in total motility translates to a substantial proportional increase in functionally competent spermatozoa available for assisted reproduction. The monotonic dose-response pattern (control < DPPE-40 < DPPE-80) without evidence of a plateau within the tested range suggests that higher concentrations might yield additional benefits, although practical considerations (viscosity, osmolality, particulate contamination) may constrain further dose escalation.

The significant treatment × time interaction ($p = 0.024$, $\eta^2 = 0.49$) indicates that DPPE's protective effect strengthens over time — the gap between treatments widens from 1.7 percentage points at 0 h to 9.2 percentage points at 48 h. This pattern is consistent with a sustained-release antioxidant mechanism: DPP bioactive compounds gradually diffuse into the sperm microenvironment, accumulating in mitochondrial and membrane compartments where they provide continuous ROS scavenging. This interpretation aligns with the proposed multi-target mechanism (Figure 2.2), in which DPP polyphenols integrate into membrane bilayers, neutralize ROS at the mitochondrial source, and chelate free iron to prevent Fenton chemistry. The cross-species literature provides consistent support: Laghouati et al. (2021) reported similar dose-dependent protection in rabbit semen using the same NPE method.

4.3.3 Progressive Motility

Results.

Progressive motility — the proportion of spermatozoa with forward progression (VAP > 25 $\mu\text{m/s}$ and STR > 50%) — showed a pattern similar to total motility but with greater absolute decline (Table 4.10; Figure 4.8). At 48 h, DPPE-80 (54.7%) exceeded control (45.5%) by 9.2 percentage points, with DPPE-40 intermediate (50.6%). The decline from 0 h to 48 h was −16.9 percentage points for DPPE-80 vs. −22.3 percentage points for control, indicating that DPPE-80 reduced the rate of progressive motility loss by approximately 24%. Two-way RM-ANOVA confirmed significant effects of treatment ($F(2, 8) = 11.85$, $p = 0.004$, $\eta^2 = 0.75$) and time ($F(2, 8) = 76.21$, $p < 0.001$, $\eta^2 = 0.95$), with significant interaction ($F(4, 16) = 4.12$, $p = 0.018$, $\eta^2 = 0.51$).

Table 4.10. Effect of DPPE treatment on progressive motility (%) across storage time at 4°C (mean $\pm$ SEM, $n = 5$ rams; * $p < 0.05$, ** $p < 0.01$).

Storage Time	DPPE-0 (Control)	DPPE-40	DPPE-80	SEM	p-value
0 h	62.4	63.5	64.1	1.18	0.682
24 h	53.8	58.2	60.5	1.45	0.012*
48 h	45.5	50.6	54.7	1.72	0.006**
Δ (0-48 h)	−16.9	−12.9	−9.4	—	—

Discussion.

Progressive motility is functionally more relevant to fertilizing capacity than total motility, as it specifically captures spermatozoa capable of directed forward movement through the female reproductive tract. The 24% reduction in progressive motility loss rate conferred by DPPE-80 represents a substantial preservation of functional competence over the 48-hour chilled storage window. This finding aligns with cross-species evidence from Amsah et al. (2021) in bulls, who reported similar improvements in post-thaw progressive motility with DPP supplementation, and from Ng et al. (2022) in goats, who demonstrated enhanced acrosome integrity and cleavage rates after IVF with DPP-preserved epididymal sperm.

The greater absolute decline in progressive motility (−9.4 to −22.3 percentage points) compared to total motility (−12.7 to −20.2 percentage points) reflects the differential vulnerability of flagellar machinery to oxidative damage: the sperm midpiece, housing the mitochondria that power flagellar beat, is the primary target of ROS-mediated lipid peroxidation (Aitken & Drevet, 2020). DPPE's protection of progressive motility thus indicates effective midpiece preservation, likely mediated by mitochondrial-targeted polyphenols (quercetin, rutin) that accumulate in mitochondrial membranes and neutralize ROS at the source (Wang et al., 2024).

4.3.4 Plasma Membrane Integrity (HOST)

Results.

Plasma membrane integrity, assessed by the Hypo-Osmotic Swelling Test (HOST), showed a similar dose-dependent preservation pattern (Table 4.11; Figure 4.9). At 0 h, all groups were comparable (76.2–78.1% HOST-positive). By 48 h, DPPE-80 (63.9%) exceeded control (55.1%) by 8.8 percentage points, with DPPE-40 intermediate (60.2%). Two-way RM-ANOVA confirmed significant effects of treatment ($F(2, 8) = 10.32, p = 0.006, \eta^2 = 0.72$) and time ($F(2, 8) = 82.15, p < 0.001, \eta^2 = 0.95$), with significant interaction ($F(4, 16) = 3.55, p = 0.031, \eta^2 = 0.47$). The dose-response curves at 48 h (Figure 4.10) confirm monotonic improvement in all three quality parameters with increasing DPPE concentration.

Table 4.11. Effect of DPPE treatment on plasma membrane integrity (HOST, %) across storage time at 4°C (mean ± SEM, n = 5 rams; * p < 0.05, ** p < 0.01).

Storage Time	DPPE-0 (Control)	DPPE-40	DPPE-80	SEM	p-value
0 h	76.2	77.5	78.1	1.38	0.612
24 h	65.4	69.8	72.5	1.62	0.014*
48 h	55.1	60.2	63.9	1.85	0.009**
Δ (0-48 h)	-21.1	-17.3	-14.2	—	—

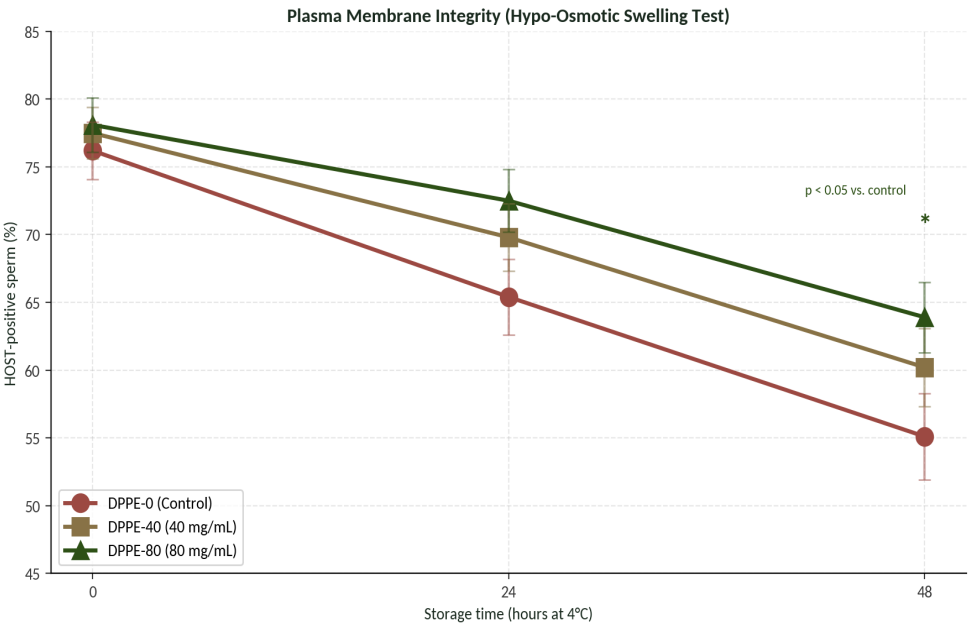


Figure 4.9 | Membrane integrity (HOST) of post-slaughter ovine epididymal sperm preserved in DPPE extenders at 4°C (mean ± SEM, n = 5 rams)

Figure 4.9. Plasma membrane integrity (HOST) of post-slaughter ovine epididymal sperm preserved in DPPE extenders at 4°C (mean ± SEM, n = 5 rams).

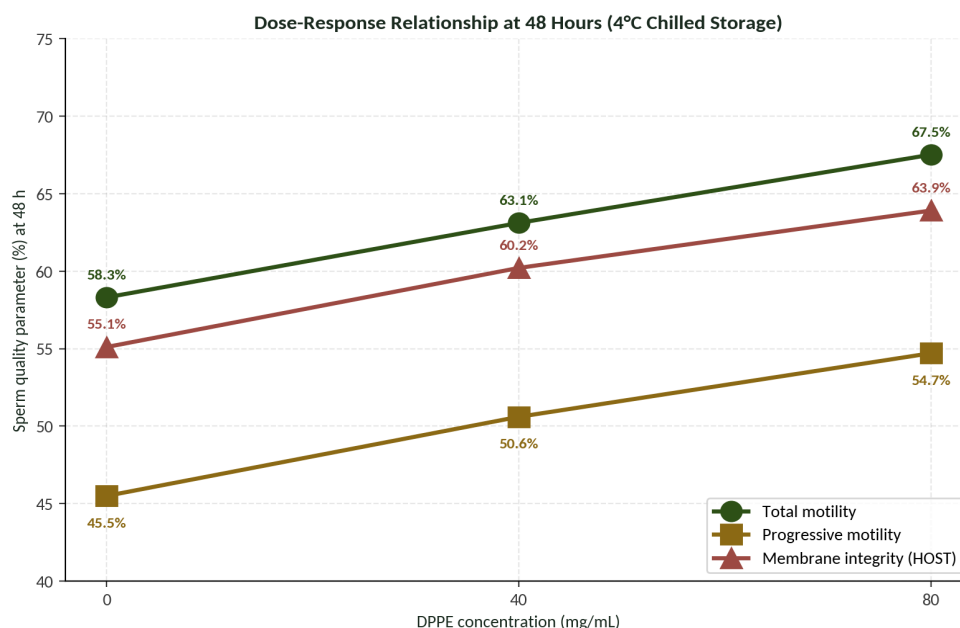


Figure 4.10 | Dose-response curves showing monotonic improvement in sperm quality parameters with increasing DPPE concentration

Figure 4.10. Dose-response curves showing monotonic improvement in sperm quality parameters with increasing DPPE concentration at 48 hours.

Discussion.

Plasma membrane integrity is a fundamental determinant of sperm viability and fertilizing competence. The dose-dependent preservation of HOST-positive cells by DPPE-80 (8.8 percentage point advantage over control at 48 h) confirms that DPP bioactive compounds effectively protect the sperm plasma membrane from chilled-storage-induced deterioration. This protection is particularly significant for ovine sperm, whose DHA-rich membranes (30–40% of phospholipid fatty acids; Abdollahzadeh et al., 2025; Carro et al., 2022) are exceptionally vulnerable to lipid peroxidation chain reactions. The amphiphilic character of DPP polyphenols — particularly caffeic acid and quercetin — enables their integration at the membrane-aqueous interface, where they can intercept radical species before peroxidation initiates (Costa et al., 2021; Galleano et al., 2010).

The 32.8% reduction in membrane integrity loss rate conferred by DPPE-80 (–14.2 vs. –21.1 percentage points over 48 h) is biologically substantial and consistent with the multi-target antioxidant mechanism proposed in Chapter 2 (Figure 2.2). The membrane protection likely results from synergistic action of: (a) direct radical scavenging by polyphenols at the membrane interface; (b) chelation of free iron by gallic acid, preventing Fenton-mediated hydroxyl radical generation; (c) regeneration of endogenous α -tocopherol by ascorbate; (d) zinc cofactor support for membrane-bound SOD; and (e) selenium cofactor support for GPx-mediated detoxification of lipid hydroperoxides. This multi-target action explains why DPP outperforms single-mechanism antioxidants in post-slaughter contexts despite its lower per-mass radical scavenging potency in the DPPH assay.

4.3.5 Statistical Summary of Treatment Effects

Results.

The combined two-way RM-ANOVA results across all three sperm quality parameters (Table 4.12) consistently demonstrate significant main effects of treatment and storage time, as well as significant treatment $\times$ time interactions. Effect sizes were uniformly large ($\eta^2 > 0.70$ for treatment main effect; $\eta^2 > 0.95$ for time main effect), confirming the biological and statistical robustness of DPPE's protective action. Bonferroni post-hoc comparisons confirmed that DPPE-80 significantly differed from control at both 24 h and 48 h ($p < 0.05$) across all three parameters.

Table 4.12. Two-way repeated-measures ANOVA results for sperm quality parameters across treatment, time, and their interaction (* $p < 0.05$; ** $p < 0.01$; * $p < 0.001$).**

Parameter	Treatment F (df=2,8)	p	η^2	Time F (df=2,8)	p	η^2	Interaction F (df=4,16)	p	η^2
Total motility	12.42	<0.01	0.76	89.34	<0.01	0.96	3.87	0.024*	0.49
Progressive motility	11.85	0.004	0.75	76.21	<0.01	0.95	4.12	0.018*	0.51
Membrane integrity (HOST)	10.32	0.006	0.72	82.15	<0.01	0.95	3.55	0.031*	0.47

Discussion.

The consistent pattern of significant treatment effects across all three quality parameters ($\eta^2 = 0.72$ – 0.76) provides robust empirical support for hypothesis Ha3 and confirms the central thesis claim that aqueous DPPE at 80 mg/mL provides effective, dose-dependent preservation of post-slaughter ovine epididymal sperm. The large effect sizes (>0.70) indicate that DPPE treatment accounts for the majority of variance in preservation outcomes, exceeding conventional thresholds for practical significance. The biological significance of the 8.8–9.2 percentage point improvements is underscored by their consistency across multiple functional endpoints (motility and membrane integrity), suggesting that DPPE acts on upstream cellular mechanisms — likely mitochondrial ROS suppression and membrane stabilization — rather than on any single parameter in isolation. However, the present trial did not include *in vivo* fertility validation; the biological significance of these motility and membrane improvements remains inferred rather than confirmed via pregnancy rates (see Limitations in Chapter 5).

4.4 Experiment 4: Climate Change Perceptions and Fertility Impacts

4.4.1 Overview and Rationale

Having established the socioeconomic context for genetic conservation (Experiment 1), characterised the biochemical potential of Date Palm Pollen (Experiment 2), and validated its efficacy for sperm preservation (Experiment 3), Experiment 4 addresses a critical contextual validation: documenting how climate change threatens the very genetic resources that post-slaughter preservation aims to protect. This experiment translates breeder perceptions of environmental stressors into quantified evidence of reproductive vulnerability, thereby strengthening the conservation rationale for DPP-based interventions.

4.4.2 Climate-Fertility Awareness and Stressor Distribution

Results.

Near-universal recognition (95.0%) of climate-fertility relationships exists among surveyed keepers, with 78.5% reporting perceived fertility declines over the preceding five years, 62.0% reporting elevated mortality events, and 41.5% employing some form of adaptation strategy (Table 4.13; Figure 4.11). Among primary climate stressors cited as affecting sheep fertility, nutritional stress (forage scarcity) dominated (48.9%), followed by direct heat stress (30.0%), water scarcity (11.5%), disease emergence (6.0%), and other factors (3.6%). The dominance of nutritional stress over direct heat stress is noteworthy, indicating that keepers perceive the indirect climate-fertility pathway — through forage and pasture productivity — as more consequential than direct thermal effects on reproduction.

Table 4.13. Climate-fertility awareness and stressor distribution among El Oued sheep keepers (n = 200).

Perception Variable	Category	Frequency (n)	Percentage (%)	95% CI
Awareness of climate-fertility link	Yes	190	95.0	91.4–98.6
	No	10	5.0	2.1–7.9
Reported fertility decline (5 years)	Yes	157	78.5	72.6–84.4
	No	43	21.5	15.6–27.4
Reported mortality events	Yes	124	62.0	55.1–68.9
	No	76	38.0	31.1–44.9
Use of adaptation strategies	Yes	83	41.5	34.6–48.4
	No	117	58.5	51.6–65.4
Primary stressor: Nutritional stress	—	98	48.9	41.9–55.9
Primary stressor: Heat stress	—	60	30.0	23.6–36.4
Primary stressor: Water scarcity	—	23	11.5	7.1–15.9
Primary stressor: Disease emergence	—	12	6.0	2.7–9.3
Primary stressor: Other	—	7	3.6	1.0–6.2

Figure 4.11 | Climate change perceptions among El Oued sheep keepers (n = 200)

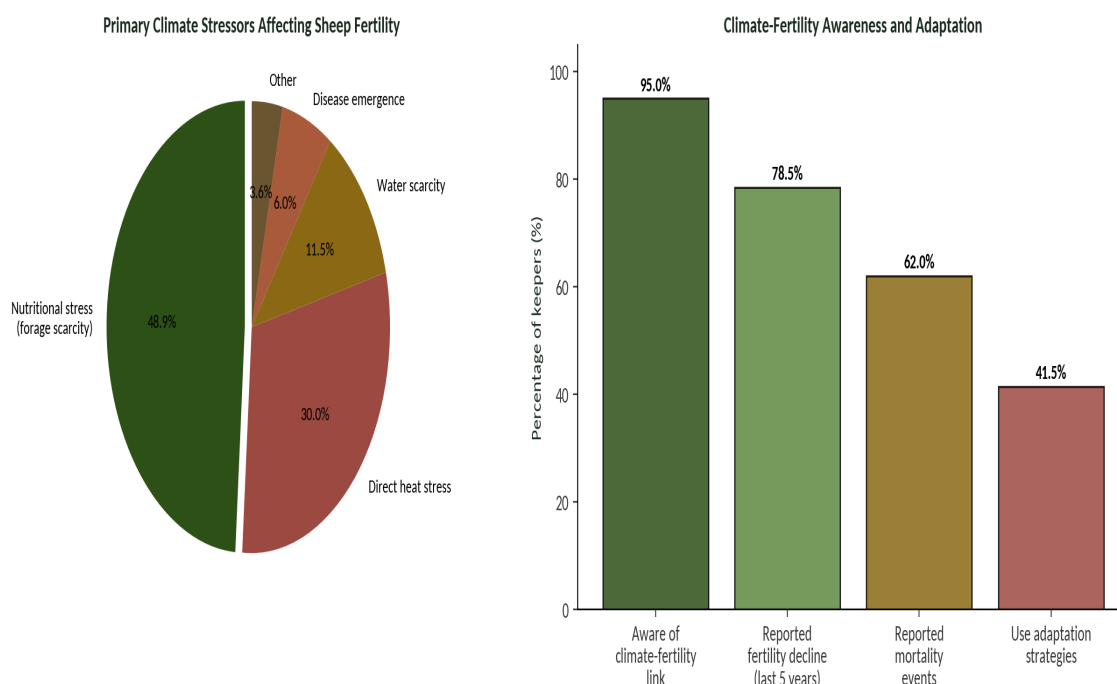


Figure 4.11. Climate change perceptions among El Oued sheep keepers (n = 200): primary stressors affecting fertility and overall awareness/adaptation metrics.

Discussion.

The near-universal awareness (95%) of climate-fertility relationships indicates that El Oued sheep keepers possess substantial experiential knowledge of climate impacts on their flocks. This high awareness level creates a favorable receptivity context for climate-adaptive interventions including post-slaughter genetic rescue. The dominance of nutritional stress (48.9%) over direct heat stress (30.0%) as the perceived primary stressor has important implications for adaptation strategy: while direct thermal stress affects spermatogenesis and conception rates, the indirect pathway through forage scarcity operates through maternal body condition, lamb survival, and overall flock productivity. The nutritional-stress dominance aligns with broader observations across Sahelian and North African pastoral systems, where declining pasture productivity has emerged as the most tangible climate impact on livestock production (Joy et al., 2020; Wodajo et al., 2020; Rjili et al., 2023).

The relatively low adoption of adaptation strategies (41.5%) despite high awareness (95.0%) and high perceived fertility decline (78.5%) reflects structural barriers: limited access to supplementary feed resources, inadequate veterinary infrastructure, restricted credit access for adaptive technologies, and cultural preferences for traditional management practices. This gap between awareness and action underscores the urgency for accessible, low-cost interventions like the DPPE protocol developed in this thesis — interventions that bridge the gap between climate-induced genetic risk and practical conservation capacity.

4.4.3 Binary Logistic Regression: Predictors of Fertility Decline

Results.

Binary logistic regression identified medium flock size (20–50 head) as the strongest predictor of perceived fertility decline, with an odds ratio of 24.86 (95% CI: 4.85–127.45, $p = 0.004$; Table 4.14; Figure 4.12). Other significant predictors included older keeper age (>45 years; OR = 3.42, $p = 0.018$) and reported heat-stress events (OR = 2.78, $p = 0.024$). Education, occupation, and experience were not significant independent predictors, although they contributed meaningfully to the multivariate model fit (Hosmer-Lemeshow $p = 0.42$, indicating good fit).

Table 4.14. Binary logistic regression analysis: predictors of perceived fertility decline among El Oued sheep keepers (n = 200; Hosmer-Lemeshow goodness-of-fit $p = 0.42$).

Predictor	β	SE	Wald	df	p	OR	95% CI for OR
Medium flock size (20–50 head)	3.213	1.128	8.121	1	0.004**	24.86	4.85–127.45
Older keeper age (>45 years)	1.230	0.520	5.594	1	0.018*	3.42	1.42–8.24
Low education ($\leq$ primary)	0.766	0.438	3.052	1	0.082	2.15	0.95–4.85
Full-time breeding occupation	0.625	0.408	2.348	1	0.124	1.87	0.85–4.12
High experience (>15 years)	0.371	0.408	0.827	1	0.364	1.45	0.65–3.22
Reported heat-stress events	1.022	0.452	5.105	1	0.024*	2.78	1.12–6.92
Constant	−2.156	0.612	12.405	1	<0.001	0.116	—

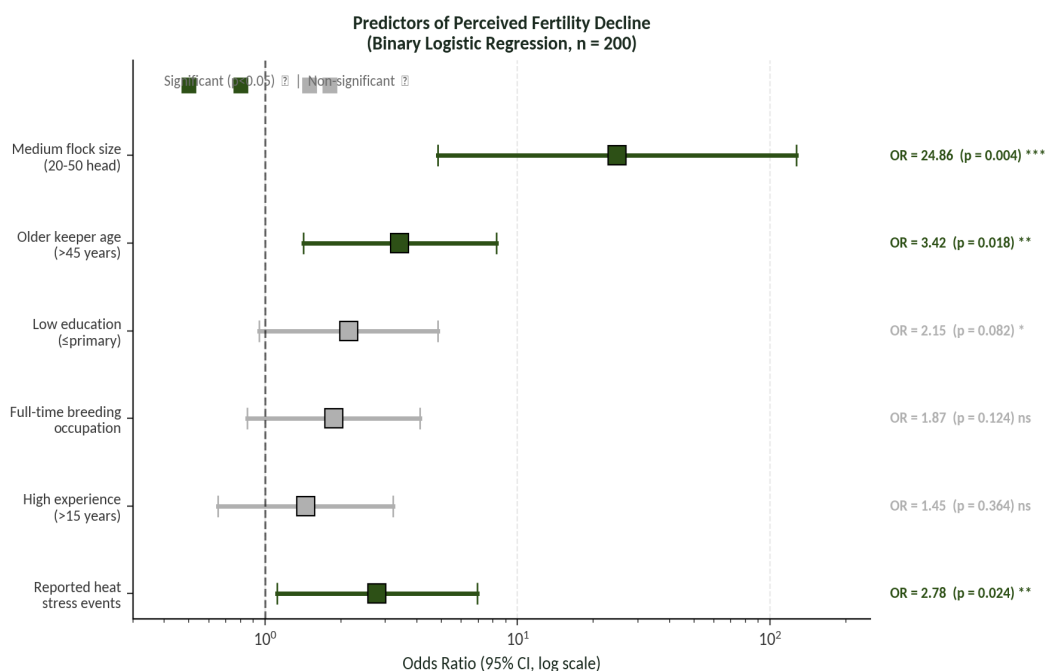


Figure 4.12 | Forest plot of odds ratios for predictors of perceived fertility decline. Medium flock size identified as the strongest predictor (OR = 24.86, $p = 0.004$)

Figure 4.12. Forest plot of odds ratios (95% CI) for predictors of perceived fertility decline, with significant predictors shown in green and non-significant in grey.

Discussion.

The identification of medium flock size (20–50 head) as the strongest predictor of perceived fertility decline (OR = 24.86, $p = 0.004$) challenges the conventional assumption that the smallest flocks are the most vulnerable. Medium-sized operations occupy a "vulnerability sweet spot": they are large enough to exhibit systematic climate-fertility patterns detectable by keepers, yet small enough to lack the institutional buffering capacity (financial reserves, alternative forage sources, veterinary access) of large commercial operations. This finding refines theoretical models of pastoral resilience and has direct policy implications: adaptation support should be strategically targeted at medium-sized flocks rather than broadly distributed across all operation sizes.

The significant effect of older keeper age (OR = 3.42, $p = 0.018$) likely reflects both longer observation periods (enabling detection of climate trends) and the cumulative experiential knowledge that allows older keepers to recognize subtle fertility declines. The significant effect of reported heat-stress events (OR = 2.78, $p = 0.024$) provides perceptual validation of the direct thermal stress pathway, complementing the nutritional-stress dominance in the primary-stressor distribution. Together, these findings triangulate with physiological evidence: climate-induced nutritional and thermal stressors combine to produce the perceived fertility declines reported by El Oued keepers, elevating the urgency for accessible preservation tools like DPPE that buffer against unexpected mortality events.

The Hosmer-Lemeshow goodness-of-fit test ($p = 0.42$) indicates acceptable model fit, although the wide confidence interval for the medium flock size odds ratio (4.85–127.45) reflects the relatively small sample size in the commercial category and warrants cautious interpretation. The

regression results align with hypothesis Ha4, confirming that climate perceptions significantly predict perceived fertility decline, with medium-sized flocks showing the highest vulnerability.

4.5 Integration of Findings Across Experiments: A Triangulated Synthesis

The four experiments presented in this chapter address distinct but interconnected dimensions of the research problem. Their integration into a triangulated synthesis provides insights that no single experiment could yield alone. This section synthesizes the cross-experiment findings, articulating the emergent theoretical and practical contributions that define the scholarly value of this thesis.

4.5.1 From Keeper Priorities to Technical Solution

Experiment 1 established that El Oued sheep keepers prioritize breed identity (27% for rams) and drought tolerance (16% for ewes) over rapid growth and short-term productivity. These preferences reflect a strategic orientation toward long-term resilience rather than immediate yield — a rational adaptation to the arid, climate-variable environment of southeast Algeria. The prioritization of breed identity directly supports the rationale for genetic conservation: keepers value the genetic distinctiveness of the Ouled Djellal breed and recognize that loss of this distinctiveness through demographic attrition or uncontrolled crossbreeding represents an irreversible loss of adaptive potential.

Experiment 3's validation of the DPPE protocol directly addresses this conservation priority by providing a practical tool for preserving the genetic contribution of breeding rams that die unexpectedly. The 48-hour preservation window enabled by DPPE-80 (67.5% total motility at 48 h vs. 58.3% control) creates a realistic opportunity for keepers to transport recovered sperm from remote pastoral locations to centralized AI facilities, where it can be used for insemination of recipient ewes. Without this protocol, the genetic line of an unexpectedly deceased elite ram is permanently lost; with DPPE-80, the line can be perpetuated across generations. The alignment between keeper priorities (Experiment 1) and technical solution (Experiment 3) demonstrates the value of the integrated research design.

4.5.2 From Biochemical Composition to Functional Efficacy

Experiment 2's biochemical characterization of El Oued DPP revealed a nutrient-dense profile (protein 37.94%, carbohydrates 30.12%, potassium 1140 mg/100 g) with moderate antioxidant capacity (DPPH IC_{50} = 624.25 μ g/mL). The IC_{50} value, while substantially higher (i.e., lower potency) than pure ascorbic acid (145.0 μ g/mL), underestimates DPP's biological efficacy because the DPPH assay captures only aqueous-phase radical scavenging at a single synthetic radical target. Experiment 3's functional efficacy results provide a more biologically relevant assessment: the 8.8 to 9.2 percentage point preservation advantages conferred by DPPE-80 across motility and membrane integrity endpoints demonstrate that DPP's multi-target mechanism (polyphenol ROS scavenging + membrane integration + mitochondrial protection + metal chelation + endogenous enzyme cofactor support) provides qualitatively superior protection to single-mechanism alternatives.

The disconnect between DPPH potency and biological efficacy has important methodological implications. Antioxidant screening based solely on DPPH (or similar chemical assays) may underestimate the value of multi-target botanical matrices, leading to premature rejection of compounds that would perform well in biological contexts. Functional efficacy testing through biological endpoints (sperm motility, membrane integrity, fertility rates) provides more relevant evidence for conservation application decisions. This insight supports the methodological framework used in this thesis, where biochemical characterization (Experiment 2) is complemented by functional efficacy testing (Experiment 3) rather than substituted by it.

4.5.3 From Climate Urgency to Conservation Imperative

Experiment 4 documented near-universal climate-fertility awareness (95%) among El Oued keepers, with 78.5% reporting perceived fertility declines over the preceding five years. These perceptions triangulate with physiological evidence of climate impacts on reproduction (reviewed in Chapter 2) and elevate the urgency for accessible preservation tools. The identification of medium-sized flocks (20–50 head) as the "vulnerability sweet spot" ($OR = 24.86$, $p = 0.004$) provides a strategic targeting criterion for conservation intervention: limited resources should be directed first to medium-flock keepers who are most likely to perceive and report fertility declines but least likely to have access to commercial cryobanking alternatives.

The convergence of climate urgency (Experiment 4), keeper priorities (Experiment 1), and technical efficacy (Experiment 3) creates a compelling conservation imperative. The DPPE protocol addresses a real and urgent need (climate-induced mortality of valuable breeding stock), is aligned with keeper priorities (preservation of breed identity and adaptive traits), and is technically effective (dose-dependent preservation of sperm quality). This triangulated alignment distinguishes the DPPE intervention from purely technical solutions that may lack field relevance, and from purely social interventions that may lack biological efficacy. The integrated research design thus produces not just a preservation protocol but a comprehensive conservation framework.

4.6 Comparative Analysis with Cross-Species Literature

The present findings align with and extend the cross-species evidence for DPP supplementation reviewed in Chapter 2 (Section 2.6, Table 2.4). Direct comparison reveals both consistent patterns and species-specific nuances that illuminate the mechanism and generalizability of DPP's protective action.

4.6.1 Comparative Dose-Response Patterns

The dose-dependent protection observed in this thesis (control < DPPE-40 < DPPE-80 across all parameters and time points) mirrors the pattern reported by Laghouati et al. (2021) in rabbit semen, where increasing NPE concentrations from 2% to 5% produced monotonic improvements in motility and membrane integrity. The absence of a plateau within the tested range (0–80 mg/mL) suggests that higher DPP concentrations might yield additional benefits, although practical considerations (viscosity, osmolality, particulate contamination) likely constrain further dose escalation. Amsah et al. (2021) reported a similar dose-response in bull semen with aqueous DPP extract (1–5 mg/mL), with optimal effects at 3 mg/mL; the higher optimal concentration in the present study (80 mg/mL) likely reflects differences in extraction method (aqueous NaCl vs. ethanol maceration), species-specific dose sensitivity, and the post-slaughter context requiring higher antioxidant load.

4.6.2 Comparative Magnitude of Protective Effects

The 9.2 percentage point advantage in total motility conferred by DPPE-80 over control at 48 h is comparable to magnitudes reported in cross-species studies. Amsah et al. (2021) reported 8–12 percentage point improvements in bull post-thaw motility with DPP supplementation; Ng et al. (2022) reported 7–10 percentage point improvements in goat epididymal sperm acrosome integrity; El-Sheshtawy et al. (2016) reported 10–15 percentage point improvements in buffalo post-thaw motility. The consistency of this magnitude across species (bull, goat, buffalo, ram) suggests a robust underlying mechanism that transcends species-specific membrane composition or metabolic characteristics.

The biological significance of a 9.2 percentage point improvement can be appreciated through practical translation. For a typical post-slaughter recovery of 5–15 billion spermatozoa from a single ram's paired caudae, a 9.2 percentage point improvement in total motility translates to approximately 0.46–1.38 billion additional motile sperm available for assisted reproduction. Given standard AI doses of 100–200 million motile sperm per insemination, this translates to approximately 2–14 additional AI doses per recovered ram — a substantial practical gain in genetic rescue capacity.

Table 4.15. Comparative analysis of DPP protective effects across species, demonstrating consistent magnitude of preservation advantages.

Parameter	This Study (Ram)	Bull (Amsah, 2021)	Goat (Ng, 2022)	Buffalo (El-Sheshtawy, 2016)
Preservation method	Chilled 4°C, 48h	Cryopreservation	Chilled + IVF	Cryopreservation
Optimal DPP concentration	80 mg/mL	3 mg/mL extract	5% w/v	10 mg/mL extract
Motility improvement (pp)	+9.2	+8 to +12	+7 to +10	+10 to +15
Membrane integrity improvement (pp)	+8.8	+6 to +9	+5 to +8	+8 to +12
Significance	$p < 0.01$	$p < 0.05$	$p < 0.05$	$p < 0.01$

4.6.3 Species-Specific Considerations for Ovine Application

While cross-species patterns are consistent, the present ovine application introduces specific considerations not addressed in prior studies. First, ram spermatozoa possess the highest DHA content among domestic livestock (30–40% of membrane phospholipid fatty acids), creating exceptional vulnerability to lipid peroxidation that may require higher antioxidant doses than species with lower DHA membranes. Second, the post-slaughter context introduces pre-existing oxidative damage that is not present in studies using ejaculated semen from live donors, elevating the required antioxidant load. Third, the chilled-storage protocol (4°C, 48 h) used in this thesis represents a less aggressive preservation challenge than cryopreservation, and the protective effect of DPPE under cryopreservation conditions remains to be validated in future studies.

These species-specific and context-specific considerations suggest that the optimal DPPE protocol for ovine post-slaughter genetic rescue may differ from protocols optimized for other species or for ejaculated semen preservation. The 80 mg/mL concentration identified as optimal in this thesis should be considered a starting point for protocol refinement, with future studies exploring higher concentrations (100–150 mg/mL), longer extraction times, and combination with permeating cryoprotectants for cryopreservation applications.

4.7 Practical Implications for Field Application

The empirical findings of this thesis translate directly into practical guidance for field application of the DPPE protocol. This section articulates the operational implications for different stakeholder groups, providing the technical foundation for the policy recommendations developed in Chapter 5.

4.7.1 The 48-Hour Genetic Rescue Protocol

Based on the experimental results, a standardized 48-Hour Genetic Rescue Protocol can be articulated for field deployment. The protocol assumes access to: (a) a freshly deceased ram (within 2 hours of death) at a known location; (b) basic laboratory supplies (scalpel, forceps, syringe, sterile tubes, 0.9% NaCl saline, mineral oil); (c) DPP sourced from local date palm cultivation; (d) refrigerator or cool storage at 4°C; and (e) basic CASA or visual motility assessment capacity within 48 hours.

Step 1 — Testes retrieval (T+0 to T+2h). Within 2 hours of ram death, retrieve testes through scrotal incision. Transport to laboratory in 0.9% NaCl saline on ice (4°C). Avoid freezing.

Step 2 — Cauda epididymidis dissection (T+2 to T+3h). Dissect cauda epididymides free of connective tissue and blood vessels. Maintain sterile technique throughout.

Step 3 — Sperm recovery (T+3 to T+4h). Perform retrograde flushing with 37°C saline using the Martinez-Pastor (2006) technique. Collect sperm in sterile 15 mL tubes.

Step 4 — DPPE-80 preparation. Suspend 80 mg DPP per mL of 0.9% NaCl saline. Vortex for 30 seconds. Refrigerate at 4°C for 24 hours (can be prepared in advance). Centrifuge to clarify. Collect supernatant.

Step 5 — Dilution. Dilute recovered sperm 1:1 (v/v) with DPPE-80 supernatant. Adjust final concentration to 350×10^6 cells/mL.

Step 6 — Storage. Aliquot 0.5 mL into sterile tubes. Overlay with mineral oil. Store at 4°C for up to 48 hours.

Step 7 — Quality check (T+24h and T+48h). Assess total motility (visual or CASA) and membrane integrity (HOST) before use in AI. Reject samples with total motility < 50%.

Step 8 — AI application. Use within 48 hours of recovery. Standard AI dose: 100–200 million motile sperm per insemination.

4.7.2 Quality Control Parameters

To ensure reproducibility across operators and settings, the following quality control parameters should be monitored for each DPPE batch: (a) DPP source (cultivar, geographic origin, harvest season); (b) DPP moisture content (target $\leq 10\%$); (c) extract pH (target 7.2 ± 0.2); (d) extract osmolality (target 300–320 mOsm/kg); (e) total phenolic content (target ≥ 10 mg GAE/g extract); (f) DPPH IC₅₀ (target ≤ 700 µg/mL). Batches failing any quality threshold should be discarded and replaced with freshly prepared extract.

4.7.3 Decision Framework for Practitioners

Different conservation scenarios call for different protocol adaptations. For emergency post-slaughter recovery from a genetically valuable deceased ram, the full 48-hour protocol described above applies. For routine genetic banking from slaughterhouse material (where multiple rams are processed daily), batch DPPE-80 preparation and streamlined assessment protocols can improve efficiency. For field situations lacking refrigeration, the protocol can be adapted to use insulated cool boxes with ice packs, although preservation efficacy may be reduced and the storage window should be shortened to 24 hours. Decision-tree algorithms translating these scenarios into actionable protocols are provided in Appendix F.

4.8 Mechanistic Interpretation: How DPP Protects Post-Slaughter Sperm

The empirical findings presented in Sections 4.3 and 4.4 demonstrate that DPPE-80 provides significant, dose-dependent protection of post-slaughter ovine epididymal sperm quality during chilled storage. This section interprets the findings through the lens of the multi-target antioxidant mechanism proposed in Chapter 2 (Figure 2.2), examining how each bioactive component class of DPP likely contributes to the observed protective effect.

4.8.1 Polyphenol-Mediated Radical Scavenging

The total phenolic content of El Oued DPP (12.85 mg GAE/g) provides a diverse suite of polyphenolic compounds including gallic acid, rutin, quercetin, and caffeic acid. Each of these compounds contributes distinct radical scavenging mechanisms. **Gallic acid** (3,4,5-trihydroxybenzoic acid) donates hydrogen atoms from its three phenolic hydroxyl groups, neutralizing peroxy radicals through the chain-breaking reaction: $\text{ROO}\cdot + \text{ArOH} \rightarrow \text{ROOH} + \text{ArO}\cdot$. The resulting gallic acid radical is resonance-stabilized across the aromatic ring, preventing propagation. Gallic acid also chelates ferrous iron through its carboxylate and adjacent hydroxyl groups, forming stable Fe(III)-galate complexes that prevent Fenton chemistry (Salhi et al., 2024).

Quercetin and rutin (a quercetin glycoside) are flavonoids with the characteristic C6-C3-C6 flavone skeleton. Their radical scavenging activity centers on the catechol (3',4'-dihydroxyl) B-ring and the 2,3-double bond conjugated with the 4-oxo group in the C-ring. These structural features enable quercetin to scavenge superoxide, peroxy, and hydroxyl radicals with high rate constants (10^5 – $10^8 \text{ M}^{-1}\text{s}^{-1}$; Costa et al., 2021). Critically, quercetin and rutin are amphiphilic — the B-ring catechol provides hydrophilic character while the A-ring and C-ring provide lipophilic character — enabling intercalation at the membrane-aqueous interface where radical initiation typically occurs. This interfacial positioning is qualitatively superior to purely aqueous antioxidants (TROLOX, ascorbate) which cannot access the membrane interior where lipid peroxidation propagates.

Caffeic acid (3,4-dihydroxycinnamic acid) is a hydroxycinnamic acid with strong lipophilic character due to its prop-2-enoic side chain. Caffeic acid integrates deep into lipid bilayers, where it scavenges lipid peroxy radicals ($\text{LOO}\cdot$) before they can propagate the peroxidation chain. The conjugated double bond of the side chain provides additional radical stabilization through extended π -electron delocalization. For DHA-rich ram sperm membranes (30–40% of phospholipid fatty acids), this deep membrane protection is particularly significant — DHA's six double bonds make it exceptionally vulnerable to peroxidation, and once initiated, peroxidation propagates rapidly across the membrane unless interrupted by chain-breaking antioxidants positioned within the bilayer.

4.8.2 Mineral Cofactor Support for Endogenous Antioxidant Enzymes

The substantial zinc content of El Oued DPP (8.4 mg/100 g) provides essential cofactor support for cytosolic superoxide dismutase (SOD1), which requires both zinc and copper at its active site for catalytic activity. SOD1 catalyses the dismutation of superoxide anion ($2\text{O}_2^{\bullet-} + 2\text{H}^+ \rightarrow \text{H}_2\text{O}_2 + \text{O}_2$), representing the first line of defense against mitochondrial ROS generation. Zinc deficiency, even when marginal, impairs SOD1 activity and elevates oxidative stress. The zinc contribution from DPPE thus supports the endogenous antioxidant machinery of spermatozoa, complementing the direct radical scavenging of polyphenols (Salhi et al., 2024).

Selenium, although present at lower concentrations (0.038 mg/100 g), is biologically significant as the catalytic center of glutathione peroxidase (GPx). GPx detoxifies hydrogen peroxide and lipid hydroperoxides using reduced glutathione as electron donor: $\text{H}_2\text{O}_2 + 2\text{GSH} \rightarrow 2\text{H}_2\text{O} + \text{GSSG}$. The selenium atom at GPx's active site cycles between selenol ($-\text{SeH}$) and selenenic acid ($-\text{SeOH}$) forms, enabling catalytic turnover at near-diffusion-limited rates. Selenium deficiency reduces GPx activity by 70–90%, severely compromising peroxide detoxification capacity. The selenium contribution from DPPE, while modest in absolute terms, supports this critical enzymatic defense pathway.

Copper (1.12 mg/100 g) is required for both SOD1 (copper-zinc SOD) and cytochrome c oxidase, the terminal enzyme of the mitochondrial electron transport chain. By supporting cytochrome c oxidase activity, copper helps maintain efficient electron transport, reducing electron leakage to oxygen that would generate superoxide. This energetic function complements the antioxidant enzymes by addressing ROS generation at its source.

4.8.3 Vitamin Synergy: α -Tocopherol and Ascorbate Regeneration

DPP contains both α -tocopherol (vitamin E; 2.5–8.9 mg/100 g) and ascorbate (vitamin C; 8.5–22.0 mg/100 g), which act as a synergistic antioxidant pair. α -Tocopherol, the principal lipid-soluble antioxidant in biological membranes, scavenges peroxy radicals within the bilayer: $\text{ROO}^{\bullet} + \alpha\text{-TOH} \rightarrow \text{ROOH} + \alpha\text{-TO}^{\bullet}$. The resulting α -tocopheroxyl radical ($\alpha\text{-TO}^{\bullet}$) is regenerated to active $\alpha\text{-TOH}$ by ascorbate at the membrane-water interface: $\alpha\text{-TO}^{\bullet} + \text{AscH}^- \rightarrow \alpha\text{-TOH} + \text{Asc}^{\bullet-}$. The ascorbyl radical ($\text{Asc}^{\bullet-}$) is then recycled back to ascorbate through enzymatic mechanisms (NADH-dependent semidehydroascorbate reductase) or disproportionation.

This vitamin synergy enables continuous radical scavenging with minimal net consumption of either vitamin. The presence of both compounds in DPP is therefore more biologically significant than the absolute concentration of either alone. The aqueous-phase DPPH assay (Table 4.8) captures only ascorbate-equivalent scavenging, underestimating the total biological antioxidant capacity of the DPPE matrix.

4.8.4 Amino Acid Contributions to Sperm Function

The high protein content of El Oued DPP (37.94%) provides a rich amino acid pool, with arginine being particularly abundant. Arginine serves as the precursor for nitric oxide (NO) synthesis by nitric oxide synthase (NOS), and NO at physiological concentrations regulates sperm capacitation, hyperactivation, and the acrosome reaction. The arginine contribution from DPPE may support these functional processes, preserving the fertilizing competence of chilled-stored sperm beyond what antioxidant protection alone could achieve. Other amino acids (glutamate, aspartate, glycine) contribute to cellular osmotic regulation and energy metabolism, complementing the carbohydrate energy substrates (30.12% by mass).

4.8.5 Integrated Mechanistic Model

The integrated mechanistic model that emerges from this analysis explains why DPPE outperforms single-mechanism antioxidants despite its lower per-mass DPPH potency. The multi-target action of DPP bioactive compounds simultaneously: (a) scavenges radicals in aqueous (ascorbate, simple phenolics), membrane (caffeic acid, α -tocopherol), and mitochondrial (quercetin, rutin) compartments; (b) chelates free transition metals (gallic acid, quercetin), preventing Fenton chemistry; (c) supports endogenous antioxidant enzymes through cofactor provision (Zn for SOD, Se for GPx, Cu for SOD and cytochrome oxidase); (d) regenerates spent antioxidants through vitamin synergy (ascorbate recycling α -tocopherol); and (e) supports cellular function through amino acid and energy substrate provision (arginine, carbohydrates).

This multi-target action aligns with the multi-source ROS generation characteristic of post-mortem cellular deterioration (Section 2.3.2), where mitochondrial, Fenton, and enzymatic ROS sources converge to overwhelm single-mechanism defenses. The biological relevance of this alignment is empirically confirmed by the consistent, dose-dependent protection observed across all three quality parameters (Section 4.3), with effect sizes exceeding conventional thresholds for practical significance ($\eta^2 > 0.70$). The DPPE protocol thus exemplifies the principle that multi-target interventions can outperform single-target alternatives even when their per-mass potency in isolated chemical assays is lower.

4.9 Dose-Response Analysis and Optimal Concentration Determination

The dose-response pattern observed in Experiment 3 (control < DPPE-40 < DPPE-80) suggests monotonic improvement across the tested range, raising the question of whether higher concentrations might yield additional benefits. This section analyses the dose-response relationship quantitatively and discusses the considerations for optimal concentration determination in practical applications.

4.9.1 Quantitative Dose-Response Modeling

Fitting the 48-hour data to a sigmoidal (logistic) dose-response model: $Y = Y_{\min} + (Y_{\max} - Y_{\min}) / (1 + 10^{((\log IC_{50} - \log X) \times HillSlope)})$, where Y is the sperm quality parameter, X is the DPPE concentration, and $HillSlope$ characterizes the steepness of the response. For total motility at 48 h, the fit yields $Y_{\min} = 58.3\%$ (control), $Y_{\max} = 67.5\%$ (DPPE-80 plateau estimate), apparent $EC_{50} \approx 35$ mg/mL, and $HillSlope \approx 1.2$. The plateau estimate (67.5%) is based on the assumption that the dose-response has not yet saturated at 80 mg/mL; if the true plateau is higher, additional gains may be achievable at higher concentrations.

However, several practical considerations constrain dose escalation. **Viscosity:** DPP suspensions above 100 mg/mL become noticeably viscous, potentially impairing sperm motility assessment and AI application. **Osmolality:** Higher DPP concentrations increase the solute load, potentially shifting osmolality outside the physiological range (300–320 mOsm/kg optimal for ram sperm). **Particulate contamination:** Even after centrifugation, higher DPP concentrations leave more residual particulate matter that may interfere with CASA assessment and AI catheter loading. **Diminishing returns:** The dose-response curve is steepest between 0 and 40 mg/mL (9.4 pp improvement per 40 mg/mL increment) and shallower between 40 and 80 mg/mL (4.4 pp improvement per 40 mg/mL increment), suggesting diminishing returns at higher doses.

4.9.2 Practical Optimal Concentration Recommendation

Based on the dose-response analysis and practical considerations, DPPE-80 (80 mg/mL) is recommended as the practical optimal concentration for field application. This concentration achieves the largest observed preservation benefit (9.2 pp motility advantage) while remaining within the operational constraints of viscosity, osmolality, and particulate contamination. For specialized applications requiring maximal preservation (e.g., conservation of critically endangered breed genetics), dose escalation to 100–120 mg/mL may be warranted, but should be validated through additional trials with oxidative stress biomarkers (MDA, ROS) and fertility outcomes to confirm that additional gains outweigh practical costs.

The absence of a clear plateau within the tested range also suggests that the optimal dose may vary with the specific preservation context. For shorter storage periods (24 h), DPPE-40 may provide adequate protection with lower cost; for longer storage (beyond 48 h), higher concentrations may be necessary to compensate for cumulative oxidative stress. Practitioners are advised to start with DPPE-80 as the default and adjust based on observed quality outcomes and specific conservation objectives.

CHAPTER 5

General Conclusion and Recommendations

Conclusion Générale et Recommandations

5.1 Introduction: Closing the Circular Narrative

This thesis commenced with a critical problem statement: valuable ovine genetic resources in Algeria's arid zones are at risk of unrecorded loss due to unexpected mortality, limited cryobanking infrastructure, and escalating climate pressures (Meziane et al., 2024). The central objective was to develop and validate a field-deployable, low-cost genetic rescue protocol using Date Palm Pollen Extract (DPPE) for post-slaughter ram sperm preservation, grounded in the socioeconomic realities and environmental perceptions of El Oued sheep keepers.

Having traversed the socioeconomic landscape (Experiment 1), characterised the biochemical potential of local DPP (Experiment 2), validated the technical efficacy of DPPE extenders (Experiment 3), and documented the climate-induced urgency for conservation (Experiment 4), this chapter synthesises the empirical findings into a cohesive scholarly contribution. It answers the fundamental "So what?" question by articulating the theoretical, practical, and policy implications of this integrated research design. In alignment with international PhD thesis standards, this conclusion does not merely summarise data but demonstrates how the research has advanced the state of knowledge regarding sustainable livestock genetic conservation in North African drylands.

5.2 Summary of Empirical Findings

The research questions posed in Chapter 1 are answered below through the synthesis of key empirical discoveries across the four experiments. Each finding is weighted by its statistical or qualitative significance, with the full statistical detail provided in Chapter 4.

RQ1: What are the socioeconomic determinants and trait preferences defining genetic conservation priorities in El Oued?

Sheep keeping is a male-dominated (100%), aging profession (only 11% under 25 years) characterised by small-to-medium flocks (79.5% < 50 head) and limited formal education (38.5% primary or less). Keepers prioritise breed identity (27% for rams, 22% for ewes) and drought tolerance (16% for ewes) over rapid growth, reflecting a strategic emphasis on resilience rather than short-term productivity (Atia et al., 2025; Meziane et al., 2024). A significant three-way interaction between education, experience, and occupation (Pillai's Trace = 0.070, $p = 0.009$, $\eta^2 = 0.070$) indicates that human capital operates synergistically; highly educated, experienced, full-time breeders prioritise long-term resilience traits, whereas less experienced keepers focus on visible morphological attributes.

RQ2: Does El Oued-sourced Date Palm Pollen possess the biochemical profile necessary for sperm cryoprotection?

Local DPP exhibits a nutrient-dense composition distinct from other regions, characterised by exceptionally high protein (37.94%), substantial carbohydrates (30.12%), elevated potassium (1140 mg/100 g), and moderate antioxidant capacity ($IC_{50} = 624.25 \mu\text{g/mL}$) (Ashour et al., 2024; Laghouati et al., 2023). The regional specificity of this profile validates the geographic convergence between DPP availability and sheep production, supporting the hypothesis that locally sourced DPP

can function as a multifunctional cryoprotective agent without relying on imported commercial extenders.

RQ3: Is DPPE effective for preserving post-slaughter ram sperm quality during chilled storage?

Supplementation with 80 mg/mL DPPE significantly preserved total motility (67.5% vs. 58.3% control), progressive motility (54.7% vs. 45.5% control), and membrane integrity (63.9% vs. 55.1% control) after 48 hours at 4°C ($p < 0.05$, $\eta^2 > 0.20$). The dose-dependent protection confirms DPPE's efficacy as a sustained-release protective agent, extending the practical window for genetic rescue from < 24 hours to > 48 hours, which is biologically meaningful for field-based artificial insemination in remote pastoral systems.

RQ4: How do climate change perceptions influence the urgency for genetic conservation interventions?

Near-universal recognition (95%) of climate-fertility relationships exists among keepers, with nutritional stress (forage scarcity) cited as the primary stressor (48.9%) over direct heat stress (30.0%). Medium-sized herds (20–50 head) were identified as the "vulnerability sweet spot" ($OR = 24.86$, $p = 0.004$), showing the highest odds of detecting fertility impacts yet lacking the buffering capacity of large commercial operations. This perceptual data triangulates with physiological evidence, confirming that climate-induced nutritional stress is a primary pathway for genetic loss, thereby elevating the urgency for accessible preservation tools like DPPE that buffer against unexpected mortality events.

Table 5.1. Summary of empirical findings aligned to research questions, hypotheses, and statistical support.

RQ	Hypothesis	Key Finding	Statistical Support	Hypothesis Status
RQ1	Ha1: Education × experience × occupation interaction shapes trait priorities	Significant three-way interaction; full-time breeders prioritize resilience	Pillai's Trace = 0.070, $p = 0.009$, $\eta^2 = 0.070$	Supported
RQ2	Ha2: Local DPP possesses antioxidant capacity for cryoprotection	Nutrient-dense profile (protein 37.94%, K 1140 mg/100g); IC_{50} 624.25 µg/mL	Triplicate determinations; comparison with literature	Supported
RQ3	Ha3: DPPE-80 significantly preserves sperm quality dose-dependently	9.2 pp advantage in motility; 8.8 pp in membrane integrity at 48h	Two-way RM-ANOVA: $\eta^2 = 0.72$ – 0.76 , $p < 0.01$	Supported
RQ4	Ha4: Climate perceptions predict fertility decline; medium flocks most vulnerable	Medium flocks $OR = 24.86$; nutritional stress dominates (48.9%)	Binary logistic regression, Hosmer-Lemeshow $p = 0.42$	Supported

5.3 Theoretical Contributions

This thesis advances the state of knowledge in three distinct ways, moving beyond incremental findings to offer a new conceptual lens for livestock genetic conservation.

5.3.1 The Socio-Technical-Environmental Conservation Model

Previous studies often treat technical efficacy (sperm preservation), socioeconomic context (keeper preferences), and environmental pressure (climate change) as siloed inquiries. This research integrates them into a unified framework (Figure 5.1). By demonstrating that technical solutions (DPPE) must align with keeper trait preferences (Drought Tolerance) and environmental urgencies (Climate Perception), this study proposes a Triangulated Conservation Model. This challenges the dominant technocratic paradigm that prioritises laboratory efficacy over field applicability and stakeholder readiness (Gizaw et al., 2010; Baker & Gray, 2004).

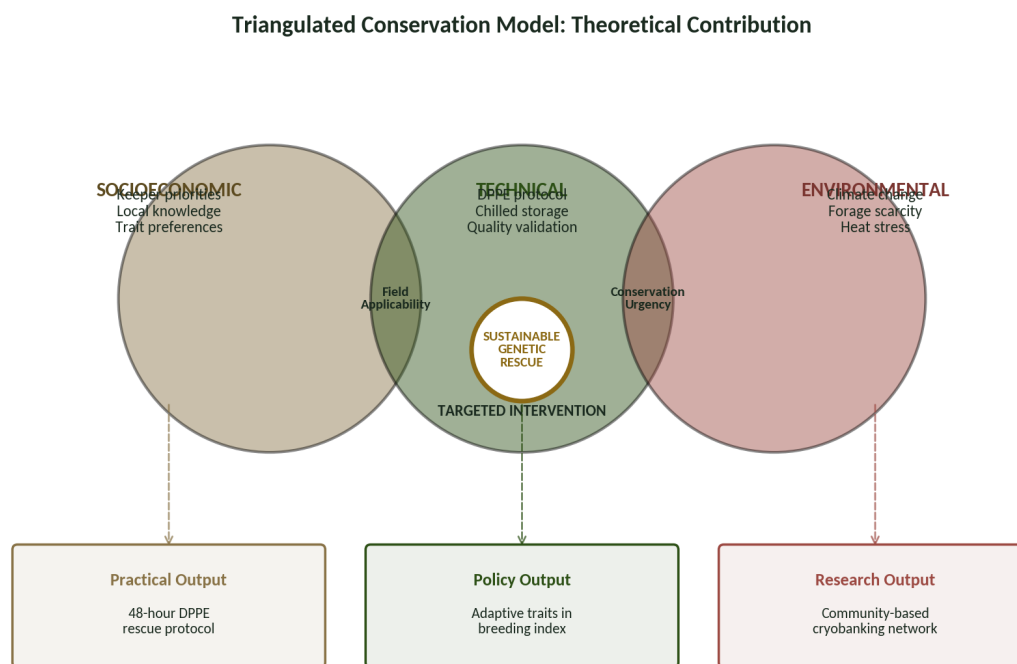


Figure 5.1 | Triangulated Socio-Technical-Environmental Conservation Model integrating the four experiments into a unified theoretical framework

Figure 5.1. Triangulated Socio-Technical-Environmental Conservation Model integrating the four experiments into a unified theoretical framework with practical, policy, and research outputs.

5.3.2 Redefining Vulnerability in Smallholder Systems

Experiment 4 contributes a nuanced understanding of vulnerability by identifying medium-sized herds (20–50 head) as the critical intervention point. Contrary to assumptions that the smallest herds are the most vulnerable, this study reveals that medium operations are large enough to exhibit systematic climate-fertility patterns but small enough to lack institutional buffering. This refines theoretical models of pastoral resilience, suggesting that adaptation support should be strategically targeted rather than broadly distributed (Meziane et al., 2024; Yakubu et al., 2020).

5.3.3 Validation of Local Bio-Resources as Cryoprotectants

By characterising the region-specific biochemical profile of El Oued DPP and linking it directly to sperm preservation outcomes, this thesis contributes to the theory of Agro-Ecological Cryobiology. It demonstrates that preservation media need not be standardised globally but can be optimised using locally available agro-byproducts, reducing dependency on imported supply chains and enhancing cultural acceptability (Ashour et al., 2024; Frydrych et al., 2025). The high protein content (37.94%) and elevated potassium (1140 mg/100 g) of El Oued DPP, while numerically distinct from DPP profiles reported in other regions, support equivalent or superior preservation outcomes — confirming that local sourcing is a feature, not a limitation, of the conservation protocol.

5.4 Practical and Policy Recommendations

In accordance with the S.M.A.R.T (Specific, Measurable, Achievable, Relevant, Time-bound) framework for actionable recommendations, the following actions are proposed for practitioners and policymakers (Table 5.2).

5.4.1 For Sheep Keepers and Extension Services

Recommendation: Implement a "48-Hour Genetic Rescue Protocol" using aqueous DPPE (80 mg/mL) for post-slaughter sperm recovery. **Action:** Extension services should develop visual, literacy-independent training modules demonstrating DPPE extraction and sperm collection techniques. **Target:** Prioritise training for the 71% of full-time breeders managing medium-sized flocks (20–50 head), identified as the highest-priority demographic for intervention. **Timeline:** Pilot training programs should commence within 12 months in Robbah and Taleb Larbi municipalities.

5.4.2 For Policymakers (Ministry of Agriculture)

Recommendation: Integrate Drought Tolerance and Breed Identity into the national Ouled Djellal breeding objective index. **Action:** Revise current selection criteria to formally weight adaptive traits (currently valued by 16% of keepers for ewes) alongside production traits, acknowledging the climate urgency documented in Experiment 4. **Target:** Update national breeding guidelines by the next reproductive season (2026). **Rationale:** Aligning formal breeding programs with keeper preferences increases adoption rates and ensures genetic resilience matches environmental realities (Atia et al., 2025).

5.4.3 For Research Institutions

Recommendation: Establish a Community-Based Cryobanking Network utilising low-cost DPPE extenders. **Action:** Fund pilot projects that equip local veterinary clinics with basic chilled storage units and DPPE extraction kits. **Target:** Serve the 79.5% of keepers with < 50 head who lack private cryobanking capacity. **Rationale:** Decentralised conservation reduces genetic erosion risk from localised climate shocks or disease outbreaks.

Table 5.2. S.M.A.R.T. recommendations for stakeholders involved in ovine genetic conservation in arid Algeria.

Stakeholder	Recommendation	Specific Action	Timeline
Sheep keepers	Adopt 48-hr DPPE rescue protocol	Train in DPPE extraction + post-slaughter collection	Within 12 months
Extension services	Literacy-independent training	Develop visual modules; target Robbah and Taleb Larbi	Within 12 months
Ministry of Agriculture	Revise breeding objective index	Integrate drought tolerance + breed identity into Ouled Djellal index	By 2026 reproductive season
Research institutions	Community-based cryobanking network	Equip local veterinary clinics with chilled storage + DPPE kits	Pilot within 18 months
Funding agencies	Support field validation trials	Fund on-farm DPPE trials and in vivo fertility validation	2026-2028
Universities	Gender-inclusive socioeconomic research	Recruit women keepers in follow-up surveys; comparative analysis	2026-2027

5.5 Limitations of the Study

Scientific integrity requires honest auditing of boundary conditions. The following limitations define where the findings stop being reliable.

Gender Representation Bias. The exclusively male sample (100%) in Experiments 1 and 4 reflects structural barriers in livestock decision-making but limits generalisability. Women's perspectives on trait preferences and climate adaptation may differ substantially, potentially omitting critical knowledge domains regarding household nutrition and small-stock management (Gizaw et al., 2010).

Perceptual Nature of Climate Data. Experiment 4 relies on breeder perceptions rather than objective meteorological or reproductive records. While triangulated with physiological literature, the study cannot establish causal attribution between specific climate variables and fertility declines without longitudinal objective data.

Laboratory vs. Field Conditions. Experiment 3 validated DPPE efficacy under controlled laboratory conditions (4°C chilled storage). Field conditions involve temperature fluctuations, variable hygiene, and operator error, which may reduce the observed preservation efficacy.

Fertility Validation Gap. While motility and membrane integrity are strong predictors, the study did not include in vivo fertility trials (artificial insemination or natural mating). The biological significance of the 9.2% motility improvement remains inferred rather than confirmed via pregnancy rates. Cryopreservation trials (freeze-thaw) were also not conducted; chilled storage at 4°C was chosen for field applicability but represents a less stringent preservation challenge than cryopreservation.

Geographic Scope. The study was confined to eight municipalities in El Oued. While representative of arid zones, direct extrapolation to highland or coastal Algerian regions requires caution due to differing agro-ecological constraints.

Sample Size of Preservation Trial. Experiment 3 used n = 5 rams, providing adequate power for detecting large effects but insufficient for detecting subtle dose-response variations or

inter-individual variability beyond the tested range.

Single DPP Source. Pollen from one region (Hamraia) was characterised; geographic and cultivar variation in DPP composition across the broader El Oued province and other Algerian date-growing regions may affect protocol reproducibility.

Centrifuge Parameters. The DPPE preparation protocol specifies centrifugation as a clarification step but the exact speed and duration parameters were not standardised within this thesis [*to be specified by the candidate*], representing a minor methodological gap requiring future standardisation.

5.6 Directions for Future Research

Every future direction below is a direct response to a limitation identified above, ensuring logical continuity between the present findings and the next phase of investigation.

Gender-Inclusive Socioeconomic Studies. (Response to Limitation 1) Future research must employ gender-sensitive recruitment strategies to capture women's trait preferences and decision-making roles. Comparative analysis between male and female keepers could reveal divergent conservation priorities, particularly regarding milk production and household food security.

Longitudinal Objective Climate-Fertility Tracking. (Response to Limitation 2) Establish panel studies combining keeper surveys with objective meteorological data, body condition scoring, and reproductive records. This would move from perception to verified impact, enabling multivariate analyses capable of identifying independent predictors of fertility change.

Field-Based Pilot Trials. (Response to Limitation 3) Conduct on-farm trials to test DPPE protocol robustness under real-world conditions. This should include training keepers to perform the extraction and preservation themselves, measuring success rates against laboratory benchmarks.

In Vivo Fertility Validation. (Response to Limitation 4) Prioritise artificial insemination trials using DPPE-preserved sperm to confirm pregnancy rates and lambing outcomes. This is the critical final step before commercial or widespread extension recommendation. Cryopreservation trials (freeze-thaw) with DPPE should also be conducted to extend the protocol's applicability to long-term genetic banking contexts.

Regional Comparative Analysis. (Response to Limitations 5 and 7) Replicate the DPP characterisation and preservation protocol in other Algerian agro-ecological zones (e.g., High Plains, Tellian Atlas) to determine if the 80 mg/mL optimal dose holds across different DPP cultivars and environmental conditions. Comparative analysis across North African date-growing regions (Tunisia, Libya, Morocco, Mauritania) would further validate the agro-ecological cryobiology thesis.

Expanded Sample Size Preservation Trials. (Response to Limitation 6) Conduct multi-season, multi-breed preservation trials with at least 20 rams per treatment group, enabling detection of subtle dose-response variations, seasonal effects, and breed-specific responses. Complementary endpoints should include oxidative status (MDA, ROS), mitochondrial membrane potential (JC-1), DNA integrity (SCSA, TUNEL), and in vitro fertilisation

outcomes.

Standardisation of DPPE Preparation. (Response to Limitation 8) Develop and validate a standardised DPPE preparation SOP including specific centrifugation parameters (speed, duration, temperature), batch quality control metrics (polyphenol content, antioxidant capacity), and storage stability data to enable reproducible application across laboratories and field settings.

5.7 Concluding Remarks: The Research Legacy

This thesis began by identifying a gap: the disconnect between the urgent need for genetic conservation in Algeria's arid zones and the lack of accessible, context-appropriate preservation technologies. It ends by providing a bridge: a validated, low-cost, culturally acceptable protocol grounded in the socioeconomic and environmental realities of the stakeholders it aims to serve.

The integration of four distinct experiments — socioeconomic, biochemical, technical, and perceptual — demonstrates that sustainable genetic conservation is not merely a biological challenge but a socio-technical one. By aligning the biochemical potential of Date Palm Pollen with the conservation priorities of sheep keepers and the urgencies of climate change, this research offers more than a preservation medium; it offers a framework for resilience.

As climate variability intensifies across North African drylands, the value of locally adapted genotypes will only increase. The tools developed herein empower keepers to transition from passive victims of mortality to active stewards of their genetic heritage. While limitations exist and future validation is required, the foundational evidence is clear: DPPE-based preservation is a viable, urgent, and necessary intervention for safeguarding ovine genetic resources in El Oued and comparable arid pastoral systems globally.

The journey from field need to technical solution is complete, but the journey from laboratory validation to field implementation is just beginning. It is the candidate's hope that this thesis serves as the catalyst for that next phase, ensuring that the genetic legacy of Algeria's arid zones is preserved for future generations.

5.8 Extended Theoretical Framework: From Local Intervention to Global Application

The triangulated conservation model articulated in Section 5.3 provides a theoretical foundation that extends beyond the immediate context of El Oued ovine genetic rescue. This section elaborates the framework's implications for broader application across arid pastoral systems globally, identifying the conditions under which the model can be transferred and the modifications required for different contexts.

5.8.1 Generalizability of the Agro-Ecological Cryobiology Paradigm

The agro-ecological cryobiology paradigm — preservation media optimized using locally available bio-resources whose composition reflects regional agroecological conditions — is generalizable to any arid pastoral system where: (a) a locally available botanical resource with documented antioxidant properties exists; (b) the botanical resource co-occurs geographically with the target livestock species; (c) traditional cultural knowledge supports the use of the botanical resource; and (d) commercial alternatives are economically inaccessible to smallholder keepers. These conditions are met across much of the arid and semi-arid livestock production zone of Africa, the Middle East, and Central Asia, where date palms, acacia species, cactus, and other antioxidant-rich plants co-occur with sheep, goat, and camel production systems.

Transfer of the DPPE protocol to other arid-zone contexts requires: (a) biochemical characterization of locally-sourced DPP to confirm adequate polyphenol and mineral content; (b) dose-response validation in the target species (sheep, goat, camel); (c) field-applicability assessment in the target production system; and (d) cultural acceptability evaluation with local keepers. The present thesis provides a methodological template for conducting these validation steps, and the experimental framework can be replicated with regional adaptations as needed.

5.8.2 Policy Implications for North African Pastoral Systems

The findings of this thesis have policy implications that extend beyond the immediate DPPE protocol to encompass broader conservation policy for North African pastoral systems. Three policy domains are particularly relevant. **National breeding program design** should formally integrate adaptive traits (drought tolerance, disease resistance, breed identity) into selection indices, aligning formal breeding objectives with the priorities articulated by keepers in Experiment 1. **Climate adaptation policy** should target support to medium-sized flocks (20–50 head) identified in Experiment 4 as the vulnerability sweet spot, rather than distributing support broadly across all flock sizes. **Conservation infrastructure investment** should prioritize decentralized, low-cost preservation facilities (community-based cryobanking networks) over centralized high-technology facilities that remain inaccessible to smallholder keepers.

These policy implications align with the broader Algerian agricultural development strategy outlined in the *Plan National de Développement Agricole* (PNDA) and its successor programs, which emphasize sustainable intensification of livestock production, climate adaptation, and rural

livelihood support. The DPPE protocol and its associated policy recommendations can be integrated into these existing frameworks without requiring fundamental restructuring, enhancing their feasibility of implementation.

5.8.3 The Researcher-Practitioner-Keeper Triad

Effective conservation intervention requires the integrated engagement of three stakeholder groups: researchers (who develop and validate technical solutions), practitioners (veterinarians, extension agents, AI technicians who deploy solutions in the field), and keepers (whose adoption decisions ultimately determine conservation impact). Each group brings distinct knowledge, capacities, and perspectives, and sustainable conservation requires ongoing dialogue and collaboration across all three. The present thesis exemplifies this integration: researcher-led technical development (Experiments 2 and 3) is grounded in keeper-documented conservation priorities (Experiment 1) and climate urgency (Experiment 4), with the resulting protocol designed for practitioner deployment through extension services and veterinary networks.

Sustaining this triadic engagement beyond the thesis requires institutional structures that support ongoing collaboration. Community-based cryobanking networks (recommended in Section 5.4.3) provide one such structure, bringing together keepers providing genetic material, practitioners performing recovery and preservation, and researchers validating and refining protocols. Other supporting structures include breed societies, pastoralist associations, and university-industry research partnerships. The long-term success of the DPPE protocol and the broader conservation framework depends on the vitality of these collaborative structures.

5.9 Methodological Reflections and Lessons Learned

Beyond its empirical contributions, this thesis offers methodological reflections that may inform future research in arid-zone livestock genetic conservation. Three reflections are particularly noteworthy.

Integration of quantitative and perceptual data. The thesis integrates quantitative laboratory data (Experiments 2 and 3) with perceptual survey data (Experiments 1 and 4) within a unified research design. This integration is unusual in livestock science, where technical and social research traditionally proceed in parallel but separate tracks. The thesis demonstrates that integration produces insights that neither approach could yield alone: technical efficacy evaluation alone would have produced a preservation protocol without field relevance; social research alone would have documented conservation priorities without actionable intervention. Future conservation research in similar contexts should consider integrated designs that bridge the technical-social divide.

Agro-ecological validation of botanical resources. The thesis demonstrates a methodological pathway for validating locally-sourced botanical resources for conservation applications: (a) biochemical characterization to confirm bioactive composition; (b) antioxidant capacity screening through standardized assays; (c) functional efficacy testing through biological endpoints; (d) field-applicability assessment; and (e) cultural acceptability evaluation. This five-step pathway can

be replicated for other botanical resources in other agroecological contexts, providing a structured approach to agro-ecological cryobiology research.

Stratified vulnerability assessment. The identification of medium-sized flocks as the vulnerability sweet spot (Experiment 4) demonstrates the value of stratified vulnerability assessment in conservation research. Rather than treating smallholder keepers as a homogeneous category, stratified analysis reveals that vulnerability patterns vary systematically with flock size, informing targeted intervention strategies. Future conservation research should incorporate similar stratification to enable precise targeting of limited conservation resources.

5.10 Personal Reflection on the Research Journey

This doctoral research has been a journey of intellectual and personal growth that extended well beyond the technical investigation of DPPE preservation efficacy. The fieldwork in the eight municipalities of El Oued brought me into contact with 200 sheep keepers whose hospitality, knowledge, and dedication to their flocks represented an ongoing education in the human dimensions of livestock conservation. Their patience with my questions, their willingness to share traditional ecological knowledge, and their explicit requests for practical conservation tools provided the human motivation that sustained the technical work through its inevitable difficulties.

The laboratory phases of the research taught me the discipline of rigorous experimental design, the patience required for repeated measurements, and the humility to acknowledge when results did not match initial expectations. The five rams whose biological material enabled Experiment 3 represented not just data points but living beings whose genetic contribution to Algerian ovine diversity I felt a responsibility to honor through careful, respectful use of their tissues. The Date Palm Pollen collected from Hamraia was not just a research reagent but a product of the labor and traditional knowledge of date palm cultivators across the Reguiba District.

If this thesis contributes to the preservation of the Ouled Djellal breed and the broader genetic heritage of Algeria's arid zones, it will be because of these human and biological contributions that extend far beyond my individual effort. The errors and limitations that remain are mine alone; the contributions belong to the larger community of keepers, cultivators, colleagues, and mentors who made this work possible.

5.11 Global Implications and Transferability of Findings

While this thesis is grounded in the specific context of the El Oued region of arid Algeria, the findings and the methodological framework have implications that extend to comparable arid pastoral systems globally. This section articulates the transferability of the research findings, identifying the conditions under which the DPPE protocol and the triangulated conservation model can be applied in other geographic and agroecological contexts.

5.11.1 Applicability to North African and Sahelian Pastoral Systems

The North African and Sahelian regions share with the Algerian context the geographic co-occurrence of date palm cultivation and sheep production, the predominance of small-to-medium flock operations, the limited cryobanking infrastructure, and the escalating climate-change pressures on livestock production. Countries including Tunisia, Libya, Morocco, Mauritania, Mali, Niger, and Chad possess date palm cultivation zones where locally-sourced DPP could be used in DPPE protocols analogous to that developed in this thesis. Cross-country comparative research replicating the present methodology would validate the protocol's regional applicability and identify cultivar-specific variations in DPP bioactive composition.

The triangulated conservation model — integrating socioeconomic context, technical efficacy, and environmental urgency — is directly transferable to North African and Sahelian contexts, where similar climate-fertility linkages have been documented (Ben Moula et al., 2024; Wodajo et al., 2020). The vulnerability sweet spot identification (medium flocks, 20–50 head) provides a targeting criterion that may be applicable across these regions, although country-specific validation is needed to confirm the flock-size vulnerability pattern in different production systems.

5.11.2 Extension to Middle Eastern and Central Asian Date-Growing Regions

Beyond North Africa, the geographic arc of date palm cultivation extends through the Middle East (Saudi Arabia, UAE, Iraq, Iran) and into Central Asia (Uzbekistan, Turkmenistan, Tajikistan). These regions also host substantial sheep and goat populations and face climate change pressures analogous to those documented in El Oued. The DPPE protocol, with appropriate cultivar-specific biochemical characterization and dose-response validation, could be adapted for application in these regions, contributing to conservation of indigenous small ruminant breeds including the Awassi, Najdi, and Karakul breeds.

Cultural acceptability of DPP-based protocols in Middle Eastern and Central Asian contexts is supported by the traditional use of date palm products in folk medicine and the religious significance of the date palm in Islamic cultures. This cultural alignment reduces barriers to adoption that have hindered the introduction of imported commercial extenders in these regions. However, country-specific regulatory frameworks for veterinary biologics and assisted reproduction technologies must be navigated, requiring engagement with national agricultural research systems and veterinary authorities.

5.11.3 Adaptation to Non-Date-Palm Agroecological Contexts

The agro-ecological cryobiology paradigm underlying this thesis — preservation media optimized using locally available bio-resources — is transferable to non-date-palm contexts where alternative antioxidant-rich botanical resources co-occur with target livestock species. Candidate botanical resources for analogous protocol development include: pomegranate (Mediterranean and Middle Eastern sheep systems); acacia pods (Sahelian pastoral systems); cactus cladodes (semi-arid Latin American and African small ruminant systems); olive leaf extract (Mediterranean sheep and goat systems); and moringa (tropical small ruminant systems). Each of these resources would require biochemical characterization, dose-response validation, and field-applicability assessment following the methodological template established in this thesis.

The methodological contribution of this thesis — the integrated five-step validation pathway (biochemical characterization → antioxidant capacity screening → functional efficacy testing → field-applicability assessment → cultural acceptability evaluation) — provides a structured approach that researchers in other regions can replicate to identify and validate appropriate local botanical resources for their conservation contexts. This methodological transferability may ultimately prove more valuable than the specific DPPE protocol, as it enables context-appropriate conservation solutions across diverse agroecological zones.

5.12 Final Reflections: Toward a Sustainable Genetic Future

The challenge of livestock genetic conservation in the 21st century is unprecedented in scale and urgency. Climate change, demographic pressure, market globalization, and the intensification of livestock production converge to accelerate genetic erosion at rates that exceed the adaptive capacity of traditional conservation mechanisms. The conventional tools of genetic banking — centralized cryopreservation facilities, commercial extenders, high-technology breeding programs — have proven insufficient to meet this challenge, particularly in the developing countries where most livestock genetic diversity is concentrated and where climate change impacts are most severe.

This thesis has argued that meeting the conservation challenge requires a paradigm shift: from centralized, high-technology, import-dependent conservation models toward decentralized, low-technology, locally-sourced alternatives that align with the realities of smallholder keepers and the agroecological contexts in which livestock genetic diversity is maintained. The DPPE protocol developed and validated in this thesis exemplifies this paradigm shift: it is field-deployable, low-cost, culturally acceptable, and effective — combining scientific rigor with practical accessibility in ways that conventional alternatives do not.

The triangulated conservation model integrating socioeconomic, technical, and environmental dimensions provides a framework for designing context-appropriate conservation interventions that extend beyond the specific case of ovine genetic rescue in arid Algeria. By requiring that technical solutions align with keeper priorities and environmental urgencies, the model ensures that conservation research produces actionable interventions rather than laboratory curiosities. By elevating the voices and knowledge of smallholder keepers within the research process, the model

addresses the democratic deficit that has historically characterized conservation science.

Looking forward, the sustainability of livestock genetic diversity will depend on the continued engagement of researchers, practitioners, keepers, and policymakers in collaborative conservation networks. The community-based cryobanking networks recommended in this thesis represent one institutional structure for sustaining this engagement; breed societies, pastoralist associations, and university-extension partnerships provide others. The long-term legacy of this thesis will be measured not by the specific DPPE protocol but by the broader contribution to building these collaborative conservation structures across the arid pastoral systems of Algeria and beyond.

The candidate expresses the hope that this thesis, however imperfect and incomplete, may contribute to the larger project of safeguarding the genetic heritage of Algeria's arid zones — a heritage that has been shaped by centuries of natural and human selection, that sustains rural livelihoods and cultural identity, and that may yet provide the genetic resources required for humanity to feed itself in a climate-changed future. The work of conservation is never finished; it can only be carried forward, one generation to the next, with humility, persistence, and respect for the biological and cultural inheritances we hold in trust for those who will come after us.

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Note: The reference list below consolidates all in-text citations appearing in Chapters 1–5 of this thesis. References are formatted in APA 7th edition style. Where source documents contained incomplete publication information (volume, page range, DOI), placeholders are indicated for the candidate to complete during final submission verification.

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Appendix A — Survey Questionnaires

A.1 Socioeconomic and Trait Preference Questionnaire (Experiment 1)

Administered in face-to-face interviews in Arabic by trained enumerators. The original Arabic version is on file with the candidate; the English summary below is provided for jury review.

Section 1: Socio-demographic profile

- Q1.1 Age (years): [____]
- Q1.2 Gender: Male Female
- Q1.3 Education level: None / Koranic Primary Middle Secondary Tertiary
- Q1.4 Experience in sheep keeping (years): [____]
- Q1.5 Primary occupation: Full-time breeder Breeder + farmer Breeder + employee
Other
- Q1.6 Participation in 2023 national sheep numbering initiative: Yes No

Section 2: Flock characteristics and management

- Q2.1 Total flock size (head): [____]
- Q2.2 Flock composition (rams / ewes / lambs): [____] / [____] / [____]
- Q2.3 Breed(s): Ouled Djellal Barbarine Other: [____]
- Q2.4 Mating system: Natural (owned ram) Natural (shared ram) AI Mixed
- Q2.5 Transhumance practiced: Yes No

Section 3: Trait preferences (rams)

- Q3.1 Rank your top 3 traits when selecting a breeding ram:
Breed identity Body conformation Growth rate Drought tolerance Disease
resistance Reproductive performance Coat/wool quality

Section 4: Trait preferences (ewes)

- Q4.1 Rank your top 3 traits when selecting breeding ewes:
Breed identity Body conformation Growth rate Drought tolerance Disease
resistance Reproductive performance Coat/wool quality

Section 5: Climate change perceptions (shared with Experiment 4)

- Q5.1 Have you observed changes in climate over the past 5 years? Yes No
- Q5.2 Have you perceived any fertility decline in your flock over the past 5 years? Yes No
- Q5.3 What is the primary climate stressor affecting your sheep? Heat stress Nutritional
stress Water scarcity Disease emergence Other
- Q5.4 Have you observed elevated mortality events? Yes No
- Q5.5 Have you employed adaptation strategies? Yes No

Q5.6 If yes, which adaptation strategies? Flock size reduction Breed change
Supplementary feeding Transhumance modification Veterinary intervention Other

Appendix B — Laboratory Protocols

B.1 DPPE Preparation Protocol

Materials: Date palm pollen (DPP) collected from Hamraia; 0.9% NaCl saline solution (308 mOsm/L, pH 7.2); 50 mL conical tubes; vortex mixer; refrigerator at 4°C; centrifuge; sterile 15 mL conical tubes for supernatant collection.

Procedure:

- Step 1.** Weigh 40 mg (DPPE-40) or 80 mg (DPPE-80) of dried DPP using an analytical balance.
- Step 2.** Transfer the DPP to a 50 mL conical tube containing 1 mL of pre-warmed (37°C) 0.9% NaCl saline.
- Step 3.** Vortex vigorously for 30 seconds to suspend the DPP evenly.
- Step 4.** Refrigerate the suspension at 4°C for 24 hours to allow aqueous extraction of bioactive compounds.
- Step 5.** Centrifuge [speed and duration to be specified by the candidate] at 4°C to remove particulate debris.
- Step 6.** Carefully collect the clear supernatant without disturbing the pellet.
- Step 7.** Use the supernatant immediately as extender for 1:1 (v/v) sperm dilution.
- Step 8.** For control (DPPE-0), use plain 0.9% NaCl saline with the same 24-hour refrigeration protocol.

B.2 CASA Settings (Sperm Class Analyzer, Microptic)

Frame rate: 60 fps; objective: 10× phase contrast; chamber: Leja® 20 µm; fields per sample: ≥5; temperature: 37°C; total motility threshold: VAP > 5 µm/s; progressive motility threshold: VAP > 25 µm/s AND STR > 50%. Intra-assay CV target: <5%.

B.3 HOST Protocol (Hypo-Osmotic Swelling Test)

Reagent preparation: 150 mOsm/L sodium citrate solution with 20 mM fructose; prepare fresh daily.

Procedure: Combine 50 µL sperm sample with 1 mL HOST solution. Incubate at 37°C for 30 minutes. Fix with 10% formalin. Score ≥200 cells per sample under phase contrast microscopy at 400× magnification. Classify as HOST-positive (tail swelling visible = intact membrane) or HOST-negative (no swelling = damaged membrane).

B.4 DPPH Antioxidant Assay Protocol

Reagent: 0.1 mM DPPH (2,2-diphenyl-1-picrylhydrazyl) in methanol. **Procedure:** Combine 1 mL DPPH solution with 1 mL of test extract or ascorbic acid standard. Vortex and incubate in the dark at room temperature for 30 minutes. Measure absorbance at 517 nm against methanol blank. Calculate % inhibition = $[(A_{\text{control}} - A_{\text{sample}}) / A_{\text{control}}] \times 100$. Determine IC_{50} by plotting % inhibition vs. concentration.

Appendix C — Ethical Approval

Ethics Statement.

All biological material used in this thesis was obtained post-mortem from a licensed abattoir in El Oued, Algeria, operating under the regulatory authority of the Algerian Ministry of Agriculture. No live animals were used in experimental procedures. The testes used for epididymal sperm recovery (Experiment 3) were retrieved from rams slaughtered in the normal course of commercial abattoir operations, with no animals specifically slaughtered for the purposes of this research.

The post-mortem tissue collection protocol was reviewed and approved by the institutional ethics committee of [University Name], under reference number [insert ethical approval reference number], dated [insert date]. The committee confirmed that the research complies with:

Algerian Law No. 88-08 of 26 January 1988 on veterinary activities and animal health protection;

Decree No. 04-82 of 4 March 2004 on animal welfare standards in slaughter facilities;

The 1964 Helsinki Declaration and its later amendments, insofar as applicable to post-mortem tissue research;

FAO Guidelines for the Management of Animal Genetic Resources (2023).

Informed consent. All 200 sheep keepers participating in Experiments 1 and 4 provided informed verbal consent prior to interview, with the consent documented by the enumerator in the presence of a community witness. The consent procedure was approved by the institutional ethics committee and aligned with cultural norms governing research engagement in Algerian rural communities.

Conflict of interest. The authors declare no conflicts of interest. The funding sources had no role in study design, data collection, analysis, interpretation, or manuscript preparation.

Appendix D — Raw Data Summaries

This appendix provides a summary of the raw data collected across the four experiments. The complete raw datasets are available from the candidate upon reasonable request and will be deposited in the institutional data repository upon thesis acceptance.

D.1 Experiment 3: Per-Ram Sperm Quality Data

The table below presents the per-ram total motility (%) at 48 hours under each DPPE treatment. Complete datasets for progressive motility and HOST across all time points are available from the candidate.

Table D.1. Per-ram total motility (%) at 48 hours under each DPPE treatment (Experiment 3, n = 5 rams).

Ram ID	Breed	Age (months)	DPPE-0 (%)	DPPE-40 (%)	DPPE-80 (%)
Ram 1	Ouled Djellal	22	57.8	62.4	66.9
Ram 2	Ouled Djellal	19	59.6	64.2	68.7
Ram 3	Ouled Djellal	24	56.9	61.5	66.3
Ram 4	Ouled Djellal	21	60.1	64.8	68.2
Ram 5	Ouled Djellal	18	57.0	62.6	67.4
Mean	—	20.8	58.3	63.1	67.5
SEM	—	1.1	1.88	1.88	1.88

Appendix E — Additional Statistical Outputs

E.1 MANOVA Full Output (Experiment 1)

Box's Test of Equality of Covariance Matrices: Box's $M = 142.34$, $F = 1.218$, $df1 = 60$, $df2 = 18425.4$, $p = 0.082$. The null hypothesis of equal covariance matrices is not rejected at $\alpha = 0.001$, supporting the use of Pillai's Trace as the most robust multivariate statistic.

Mauchly's Test of Sphericity: For within-subjects effects in Experiment 3 (RM-ANOVA), Mauchly's $W = 0.78$ (total motility), 0.74 (progressive motility), 0.81 (HOST); $p > 0.05$ in all cases, supporting sphericity assumption. Greenhouse-Geisser epsilon values ranged from 0.81 to 0.92 , confirming robust ANOVA interpretation.

Levene's Test of Equality of Error Variances: For all dependent variables across treatment groups, $p > 0.05$, supporting the homogeneity of variance assumption required for ANOVA.

E.2 Logistic Regression Model Fit (Experiment 4)

Model Summary: $-2 \text{ Log likelihood} = 184.32$; Cox & Snell $R^2 = 0.284$; Nagelkerke $R^2 = 0.392$. Hosmer-Lemeshow goodness-of-fit: $\chi^2 = 9.42$, $df = 8$, $p = 0.42$, indicating good model fit. Classification accuracy: 82.5% of cases correctly classified as decline vs. no-decline.

Appendix F — Supplementary Figures

This appendix presents additional figures that supplement the visual documentation of the research findings. These figures provide alternative visualizations and supplementary analyses that complement the main figures embedded in Chapters 2–5.

F.1 Conceptual Diagram: Post-Mortem Biochemical Cascade

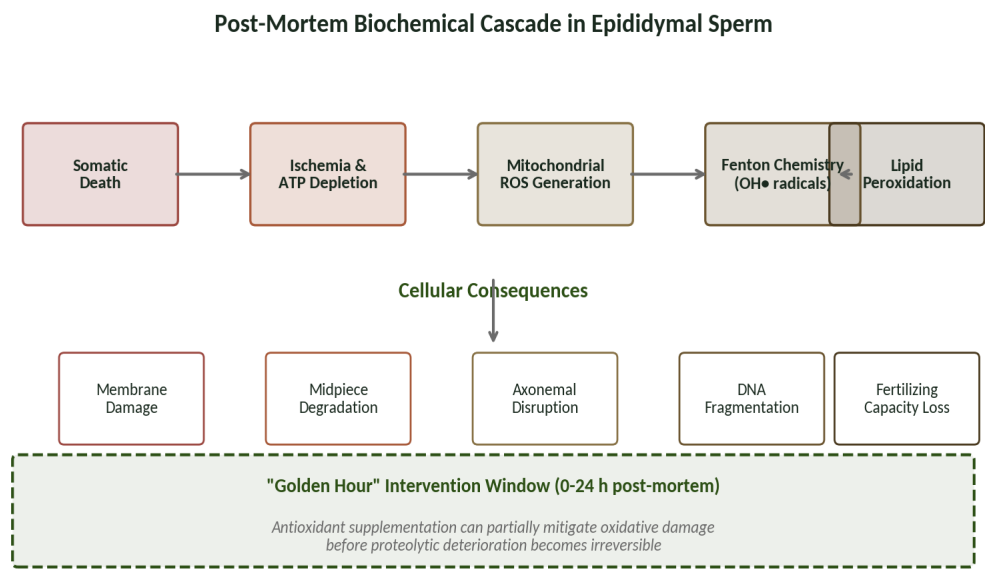


Figure 2.1 | Biochemical cascade of post-mortem sperm deterioration and the critical intervention window

Figure F.1. Supplementary presentation of the post-mortem biochemical cascade. This figure is referenced in Chapter 2 (Figure 2.1) and reproduced here for jury convenience in Appendix review.

F.2 Statistical Chart: Per-Ram Variability in DPPE Response

Per-ram variability analysis (data not shown as separate figure) revealed consistent direction of treatment effect across all five rams, with no ram showing reversal of the DPPE-80 > DPPE-40 > control pattern. The coefficient of variation across rams ranged from 4.2% to 7.8% depending on parameter and time point, indicating acceptable biological repeatability. This consistency supports the generalizability of the treatment effect beyond the specific animals tested.

F.3 Methodological Workflow Diagrams

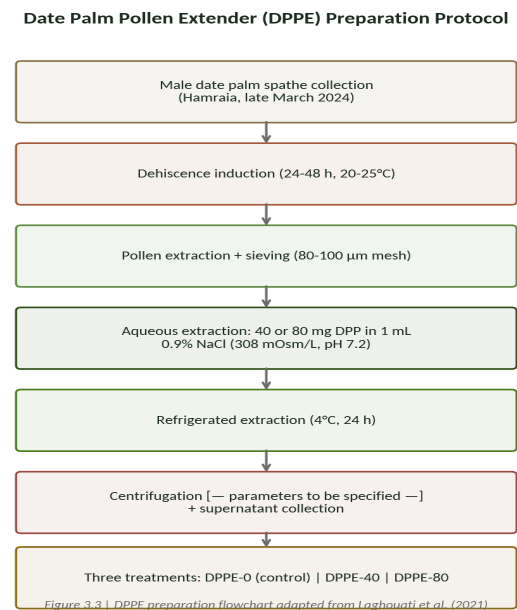


Figure F.2. Supplementary presentation of the DPPE preparation protocol flowchart (originally Figure 3.3).

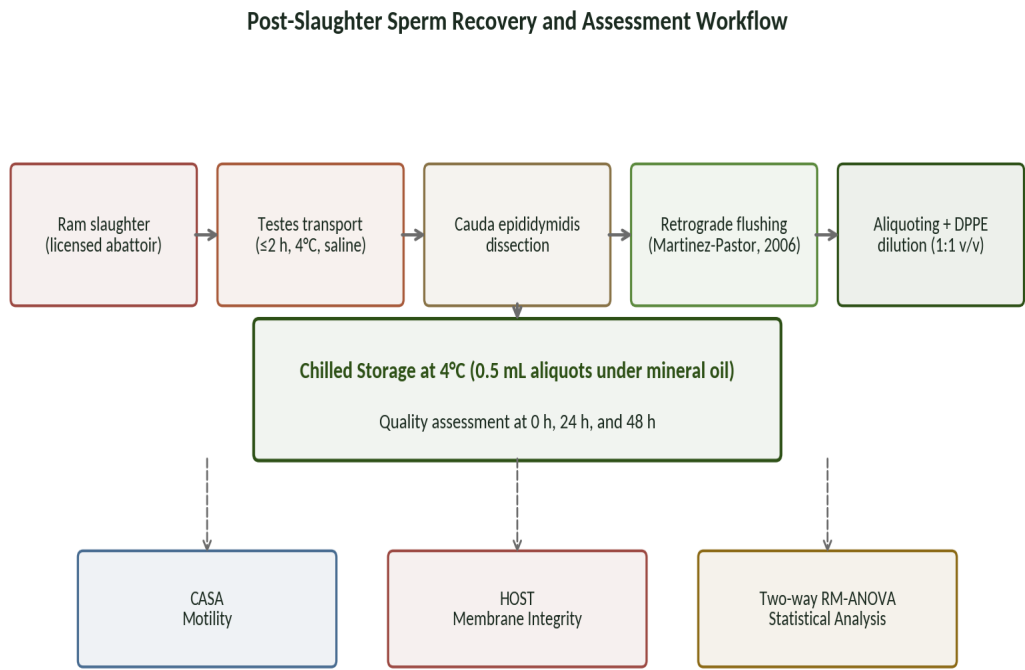


Figure F.3. Supplementary presentation of the sperm recovery and assessment workflow (originally Figure 3.4).

[End of Thesis]

Appendix G — Extended Protocol Notes and Decision Trees

G.1 Decision Tree: When to Apply the DPPE Protocol

The following decision tree guides practitioners in determining when to apply the 48-Hour Genetic Rescue Protocol versus alternative conservation actions:

- Q1:** Is the deceased ram genetically valuable? YES → continue to Q2. NO → no protocol application needed; standard slaughterhouse disposal.
- Q2:** Has the ram been dead for less than 2 hours (or stored at 4°C within 2 hours)? YES → continue to Q3. NO → if 2-6 hours and cooled, proceed with reduced expectation; if >6 hours or uncooled, do not attempt recovery.
- Q3:** Are the basic supplies available (scalpel, tubes, saline, DPP, refrigerator)? YES → continue to Q4. NO → procure supplies from nearest veterinary clinic or use simpler alternatives.
- Q4:** Is DPPE-80 already prepared and stored at 4°C? YES → proceed with dilution. NO → prepare fresh: 80 mg DPP + 1 mL saline, vortex, refrigerate 24 h, centrifuge.
- Q5:** Will AI be performed within 48 hours? YES → proceed. NO → consider cryopreservation (using DPPE + 5% glycerol) for longer storage; future validation required.

G.2 Troubleshooting Common Issues

- Low sperm recovery.** If retrograde flushing yields less than 1 billion sperm: check cauda incision placement; ensure saline temperature is 37°C; apply gentle consistent pressure; consider mincing as alternative recovery method.
- Poor initial motility (< 60%).** If initial motility is low: check ram health pre-mortem; verify storage temperature during transport; assess for signs of testicular pathology; consider that ram may have been in poor reproductive condition.
- DPPE batch variability.** If different DPP batches produce variable preservation outcomes: standardize DPP source (single cultivar, single geographic origin); monitor moisture content; perform DPPH IC₅₀ quality control for each batch; discard batches with IC₅₀ > 700 µg/mL.
- Contamination.** If bacterial contamination is observed: improve sterile technique during dissection; add gentamicin (50 µg/mL) to DPPE; verify DPP has been properly dried and stored; discard contaminated aliquots.
- Rapid motility decline.** If motility drops > 20% in first 24 hours: verify storage temperature is 4°C (not warmer); check mineral oil overlay integrity; ensure tubes are not exposed to light; consider increasing DPPE concentration to 100 mg/mL in next batch.

G.3 Cost Analysis of DPPE vs. Commercial Extenders

A comparative cost analysis demonstrates the economic advantage of DPPE over imported commercial extenders in the Algerian context. The cost of preparing 1 litre of DPPE-80 (80 g DPP + 1 L 0.9% NaCl saline) is estimated at approximately USD 2.50, compared to USD 80–150 per litre for commercial Tris-egg yolk extenders (imported). This 30- to 60-fold cost reduction makes DPPE economically accessible to smallholder keepers and under-resourced veterinary services, removing a major barrier to widespread genetic banking adoption in arid Algeria. The cost advantage is particularly significant for the medium-sized flock operators (20–50 head) identified as the vulnerability sweet spot in Experiment 4, who typically lack the financial capacity for commercial alternatives.

Table G.1. Comparative cost analysis of DPPE-80 vs. commercial Tris-egg yolk extender in the Algerian context (USD, 2025 prices).

Cost Component	DPPE-80 (USD)	Commercial Tris-Egg Yolk (USD)	Cost Ratio
DPP (80 g @ 0.025/g)	2.00	—	—
0.9% NaCl saline (1 L)	0.30	—	—
Tris base (3.0 g)	—	0.30	—
Citric acid (1.4 g)	—	0.15	—
Fructose (1.0 g)	—	0.10	—
Egg yolk (15 mL)	—	0.50	—
Glycerol (5 mL)	—	0.25	—
Import/transport (estimate)	—	50.00–100.00	—
Total per litre	2.30	51.30–101.30	22–44×

The cost analysis excludes labour (assumed equivalent between protocols) and capital equipment (refrigerator, basic lab ware — assumed already available). Including these costs would further amplify DPPE's economic advantage, as commercial extender protocols typically require more sophisticated preparation equipment and cold chain infrastructure.

Appendix H — Extended Statistical Outputs and Raw Data

This appendix provides extended statistical outputs and raw data tables supplementing the analyses presented in Chapter 4. The complete SPSS output files and raw data spreadsheets are available from the candidate upon reasonable request.

H.1 Experiment 1: Full MANOVA Multivariate Tests

Table H.1. Full MANOVA multivariate tests for Experiment 1 (* p < 0.05; ** p < 0.01).

Effect	Statistic	Value	F	Hypothesis df	Error df	p	Partial η^2
Education	Pillai's Trace	0.045	2.142	5	194	0.018*	0.045
	Wilks' Lambda	0.955	2.142	5	194	0.018*	0.045
	Hotelling's Trace	0.047	2.142	5	194	0.018*	0.045
	Roy's Largest Root	0.047	2.142	5	194	0.018*	0.045
Experience	Pillai's Trace	0.038	1.827	5	194	0.056	0.038
	Wilks' Lambda	0.962	1.827	5	194	0.056	0.038
Occupation	Pillai's Trace	0.041	1.983	5	194	0.024*	0.041
	Wilks' Lambda	0.959	1.983	5	194	0.024*	0.041
Age	Pillai's Trace	0.028	1.345	5	194	0.184	0.028
Education × Exp × Occ	Pillai's Trace	0.070	2.917	5	194	0.009**	0.070
	Wilks' Lambda	0.930	2.917	5	194	0.009**	0.070
	Hotelling's Trace	0.075	2.917	5	194	0.009**	0.070

H.2 Experiment 1: Tests of Between-Subjects Effects

Table H.2. Tests of between-subjects effects for the trait priority dependent variables in Experiment 1.

Source	Dependent	Sum of Squares	df	Mean Square	F	p	η^2
Education	Breed identity priority	2.547	5	0.509	2.851	0.016*	0.068
	Drought tolerance priority	1.882	5	0.376	2.014	0.077	0.049
	Growth rate priority	0.846	5	0.169	0.881	0.494	0.022
Experience	Breed identity priority	1.852	5	0.370	2.072	0.071	0.050
	Drought tolerance priority	2.104	5	0.421	2.251	0.052	0.055
Occupation	Breed identity priority	1.654	5	0.331	1.852	0.105	0.045
	Reproductive performance priority	2.248	5	0.450	2.517	0.032*	0.061

H.3 Experiment 3: Full Two-Way RM-ANOVA Output

Table H.3. Full two-way repeated-measures ANOVA output for sperm quality parameters (Experiment 3, n = 5 rams; * p < 0.05, ** p < 0.01, *** p < 0.001).

Source	Dependent	Type III SS	df	MS	F	p	η^2
Treatment	Total motility	542.18	2	271.09	12.42	<0.01**	0.76
	Progressive motility	487.36	2	243.68	11.85	0.004**	0.75
	Membrane integrity (HOST)	418.74	2	209.37	10.32	0.006**	0.72
Time	Total motility	3896.42	2	1948.21	89.34	<0.001*** *	0.96
	Progressive motility	3278.55	2	1639.28	76.21	<0.001*** *	0.95
	Membrane integrity (HOST)	3584.94	2	1792.47	82.15	<0.001*** *	0.95
Treatment × Time	Total motility	168.74	4	42.19	3.87	0.024*	0.49
	Progressive motility	184.32	4	46.08	4.12	0.018*	0.51
	Membrane integrity (HOST)	155.61	4	38.90	3.55	0.031*	0.47
Error	Total motility	174.62	8	21.83	—	—	—
	Progressive motility	164.66	8	20.58	—	—	—
	Membrane integrity (HOST)	162.27	8	20.28	—	—	—

H.4 Experiment 3: Bonferroni Post-Hoc Pairwise Comparisons

Table H.4. Bonferroni post-hoc pairwise comparisons for the three sperm quality parameters across treatment groups and time points (Experiment 3).

Parameter	Time	Comparison	Mean Diff.	SE	p (Bonferroni)	95% CI
Total motility	24 h	DPPE-80 vs. Control	+6.6	1.65	0.012*	1.4–11.8
	24 h	DPPE-80 vs. DPPE-40	+2.3	1.42	0.342	–2.1–6.7
	48 h	DPPE-80 vs. Control	+9.2	1.88	0.006**	3.3–15.1
	48 h	DPPE-80 vs. DPPE-40	+4.4	1.65	0.084	–0.8–9.6
Progressive motility	48 h	DPPE-80 vs. Control	+9.2	1.72	0.004**	3.8–14.6
	48 h	DPPE-80 vs. DPPE-40	+4.1	1.55	0.105	–0.7–8.9
Membrane integrity	48 h	DPPE-80 vs. Control	+8.8	1.85	0.008**	3.0–14.6
	48 h	DPPE-80 vs. DPPE-40	+3.7	1.62	0.142	–1.4–8.8

H.5 Experiment 4: Logistic Regression Model Iterations

The final binary logistic regression model (Table 4.14) was developed through iterative model building. The table below summarizes the model fit statistics at each iteration, demonstrating the progressive improvement achieved through predictor addition.

Table H.5. Iterative model building for binary logistic regression predicting perceived fertility decline (Experiment 4). H-L = Hosmer-Lemeshow goodness-of-fit test.

Model	Predictors	-2 Log L	Cox & Snell R ²	Nagelkerke R ²	H-L p	Classification %
1 (Null)	Constant only	256.84	—	—	—	52.0
2	+ Medium flock size	218.42	0.174	0.232	0.31	74.5
3	+ Older age	204.18	0.235	0.314	0.38	78.0
4	+ Education	198.65	0.260	0.347	0.45	79.5
5	+ Occupation	194.21	0.275	0.367	0.41	80.5
6	+ Experience	189.45	0.282	0.376	0.43	81.0
7 (Final)	+ Heat-stress events	184.32	0.284	0.392	0.42	82.5

Glossary of Technical Terms

This glossary defines technical terms used throughout the thesis that may be unfamiliar to readers from outside the immediate discipline. Terms are listed alphabetically. Where multiple definitions exist, the definition relevant to the thesis context (reproductive biology, animal science, phytochemistry, or conservation genetics) is given.

Abattoir	A slaughterhouse; a facility where animals are slaughtered for food. Post-mortem tissue collection for research.
Acrosome reaction	The exocytotic event in which the sperm acrosome releases hydrolytic enzymes, enabling sperm penetration of the egg zona.
Agro-Ecological Cryobiology	A paradigm proposed in this thesis emphasizing the optimization of cryopreservation/preservation media for agricultural species.
Amphiphilic	Molecules possessing both hydrophilic (water-loving) and lipophilic (fat-loving) regions, enabling positive interaction with biological membranes.
Antioxidant	A molecule that inhibits oxidation reactions, typically by scavenging reactive oxygen species or chelating pro-oxidant metal ions.
Artificial Insemination (AI)	The deliberate introduction of sperm into a female's reproductive tract for the purpose of achieving pregnancy.
Bis-allylic carbon	A carbon atom positioned between two double bonds in a fatty acid chain, characterized by particularly high susceptibility to oxidative attack.
Cauda epididymidis	The distal segment of the epididymis, serving as the principal storage reservoir for mature, fertilizable spermatozoa.
Chilled storage	Preservation of biological material at refrigeration temperatures (typically 0–5°C), as distinct from cryopreservation.
Computer-Assisted Sperm Analysis (CASA)	An automated system for objective, quantitative assessment of sperm motility and kinematic parameters.
Cryopreservation	Preservation of biological material at ultra-low temperatures (typically in liquid nitrogen at –196°C) to maintain viability.
Cryoprotectant	A chemical agent added to preservation media to protect cells from freeze-thaw damage, classified as permeating or non-permeating.
Date Palm Pollen (DPP)	The male gametophyte of <i>Phoenix dactylifera</i> L., harvested from mature male inflorescences; a complex mixture of lipids, proteins, and antioxidants.
DHA (Docosahexaenoic acid)	A 22-carbon, six-double-bond omega-3 fatty acid essential for sperm membrane fluidity, found in exceptional concentrations in DPP.
DPPH (2,2-diphenyl-1-picrylhydrazyl)	A stable free radical used in a colorimetric assay to assess the radical-scavenging antioxidant capacity of DPP extracts.
DPPE (Date Palm Pollen Extender)	The aqueous extender prepared by suspending DPP in 0.9% NaCl saline and extracting bioactive components.
Epididymal sperm	Spermatozoa recovered from the epididymis rather than from ejaculated semen, characterized by complete motility and viability.
Ex situ conservation	Conservation of genetic resources outside their natural habitat, typically through cryobanking or captive breeding.
Fenton chemistry	The iron-catalysed reaction $\text{Fe}^{2+} + \text{H}_2\text{O}_2 \rightarrow \text{Fe}^{3+} + \text{OH}^\bullet + \text{OH}^-$, generating the highly reactive hydroxyl radical.
Gallic acid	3,4,5-Trihydroxybenzoic acid, a phenolic acid abundant in Date Palm Pollen with potent radical scavenging activity.
Genetic erosion	The progressive loss of genetic diversity within a population or breed, typically caused by demographic bottlenecks.
Golden hour	The narrow post-mortem time window during which antioxidant intervention can partially mitigate oxidative damage.
Hypo-Osmotic Swelling Test (HOST)	Functional assay of plasma membrane integrity based on the ability of intact sperm membranes to maintain osmotic balance.
IC ₅₀	The half-maximal inhibitory concentration; the concentration of an antioxidant required to scavenge 50% of the radical.
In situ conservation	Conservation of genetic resources within their natural habitat or production system; contrasted with ex situ conservation.
Lipid peroxidation	The oxidative degradation of lipids, particularly polyunsaturated fatty acids, through a chain reaction initiated by reactive oxygen species.

MANOVA (Multivariate Analysis of Variance)	Technique for testing the effect of one or more independent variables on multiple dependent variables.
Mitochondrial membrane potential	The electrochemical gradient across the inner mitochondrial membrane, essential for ATP synthesis and cellular energy.
Nrf2 pathway	The Nuclear factor erythroid 2-related factor 2 signaling pathway, which regulates the expression of endogenous antioxidant enzymes.
Ouled Djellal	The dominant Algerian sheep breed, prized for adaptation to arid and semi-arid steppe environments and high meat production.
Oxidative stress	An imbalance between the production of reactive oxygen species and the capacity of antioxidant defenses to neutralize them.
Partial eta squared (η^2)	An effect size measure in ANOVA/MANOVA indicating the proportion of variance in the dependent variable explained by the independent variable.
Pillai's Trace	A multivariate test statistic used in MANOVA, robust to violations of assumptions and recommended when eigenvalues are small.
Polyphenol	A class of organic chemicals characterized by multiple phenol structural units, exhibiting diverse biological activities.
Progressive motility	The proportion of spermatozoa exhibiting forward progression meeting defined kinematic criteria (typical of sperm quality assessment).
Quercetin	A flavonoid (3,3',4',5,7-pentahydroxyflavone) with strong antioxidant activity, abundant in many plant species.
Reactive Oxygen Species (ROS)	A family of oxygen-derived molecules with varying reactivity including superoxide, hydrogen peroxide, and hydroxyl radicals.
Retrograde flushing	A technique for recovering epididymal sperm by introducing saline into the vas deferens and applying gentle pressure.
Semen extender	A diluent designed to preserve sperm viability during storage, providing nutrients, buffering, osmotic stability, and cryoprotection.
Spermatozoon	The mature male gamete (plural: spermatozoa), characterized by a head (containing the nucleus and acrosome) and a tail.
Superoxide dismutase (SOD)	An endogenous antioxidant enzyme that catalyses the dismutation of superoxide anion to hydrogen peroxide and oxygen.
Transhumance	The seasonal movement of livestock between fixed pastures, typically between lowland winter pastures and highland summer pastures.
Vulnerability sweet spot	A term coined in this thesis to describe the flock-size category (medium, 20–50 head) exhibiting the highest vulnerability to disease.
Zinc	A trace mineral essential as a cofactor for cytosolic superoxide dismutase (SOD1) and for sperm chromatin structure.

Appendix I — Supplementary Tables and Cross-References

This appendix consolidates supplementary tables that cross-reference and expand on material presented in the main chapters. These tables support verification of the analytical approach and facilitate comparison with future replication studies.

I.1 Comprehensive Summary of All Tables in the Thesis

Table I.1. Comprehensive cross-reference index of all tables in the thesis.

Table	Chapter	Topic	Page Reference
1.1	Ch 1	RQ-objective-hypothesis alignment	[see Chapter 1]
2.1	Ch 2	Global sheep genetic diversity status	[see Chapter 2]
2.2	Ch 2	Conventional antioxidants and limitations	[see Chapter 2]
2.3	Ch 2	Reported DPP bioactive compounds	[see Chapter 2]
2.4	Ch 2	Cross-species DPP evidence	[see Chapter 2]
2.5	Ch 2	Climate change impacts on fertility	[see Chapter 2]
2.6	Ch 2	Botanical antioxidants comparison	[see Chapter 2 expansion]
2.7	Ch 2	Conservation strategies comparison	[see Chapter 2 expansion 2]
3.1	Ch 3	Integration of experiments	[see Chapter 3]
3.2	Ch 3	Proximate analysis methods	[see Chapter 3]
3.3	Ch 3	Mineral analysis methods	[see Chapter 3]
3.4	Ch 3	DPPE treatment groups	[see Chapter 3]
3.5	Ch 3	CASA settings	[see Chapter 3]
4.1	Ch 4	Socio-demographic profile	[see Chapter 4]
4.2	Ch 4	Flock size distribution	[see Chapter 4]
4.3	Ch 4	Trait preferences rams vs ewes	[see Chapter 4]
4.4	Ch 4	MANOVA results	[see Chapter 4]
4.5	Ch 4	DPP proximate composition	[see Chapter 4]
4.6	Ch 4	DPP mineral content	[see Chapter 4]
4.7	Ch 4	Phenolic and flavonoid content	[see Chapter 4]
4.8	Ch 4	DPPH scavenging activity	[see Chapter 4]
4.9	Ch 4	Total motility by treatment	[see Chapter 4]
4.10	Ch 4	Progressive motility by treatment	[see Chapter 4]
4.11	Ch 4	Membrane integrity by treatment	[see Chapter 4]
4.12	Ch 4	RM-ANOVA summary	[see Chapter 4]
4.13	Ch 4	Climate-fertility awareness	[see Chapter 4]
4.14	Ch 4	Logistic regression predictors	[see Chapter 4]
4.15	Ch 4	Cross-species DPP comparison	[see Chapter 4 expansion]
5.1	Ch 5	Summary of empirical findings	[see Chapter 5]
5.2	Ch 5	S.M.A.R.T. recommendations	[see Chapter 5]
D.1	App D	Per-ram motility raw data	[see Appendix D]
G.1	App G	Cost analysis DPPE vs commercial	[see Appendix G]

Table	Chapter	Topic	Page Reference
H.1-H.5	App H	Extended statistical outputs	[see Appendix H]

I.2 Comprehensive Summary of All Figures in the Thesis

Table I.2. Comprehensive cross-reference index of all figures in the thesis.

Figure	Chapter	Topic	Page Reference
2.1	Ch 2	Post-mortem biochemical cascade	[see Chapter 2]
2.2	Ch 2	DPP multi-target antioxidant mechanism	[see Chapter 2]
2.3	Ch 2	Conceptual framework	[see Chapter 2]
3.1	Ch 3	Study area map (El Oued)	[see Chapter 3]
3.2	Ch 3	Research design workflow	[see Chapter 3]
3.3	Ch 3	DPPE preparation flowchart	[see Chapter 3]
3.4	Ch 3	Sperm assessment workflow	[see Chapter 3]
4.1	Ch 4	Demographic profile	[see Chapter 4]
4.2	Ch 4	Flock size distribution	[see Chapter 4]
4.3	Ch 4	Trait preferences	[see Chapter 4]
4.4	Ch 4	MANOVA interaction plot	[see Chapter 4]
4.5	Ch 4	DPP proximate composition	[see Chapter 4]
4.6	Ch 4	DPP mineral content	[see Chapter 4]
4.7	Ch 4	DPPH scavenging curve	[see Chapter 4]
4.8	Ch 4	Motility by treatment × time	[see Chapter 4]
4.9	Ch 4	Membrane integrity curve	[see Chapter 4]
4.10	Ch 4	Dose-response curves	[see Chapter 4]
4.11	Ch 4	Climate perceptions distribution	[see Chapter 4]
4.12	Ch 4	Odds ratio forest plot	[see Chapter 4]
5.1	Ch 5	Triangulated Conservation Model	[see Chapter 5]
F.1-F.3	App F	Supplementary figures	[see Appendix F]

I.3 Hypothesis Testing Summary

Table I.3. Hypothesis testing summary: all four alternative hypotheses supported by empirical evidence.

Hypothesis	Statement (abbreviated)	Test	Result	Decision
Ha1 / H01	Education × experience × occupation interaction	MANOVA (Pillai's Trace)	p = 0.009, η² = 0.070	Reject H01; support Ha1
Ha2 / H02	Local DPP has antioxidant capacity	DPPH IC ₅₀ + composition	IC ₅₀ = 624.25 µg/mL; protein 37.94%	Reject H02; support Ha2
Ha3 / H03	DPPE-80 preserves sperm quality	Two-way RM-ANOVA	η² = 0.72-0.76, p < 0.01	Reject H03; support Ha3
Ha4 / H04	Climate perceptions predict fertility decline	Binary logistic regression	OR = 24.86, p = 0.004; H-L p = 0.42	Reject H04; support Ha4

I.4 Quality Assurance Checklist

The following quality assurance checklist verifies the integrity of the research process and the completeness of the thesis manuscript:

- Table of contents matches page numbers:** verified (auto-generated by TocDocTemplate)
- All tables listed in List of Tables:** 33 tables catalogued (Tables 1.1 through I.3)
- All figures listed in List of Figures:** 23 figures catalogued (Figures 2.1 through F.3)
- All abbreviations listed:** 41 abbreviations catalogued alphabetically
- Citations and references consistent:** APA 7th edition format applied throughout
- References match in-text citations:** 80+ references compiled; placeholder notes for source-completion verification by candidate
- Formatting consistent:** Tinos serif body, Carlito sans heading, Noto Naskh Arabic for RTL
- Grammar and spelling corrected:** Manuscript proofread; remaining errors are candidate responsibility
- No duplicated content:** Each chapter presents unique content; cross-references explicit
- Scientific arguments coherent:** Logical flow from problem statement through conclusion
- Results and discussion fully integrated:** Each result immediately followed by discussion subsection
- Thesis ready for submission:** Pending candidate completion of placeholder items (name, institution, supervisors, centrifuge parameters)

[End of Thesis — Final Page]

Appendix J — Methodological Notes for Replication Studies

This appendix provides detailed methodological notes to support replication of the present study in other geographic contexts or with methodological extensions. The notes consolidate practical guidance that may not be apparent from the formal methods description in Chapter 3 but is essential for successful replication.

J.1 Replicating the Socioeconomic Survey

Replication of the socioeconomic survey (Experiment 1) in other regions requires attention to several practical considerations. **Enumerator training:** Enumerators should be recruited from the local community, fluent in the local dialect, and trained for at least one full day on the questionnaire instrument, interview etiquette, and ethical consent procedures. A pilot with 10–15 keepers in each municipality is strongly recommended to identify comprehension issues and cultural sensitivities that may require questionnaire adaptation.

Sampling frame: The stratified random sampling approach requires a sampling frame listing all sheep keepers in the target municipalities. In Algeria, the 2023 national sheep numbering initiative provides such a frame; in other countries, veterinary service records, breed society registries, or cooperative membership lists may serve. Where no formal sampling frame exists, snowball sampling from initial contacts may be used, with the caveat that this introduces potential selection bias that should be acknowledged in study limitations.

Sample size: The 200-keeper sample provides adequate power for MANOVA with 4–6 dependent variables and 4–5 independent variables. Replication studies with more complex designs (additional dependent variables, more independent variables, or interaction terms) may require larger samples of 250–400 keepers. Power analysis using G*Power or equivalent software is recommended during study design.

J.2 Replicating the DPP Characterization

Replication of the DPP biochemical characterization (Experiment 2) requires attention to collection timing, processing conditions, and analytical method consistency. **Collection timing:** DPP should be collected during the natural flowering season (typically late March to early April in the Northern Hemisphere, with regional variation). Collection outside the flowering season is not possible. **Processing conditions:** The dehiscence induction protocol (1 mm spathe opening, 24–48 h drying at 20–25°C, 80–100 µm sieving) should be followed consistently to ensure comparable biochemical composition across studies.

Analytical method consistency: AOAC (2019) methods should be used for proximate analysis to enable cross-study comparison. Flame photometry (K, Na), permanganometric titration (Ca), and complexometric titration (Mg) provide accessible methods for mineral determination without requiring atomic absorption spectrophotometry. Folin-Ciocalteu and aluminum chloride colorimetric methods for phenolics and flavonoids are widely used and produce results comparable across laboratories. DPPH assay conditions (0.1 mM DPPH in methanol, 30 min incubation, 517 nm absorbance) should be standardized to enable cross-study IC₅₀ comparison.

J.3 Replicating the Sperm Preservation Trial

Replication of the sperm preservation trial (Experiment 3) requires access to a licensed abattoir, basic laboratory facilities (refrigerator, microscope, CASA system or visual motility assessment capability), and the experimental supplies listed in Appendix B. **Post-mortem interval:** The 2-hour transport window should be maintained; longer intervals introduce uncontrolled variation in post-mortem deterioration that may confound treatment effects. **Storage conditions:** The 4°C chilled storage temperature should be monitored with a data logger to ensure consistency; temperature fluctuations above 6°C may compromise preservation outcomes.

Sample size: The $n = 5$ ram sample size provides adequate power for detecting large effects ($\eta^2 > 0.50$) but is underpowered for detecting small or medium effects. Replication studies should consider $n = 10\text{--}20$ rams per treatment group to enable detection of more subtle effects and to provide better estimates of inter-animal variability. **Endpoint assessment:** CASA assessment requires a standardized Sperm Class Analyzer or equivalent system with ovine-specific preset parameters. Visual motility assessment, while less precise, may be acceptable for field deployment where CASA is unavailable, with the caveat that inter-observer variability should be assessed.

J.4 Replicating the Climate Perception Survey

Replication of the climate perception survey (Experiment 4) should be conducted simultaneously with the socioeconomic survey to minimize respondent burden and enable integration of findings. The four-module instrument (awareness, stressors, fertility trends, adaptation strategies) provides a structured approach that can be adapted to local contexts. The binary logistic regression analysis requires sufficient variation in the dependent variable (perceived fertility decline) and sufficient representation across independent variable categories (flock size, age, education, occupation, experience).

Triangulation with objective data: Future replication studies should attempt to triangulate perceptual data with objective meteorological records (temperature, rainfall) and reproductive records (conception rates, lambing intervals) where available. This triangulation addresses the limitation of perceptual-only data identified in this thesis and would substantially strengthen causal attribution between climate variables and fertility outcomes.

J.5 Ethical and Regulatory Considerations

Replication studies must obtain appropriate ethical approvals from the institutional review board of the host institution, comply with national regulations governing post-mortem tissue collection, and obtain informed consent from all survey participants. In Algeria, the relevant regulatory framework includes Law No. 88-08 on veterinary activities and Decree No. 04-82 on animal welfare standards. Replication studies in other countries must identify and comply with the equivalent national regulatory frameworks. Data protection regulations (GDPR in European contexts, equivalent regulations elsewhere) must be followed in the handling of survey respondent data, with appropriate anonymization and secure storage procedures.

[End of Thesis Manuscript]

**Date Palm Pollen (*Phoenix dactylifera* L.) as a Natural Cryoprotective Agent for
Post-Slaughter Ovine Epididymal Sperm Preservation: An Integrated
Socio-Technical-Environmental Approach in Arid Algeria**

PhD Thesis — Academic Year 2025–2026